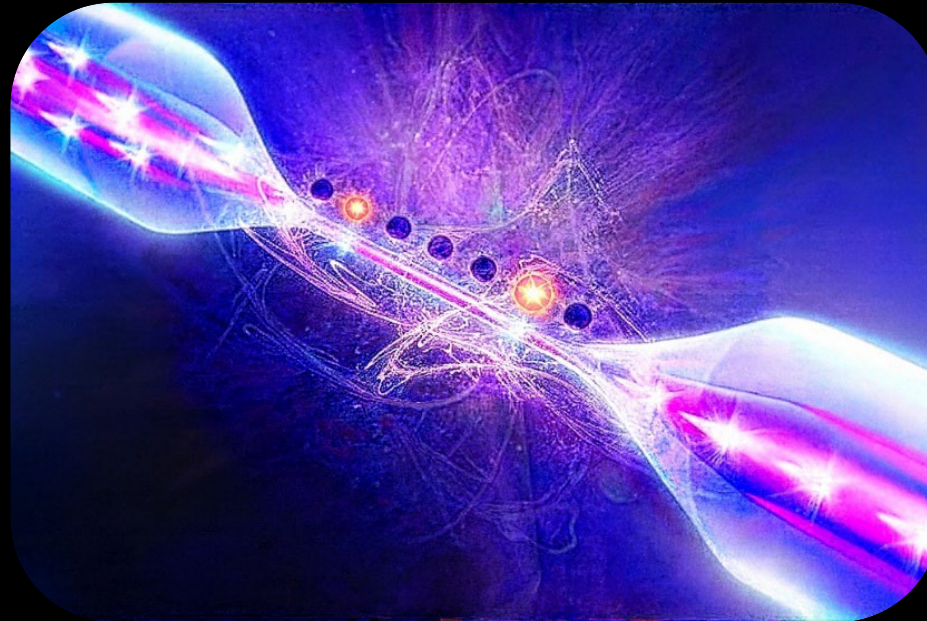


Quantum Interfaces



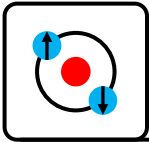
Transferring quantum information
between light and matter



Shayne Bennetts

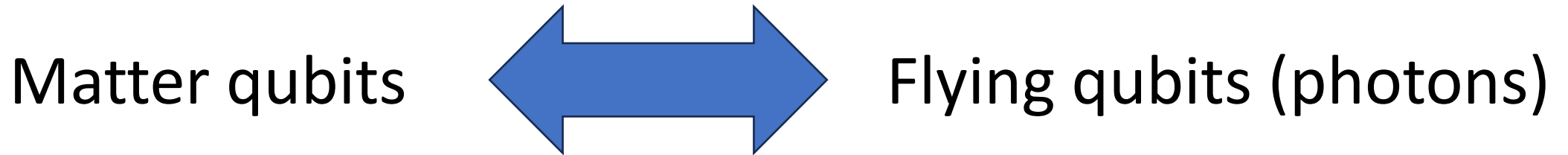
Quantum Technology Based on Atomic, Molecular, and Optical Systems

25 November 2025



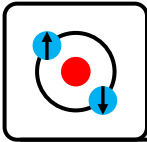
Quantum Interfaces

This lecture is about how to transfer quantum information between stationary and flying qubits.



This is a key tool for:

1. Long-distance quantum communication
2. Entanglement distribution
3. Quantum repeaters
4. Modular architectures
(multiple quantum processors linked optically)

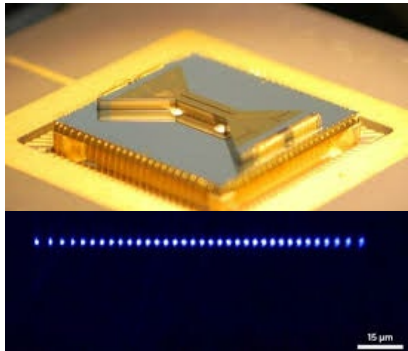


The challenge: Linking different quantum platforms

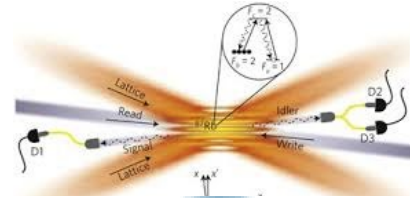
Optical interface

Microwave interface

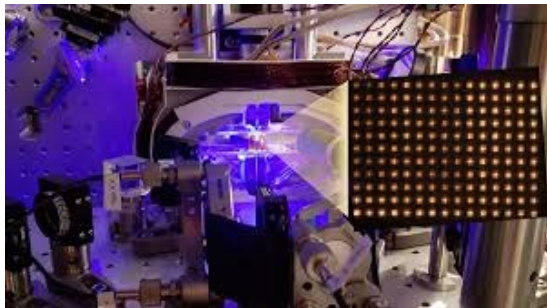
Ions



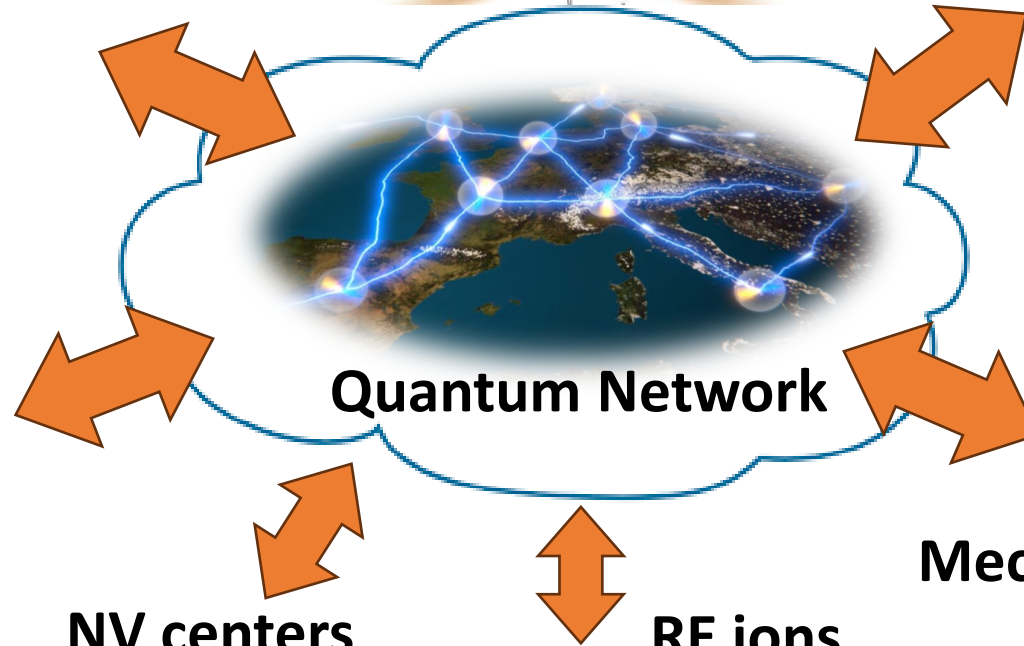
Neutral atom clouds



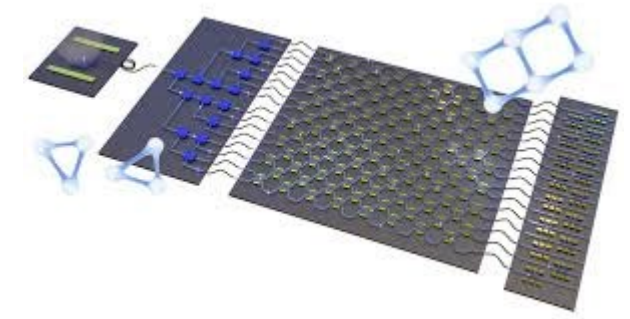
Neutral atoms



Quantum Network



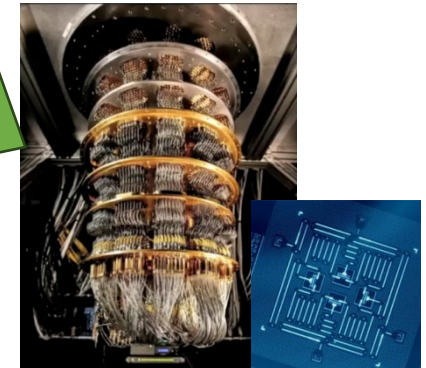
Photonics



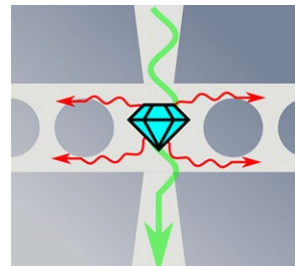
Rydberg Atoms

Mechanical

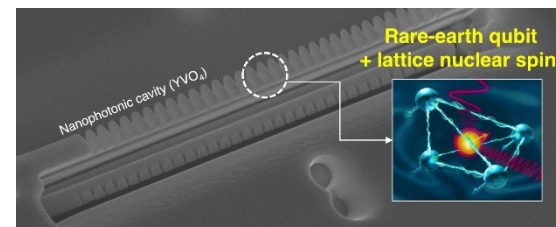
Superconducting

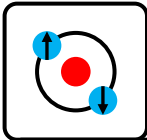


NV centers



RE ions

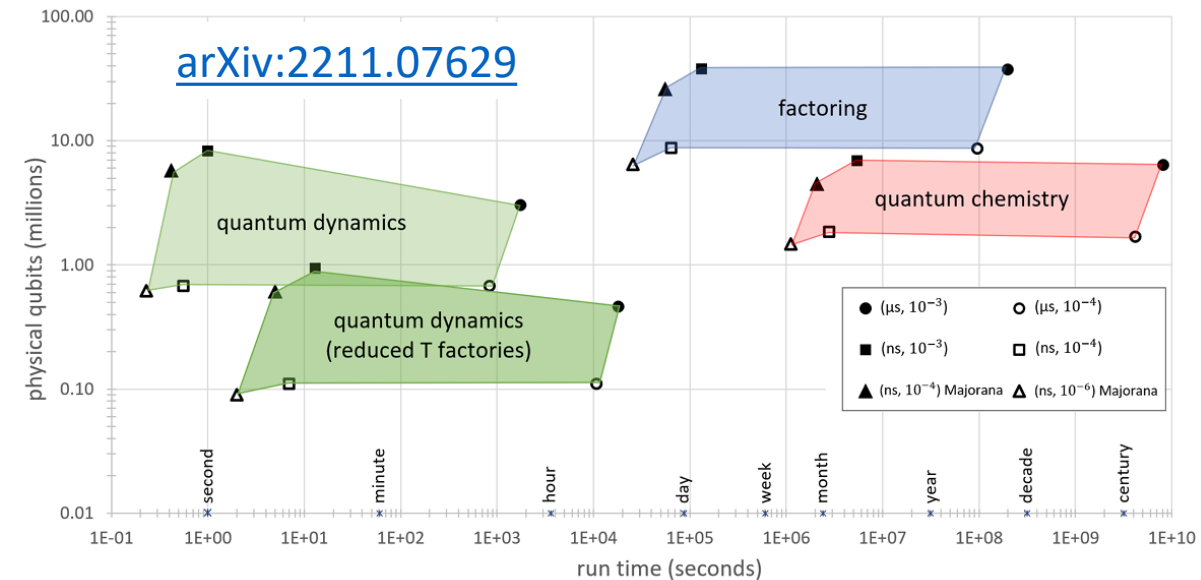




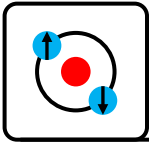
Today there is no “winner” for quantum computing

	Superconducting	Photonic	Neutral atom (cavity)	Neutral atom (Rydberg)	Ions
Speed	~ 50 ns	~100 ps	~1 ns	~1 us/100us	~ 100us
Qbits	1e3	Emerging	Emerging	1e3/1e4	1e1/1e2
Memory		0			
Errors	1e-3	Emerging	Concept	1e-2/1e-3	1e-4
Network	Concept	Native	Advanced	Emerging	Advanced

Quantum resource requirements:

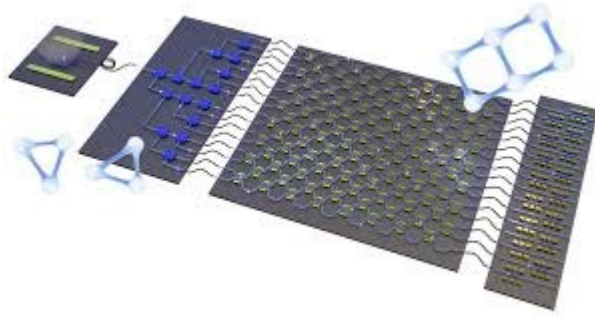


There is no winner: Quantum interfaces allow combining the advantages of different technologies.

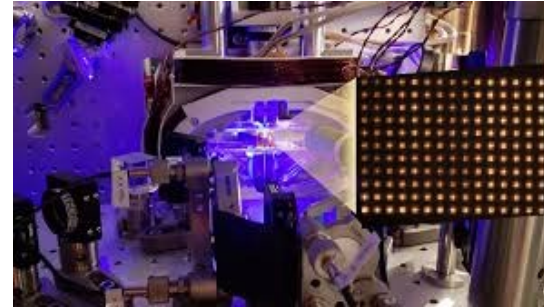


Potential benefits from combining quantum platforms

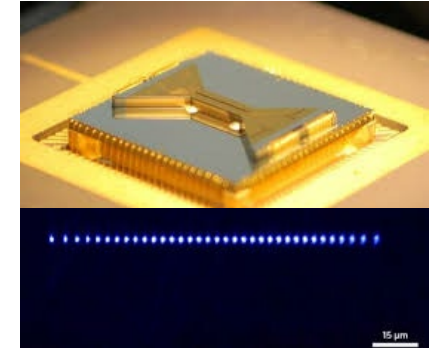
Photonics



Neutral atoms



Ions

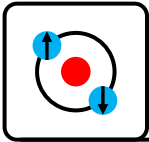


Memory: Atoms and Ions can provide long coherence, quantum memories with robust, deterministic error correction.

Deep circuits: Atoms/ions can enable deep, deterministic, high fidelity complex circuits with error correction.

Scaling/Distributed Quantum: Photonics allows linking of multiple processors enabling distributed quantum systems and scaling.

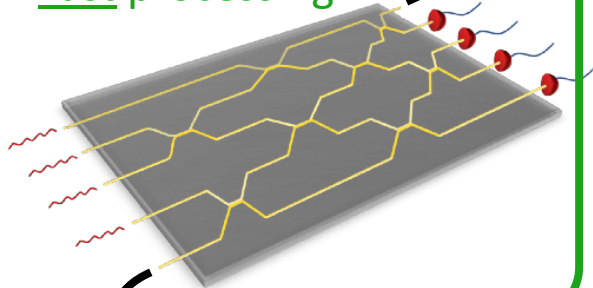
Fast: Photonic quantum coprocessors can execute specific circuits/algorithms with high speed.



The challenge: Interfacing

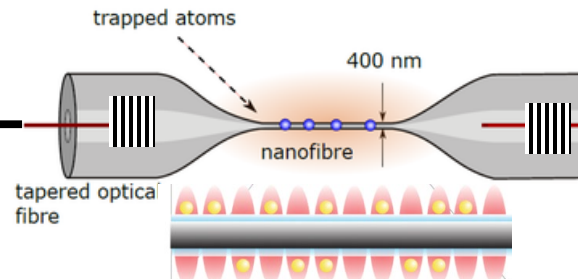
Photonic QPU

- Fast processing



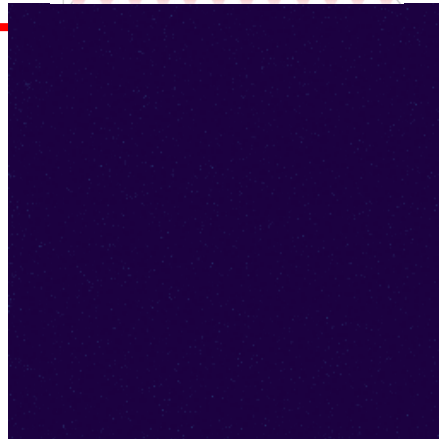
Optical nanofiber cavity

- Interfacing light and matter
- 10s-100s of qubits Q-RAM



Neutral atom QPU + memory

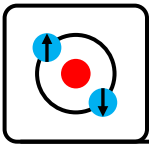
- 10,000-100,000 qubits/QPU
- Coherent, high fidelity, persistent
- Logical qubits
- High fidelity (slow/deep) processing



Distributed computing and quantum resource distribution



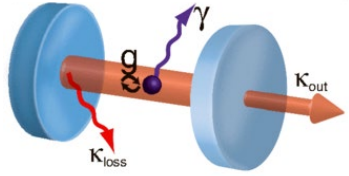
Future quantum systems will combine strengths from multiple technology platforms



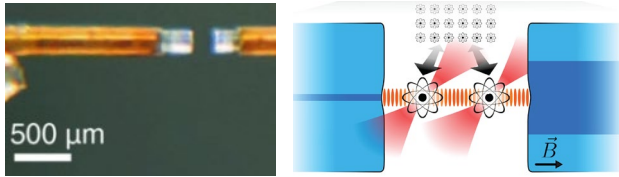
A “zoo” of photonic-atom interfaces

Bulk cavities

Free space cavities

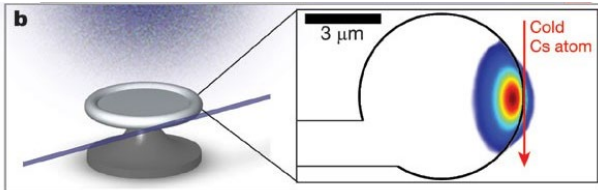


Rempe: Rev. Mod. Phys. 87, 1379 (2015).

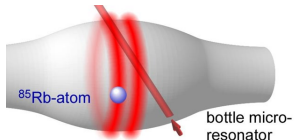


Blatt/Northup: Rev. Sci. Instr. 84, 123104 (2013).
Lukin/Vuletic groups, Science **387**, 6740, 1301 (2025).

Evanescently coupled cavities



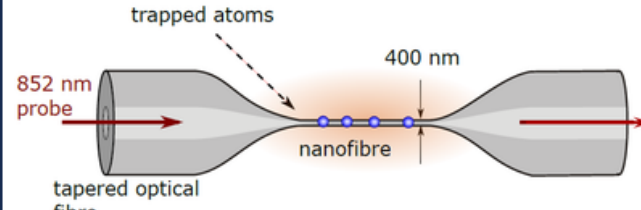
Kimble, Aoki, Nature **443**, 671–674 (2006)



Rauschenbeutel group:
PRL **104**, 203603 (2010).

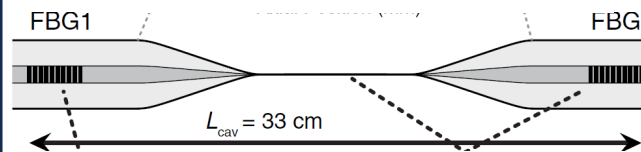
Nanofibers

Atoms evanescently coupled to optical nanofibers



Course objective: Understand quantum interfacing requirements, solutions and challenges.

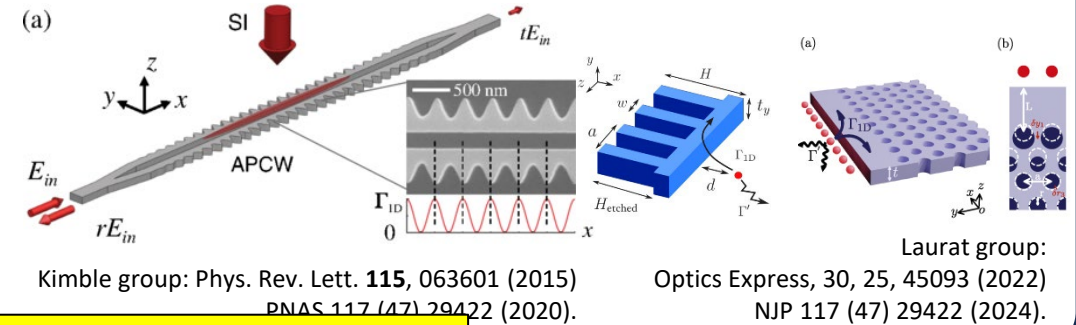
Atoms evanescently coupled to nanofiber cavities



Aoki group/nanoQT: PRL **115**, 093603 (2015)
Optics Lett. **50**, 17 5294 (2025)

Chips

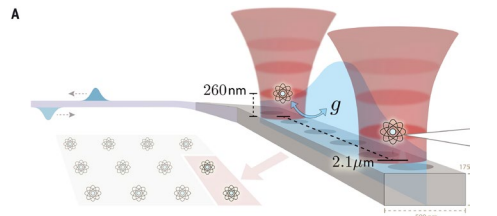
Atoms evanescently coupled to chip waveguides



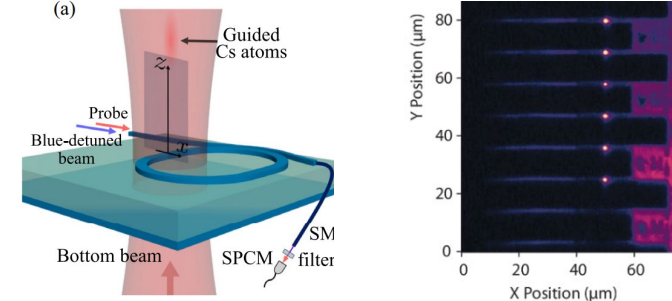
Kimble group: Phys. Rev. Lett. **115**, 063601 (2015)
PNAS **117** (47) 29422 (2020).

Laurat group:
Optics Express, 30, 25, 45093 (2022)
NJP **117** (47) 29422 (2024).

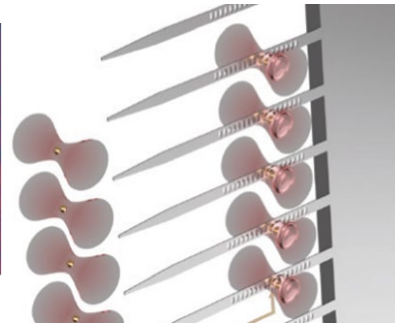
Atoms evanescently coupled to chip based cavities



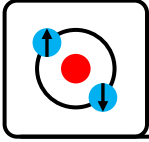
Lukin/Vuletic groups, Nature **508**, 241 (2014), Science **373**, 6562, 1511 (2021).



Hung group: PRL **130**, 103601 (2023)



Bernien group: Nature Comms **15**, 6156 (2024)



Part II: Foundations and Theory

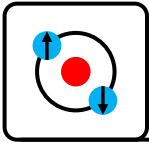
1. Atom–Light Interaction

(recap of standard equations)

2. 1D/Waveguide atom-photon interactions

3. Cavity QED





Foundations and Theory

Atom – Light Interaction

(recap)

References:

MIT Wolfgang Ketterle's course

Eg Lecture 24-25: Coherence IV-V

<https://youtu.be/Y7UsD2SNllw?list=PLUI4u3cNGP62FPGcyFJkzhqq9c5cHCK32>

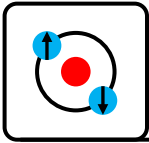
Quantum Optics (LMU Immanuel Bloch's course)

<https://youtu.be/pCa1uAvjRN4?list=PLNMgVqt8MREzCQbgPxBiHKIFbV7Xszkds>

Atomic Physics, Foot

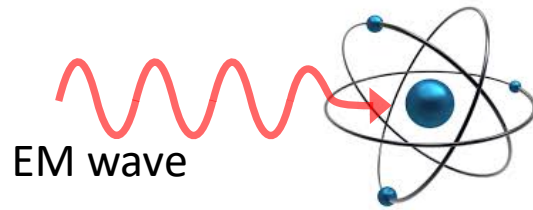
Quantum Optics, Scully and Zubary

<https://hal.science/jpa-00247657v1/document>



Atom-light interaction

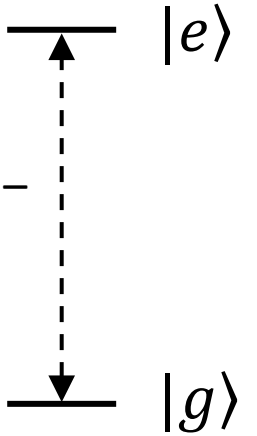
Consider an atom with a ground state $|g\rangle$ and an excited state $|e\rangle$:



EM wave

Dipole moment $\mathbf{d}_{eg}$

Microwave (GHz) –
Optical (THz)



$$\mathbf{E}(t) = \mathcal{E}_0 \cos(\omega_L t - kz)$$

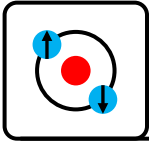
$$H_{\text{int}} = -\mathbf{d} \cdot \mathbf{E}$$

$$H_{\text{int}} = -\frac{\hbar \Omega_R}{2} (e^{-i\omega_L t} |e\rangle \langle g| + h.c.) \text{ (in the RWA)}$$

When light with frequency $\omega_L = (\omega_0 + \Delta)$, Δ the detuning, illuminates the atom, the light interacts with the atomic dipole $\mathbf{d}$:

The Rabi frequency, Ω_R quantifies strength of the coupling to the field

$$\Omega_R = \frac{\mathbf{d}_{eg} \cdot \mathcal{E}_0}{\hbar}$$



Optical Bloch Equations

For a 2 level atom the wavefunction is given by:

$$|\Psi(t)\rangle = c_a(t)|g\rangle + c_b(t)|e\rangle$$

The solutions for c_a and c_b are then given by:

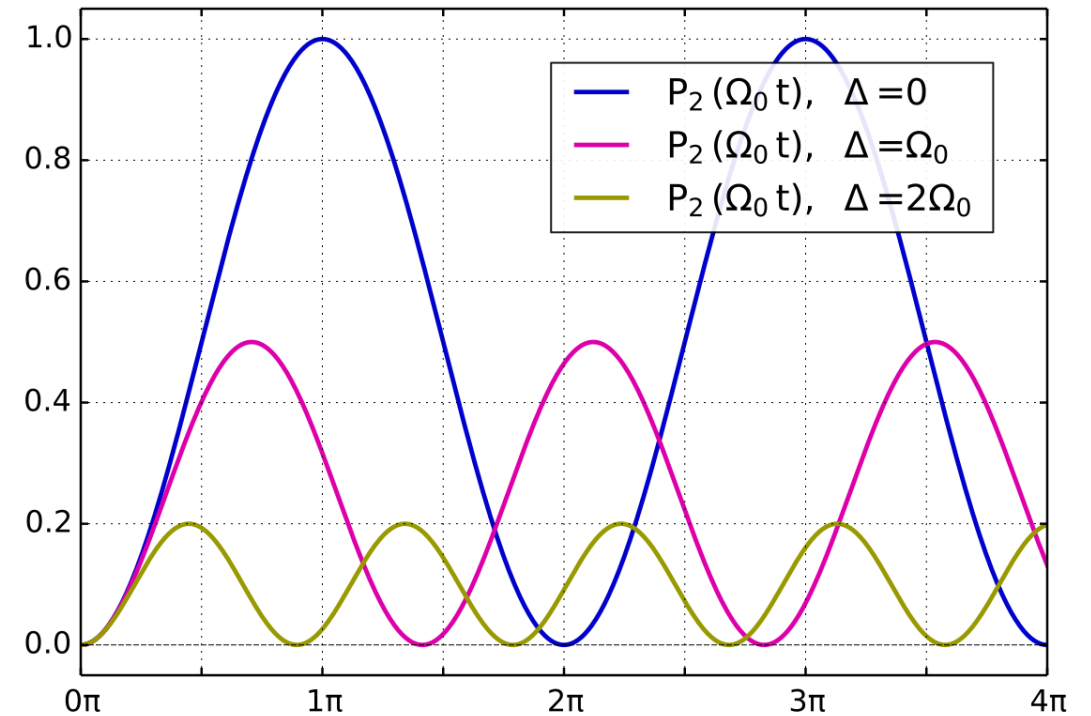
$$c_a(t) = (a_1 e^{i\Omega t/2} + a_2 e^{-i\Omega t/2}) e^{i\Delta t/2}$$

$$c_b(t) = (b_1 e^{i\Omega t/2} + b_2 e^{-i\Omega t/2}) e^{-i\Delta t/2}$$

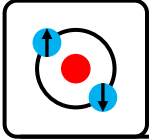
$$\Omega = \sqrt{\Omega_R^2 + \Delta^2}$$

The Optical Bloch Equations

(constants a, b defined by the initial conditions)

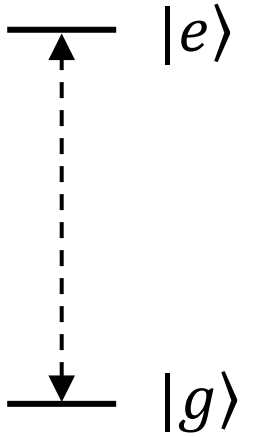


*wikipedia

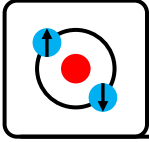


Dissipation: Spontaneous emission

Due to vacuum fluctuations an atom in an excited state $|e\rangle$ can spontaneously decay to the ground state $|g\rangle$ with a natural linewidth γ and an excited state lifetime τ .



$$\gamma = \frac{1}{\tau} = \frac{\omega_0^3 |\mathbf{d}_{eg}|^2}{3\pi\epsilon_0 \hbar c^3}$$



Atom – Photon Interaction in free space

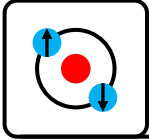
Resonant scattering cross section for a dipole in a beam of light is given by:

$$\sigma_0 = \frac{3\lambda_0^2}{2\pi}$$

The area of the “shadow” caused
by a dipole in a beam of light

If our light is detuned by Δ and the transition rate is γ the scattering cross section becomes:

$$\sigma(\Delta) = \frac{\sigma_0}{1 + 4\Delta^2/\gamma^2}$$



Many atom scattering (Beer's law)

Assume

- Optical beam of area $A=(\pi/4)W_0^2$, (W_0 the gaussian waist radius)
- Number of scatterers per unit volume n
- Probability of a single scatterer scattering a photon σ/A

If we have many dipoles we will attenuate light according to Beer's law

$$dI(z) = -\frac{\sigma}{A} n A I(z) dz$$

$$I(z) = I(0)\exp(-\alpha z) \quad \alpha = \sigma n$$

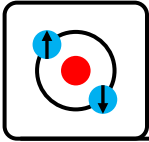
α – optical depth

$I(z)$ - light intensity vs position

n - density of absorbers

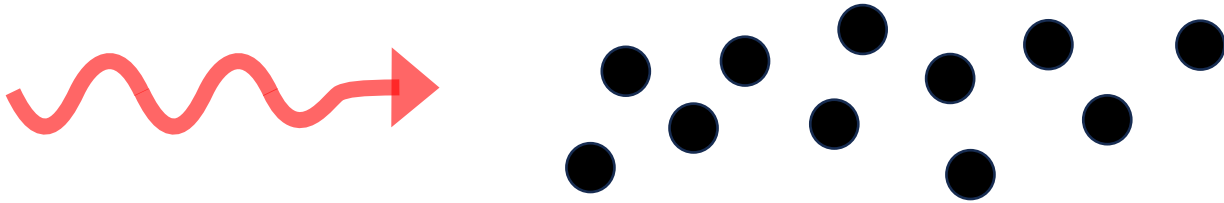
σ - cross section

$A = (\pi/4)W_0^2$ – mode area where W_0 is the gaussian waist radius

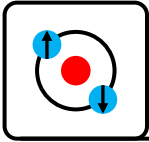


Fidelity limit for storing a single photon?

Single photon carrying quantum information



$$\text{Fidelity} \leq 1 - \frac{5.8}{OD}$$



Quantum memory fidelity limits

Rb 2DMOT based quantum memory

- Number density $n \sim 1e10-1e11$ at/cm³
- Cloud waist radius 1mm, 1cm long

Gives an atom number in cloud $\sim 1e9$ atoms

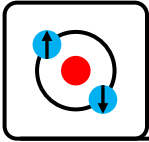
Atoms in the optical mode:

- Waist $\sim 200\mu\text{m}$, $L \sim 5\text{mm}$
- Gives an atom number in optical mode $\sim 3e7$ atoms

Optical depth:

- Resonant cross section $\sigma = 3\lambda^2/(2\pi) = 3e-9$ cm²
- $OD = n\sigma L \sim 70$ (used $3e7$ atoms)
- Fidelity $\leq 1 - 5.8/OD = 91.7\%$

	2D MOT quantum memory
N	3e7
OD	70
β	1e-6 (effective)
Fidelity	91.7%

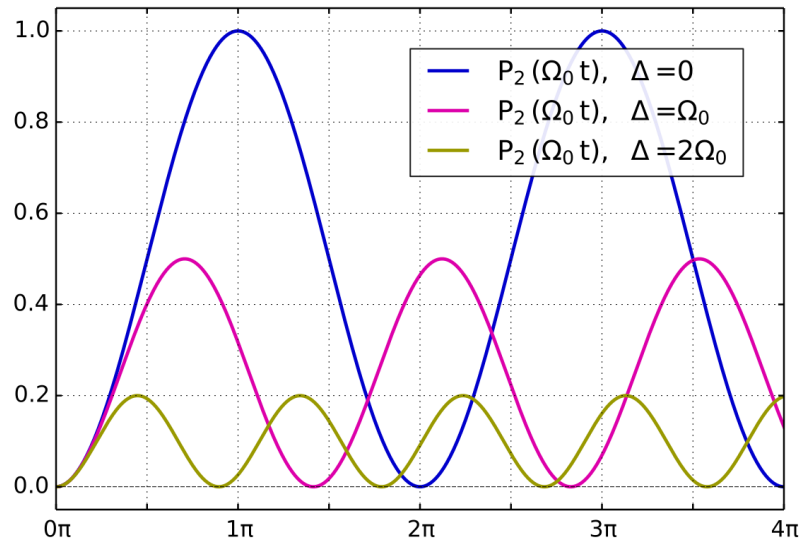


Atom – Photon Interaction regimes

Coherent interactions (strong)

Closed system, information is not lost to the environment

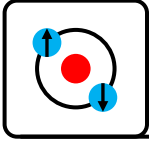
$$\Omega_R \gg \gamma$$



Absorption/scattering

Open system, information rapidly lost to the environment

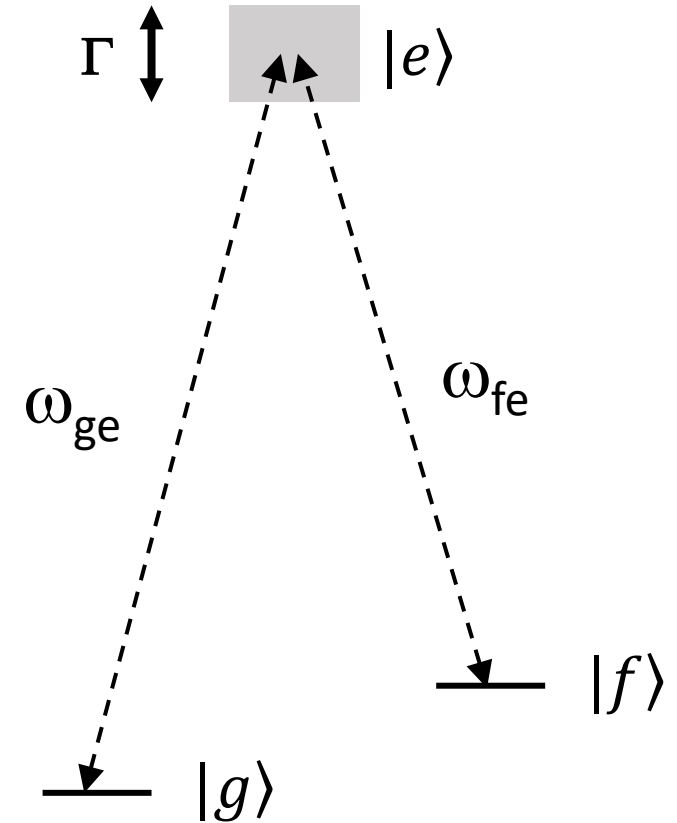
$$\Omega_R \ll \gamma$$

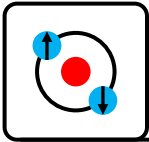


3 level atom, dark states

In a 3 level (Λ) system interference between the 2 coherent couplings ω_{ge} and ω_{fe} means there always exists a **dark state**.

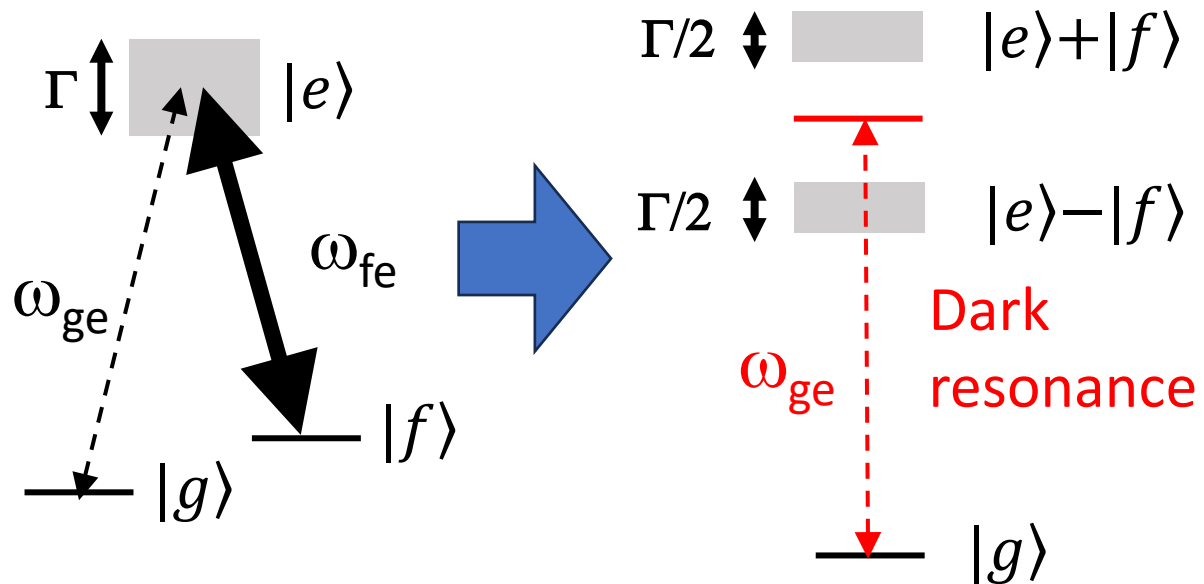
The dark state is a coherent superposition of the 3 levels with amplitudes that depend on the strength of the couplings applied.





Electromagnetically induced transparency (EIT)

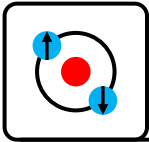
A **strong coupling** laser with Rabi frequency Ω_c will **mix** the coupled states (coupling rate must be stronger than the spontaneous emission/decoherence rate of the excited state).



$$|D\rangle = \cos \theta |g\rangle - \sin \theta |f\rangle$$

$$\tan \theta = \frac{g\sqrt{N}}{\Omega_c}$$

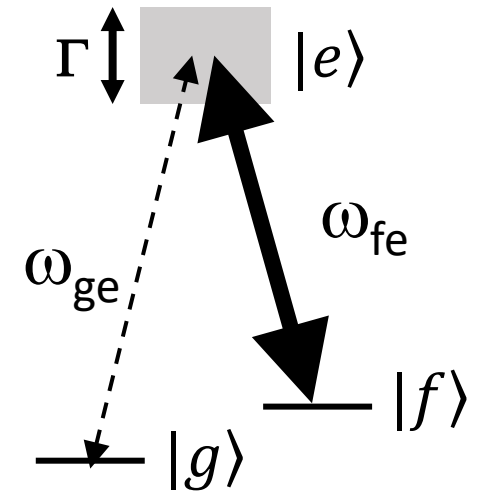
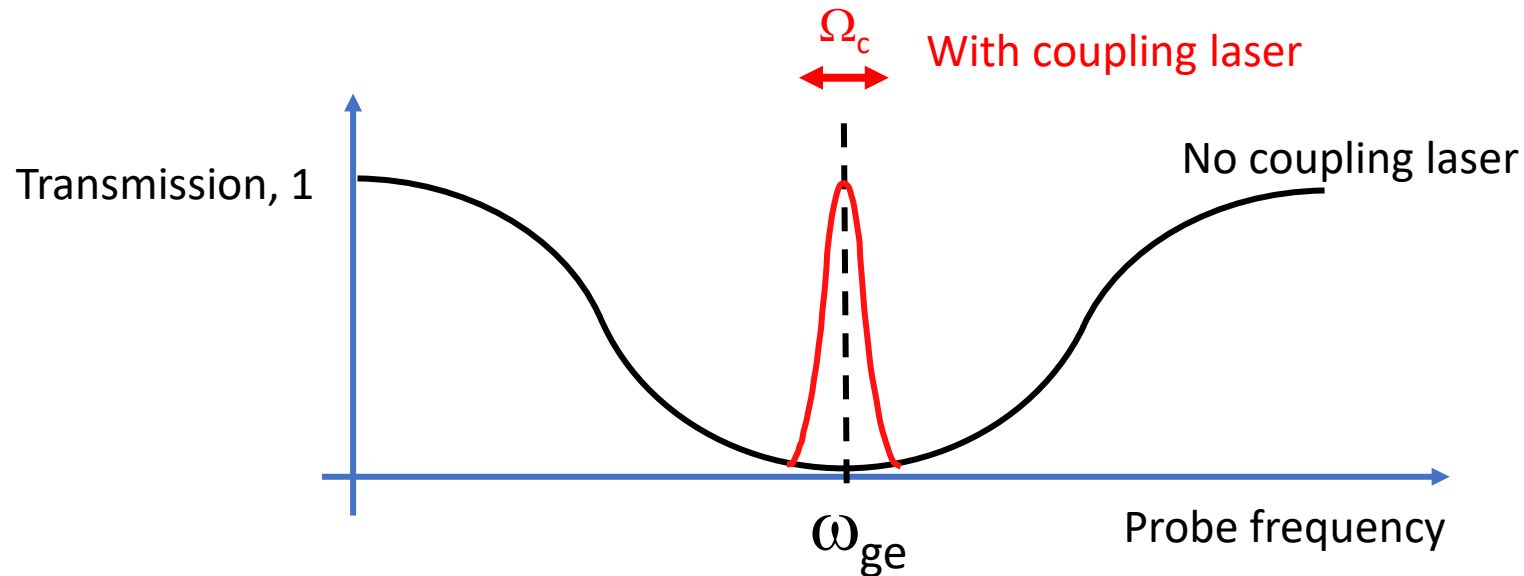
θ is the mixing angle



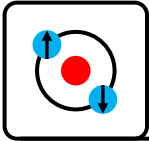
Electromagnetically induced transparency (EIT)

If a probe is applied at a frequency exactly in between the split state interference results in a **dark resonance** with no absorption (**electromagnetically induced transparency**).

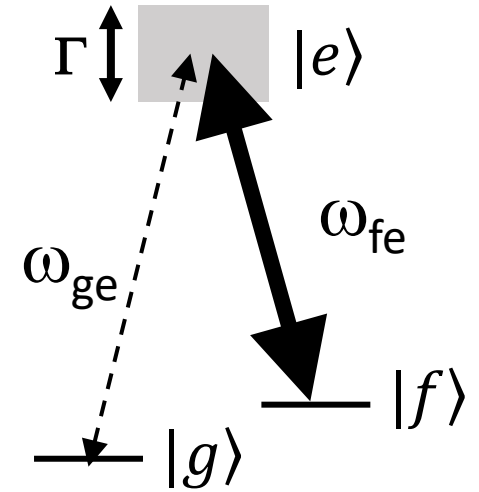
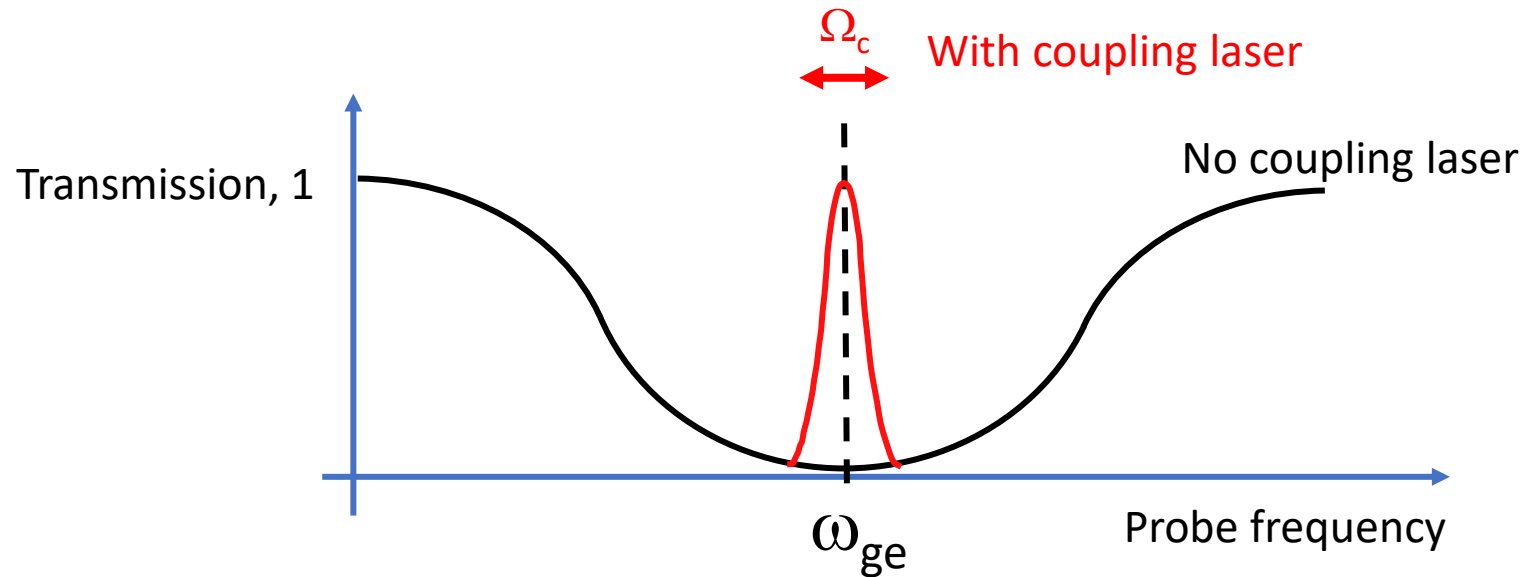
The width of the transparency feature is the Rabi frequency of the coupling laser



The sharp dispersive feature also slows light with the resulting group velocity:
$$v_g \approx c \frac{\Omega_c^2}{g^2 N}$$

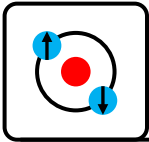


Frozen light/stopped light/quantum memory



The sharp dispersive feature also slows light with the resulting group velocity: $v_g \approx c \frac{\Omega_c^2}{g^2 N}$

If the control laser is switched off while the probe is passing Ω_c (and group velocity) gets smaller and smaller until it becomes 0 and the photon is coherently stored in the levels $|g\rangle$ and $|f\rangle$

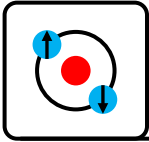


Foundations and Theory

Atom-photon interaction in a waveguide (waveguide QED)

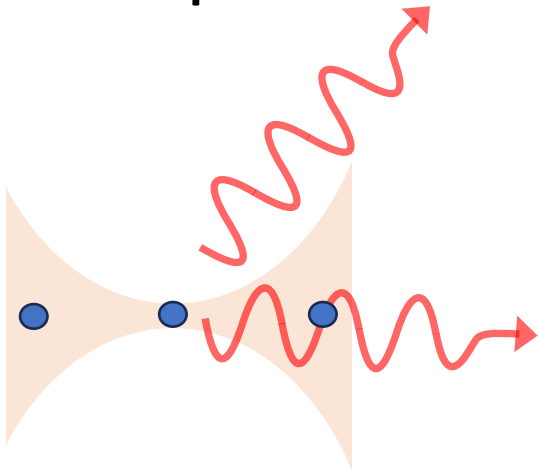
References:

- arXiv:2103.06824 “Waveguide quantum electrodynamics: collective radiance and photon-photon correlations”
Nature **541**, 473 (2017), “Chiral quantum optics”
- arXiv:1703.10533, “Optical Nanofibers: a new platform for quantum optics”
- arXiv:1912.06234, “Atomic-waveguide quantum electrodynamics”
Nature **583**, 775 (2020), “Waveguide quantum electrodynamics with superconducting artificial giant atoms”



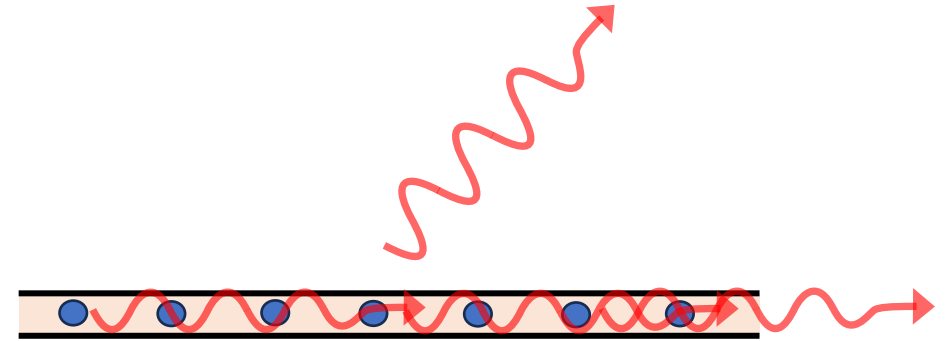
Scattering in a waveguide

Free space:

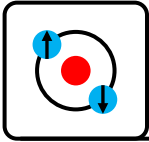


If you focus light to $A \sim \lambda^2$ you get a strong interaction but diffraction means it rapidly diverges

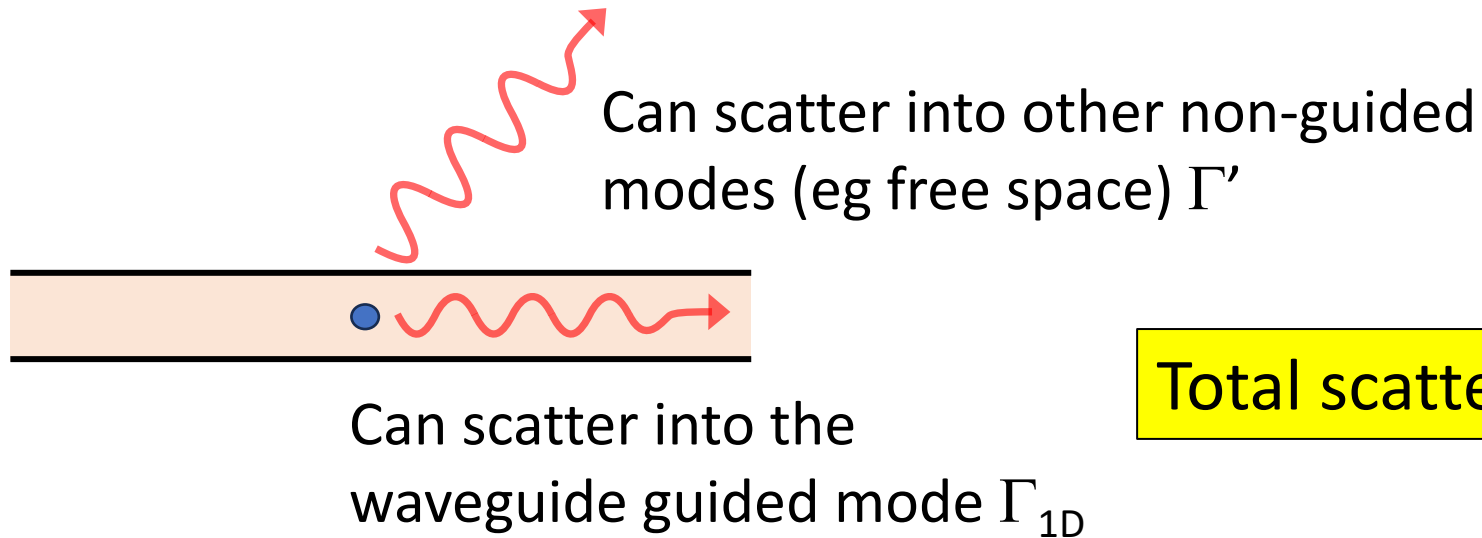
Waveguide:



In a waveguide you can have $A \sim \lambda^2$ and you can get a **strong interaction without diffraction**



Scattering in a waveguide



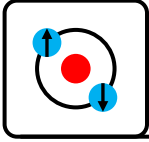
$$\text{Total scattering } \Gamma_{\text{TOT}} = \Gamma' + \Gamma_{1D}$$

β : Proportion of emission that is coupled to the waveguide mode

$$\beta = \frac{\Gamma_{1D}}{\Gamma_{1D} + \Gamma'}$$

Tricks to increase β :

1. Make waveguide **mode very small**
2. Use **subradiant states** that reduce coupling to free space, make $\Gamma' \ll \Gamma_0$



Cooperativity in a waveguide

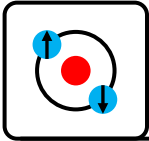
Other important metrics are:

Single atom cooperativity in a waveguide:

$$C = \frac{1}{2} \frac{\beta}{1 - \beta}$$

Purcell factor:

$$\text{Purcell factor} = \frac{\Gamma_{1D}}{\Gamma_0}$$



Optical Density in Waveguides

$$\text{Transmission} = \exp(-\text{OD}) = \exp(-2N\Gamma_{1D}/\Gamma_0)$$

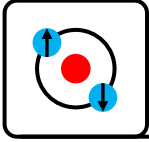
$$\text{OD} = 2N\beta$$

The power of waveguides:

$$\text{Fidelity} \leq 1 - \frac{5.8}{\text{OD}} = 1 - \frac{5.8}{2N\beta}$$

	2D MOT quantum memory	Nanofiber waveguide
N	3e7	350
OD	70	70
β	1e-6 (effective)	0.1
Fidelity	91.7%	91.7%

Very important if you want to entangle single atoms in quantum devices like QPUs!!!



Γ_{1D} : Maximizing coupling to the waveguide

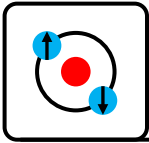
Optimize the overlap between the atomic dipole field pattern and the transverse electric field of the waveguide mode.

The coupling rate is proportional to:

$$\Gamma_{1D} \propto \left| \int \mathbf{E}_{\text{guide}}^*(\mathbf{r}) \cdot \mathbf{u}_{\text{dip}}(\mathbf{r}) d^3r \right|^2$$

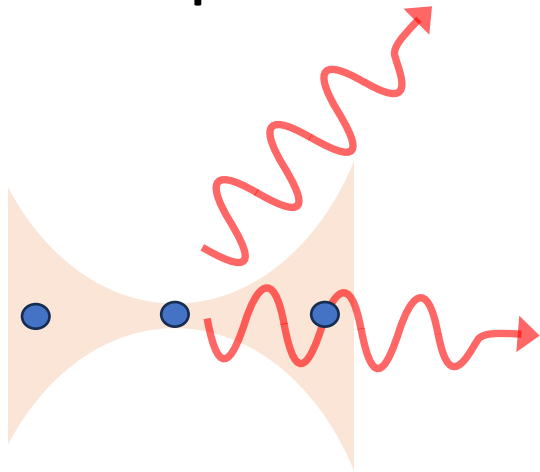
Strategies:

- Nanofibers
- Photonic crystal waveguides



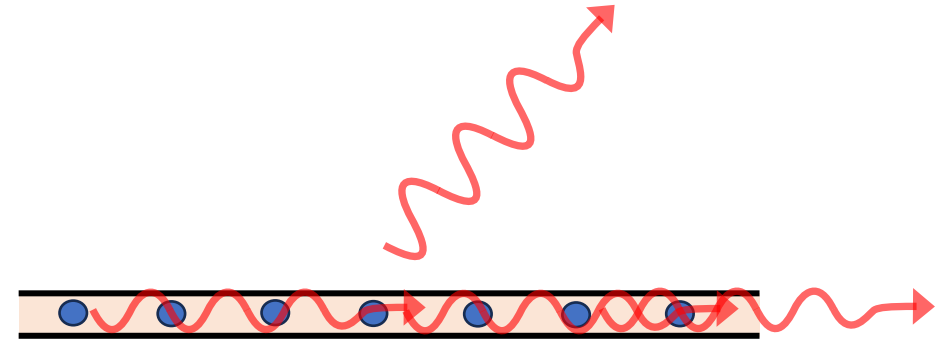
Free Space vs Waveguide: Why Waveguide Win

Free space:

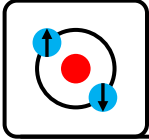


- emission spread over 4π steradians
- $\beta \approx 10^{-4}$ – 10^{-6} typically
- Realistic few atom light–matter interfaces need optical cavities to enhance coupling

Waveguide:



- emission confined to 2 directions $\rightarrow$ effectively 1D
- β can exceed 0.5 without cavities
- $\beta \approx 0.9$ – 0.99 in photonic crystal waveguides
- deterministic scattering without finesse or mirrors



“Perfect” waveguide QED (Superconducting qubits) $\beta=1$

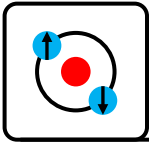
Superconducting qubits (eg transmons) are nonlinear LC oscillators with:

- huge dipole moments ($\gg 1000 e a_0$ compared with $\sim 1 e a_0$ for atoms)
- transition frequencies in the microwave regime (4-10 GHz)
- engineered strong coupling to a microwave waveguide
- has negligible coupling to free space (due to grounding and geometry)

$$\beta \sim 1$$

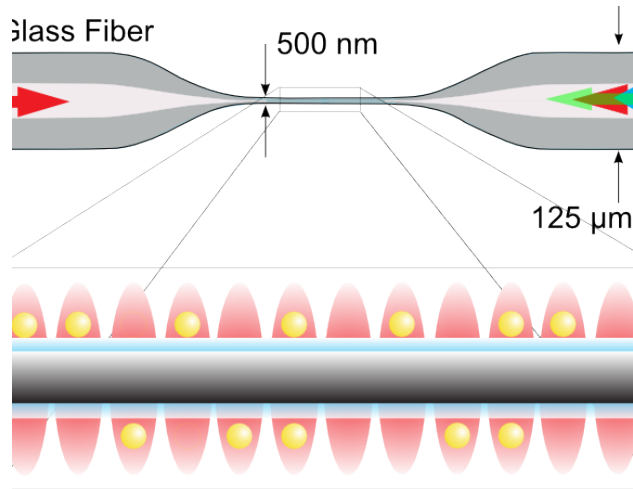
BUT...

- Thermal noise dominates at room temperature $k_B T \sim 1200 h f$
- Requires superconducting wires, temperatures $< 1K$
- Challenging to scale!

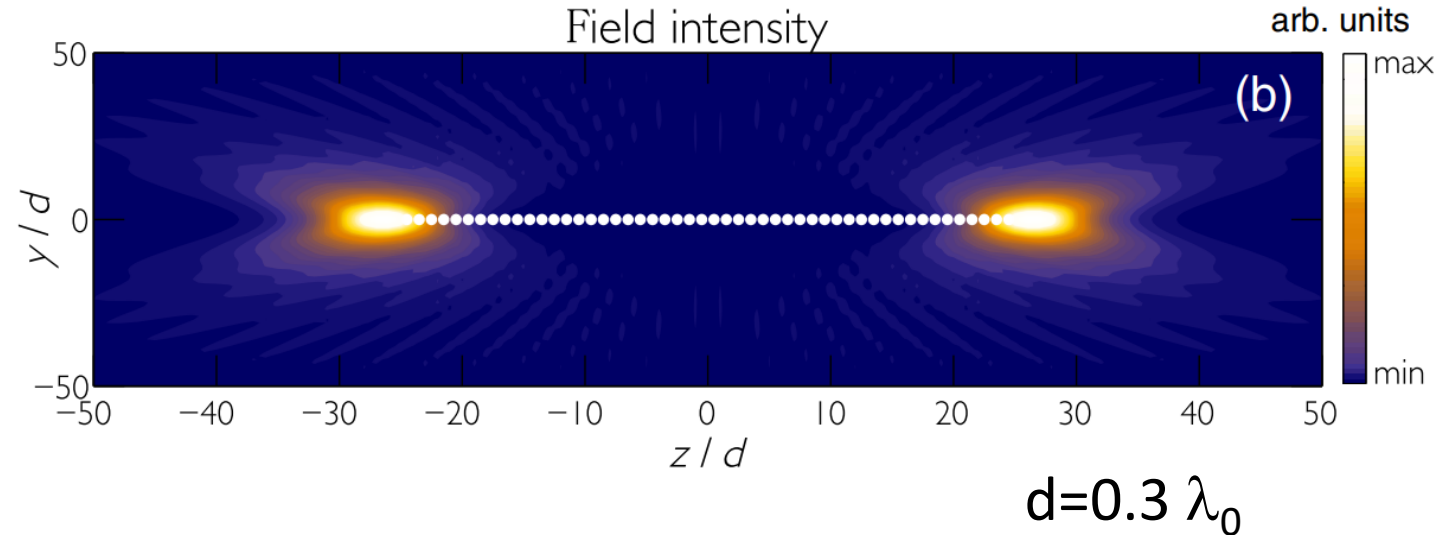


Collective waveguide QED

Random arrays

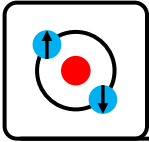


Defect free subwavelength arrays



Subwavelength defect free arrays
("phased arrays") can create collective
superradiant and subradiant modes

"Exponential Improvement in Photon Storage Fidelities Using Subradiance and "Selective Radiance" in Atomic Arrays", PRX 7 031024, (2017)



Collective waveguide QED

Superradiance

All atoms oscillate **in phase**

-> bright, enhanced coupling to guided modes

$$|B\rangle = \frac{1}{\sqrt{N}} \sum_j \sigma_j^- |g \dots g\rangle$$

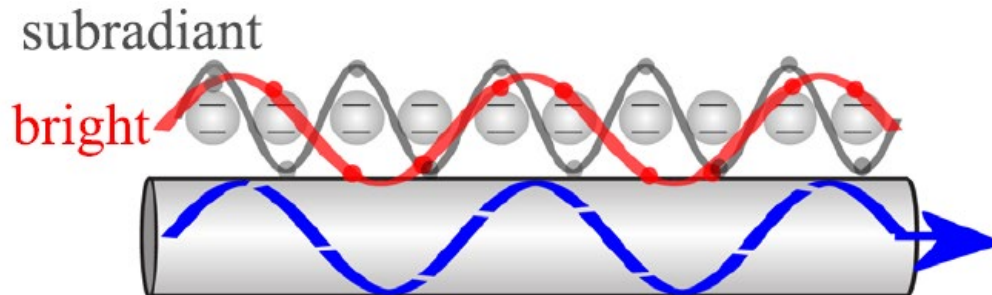
$$\Gamma_{\text{super}} \approx \Gamma' + N\Gamma_{1D}$$

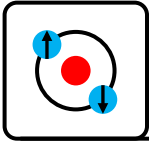
Subradiance

Atoms oscillate **out of phase**

destructive interference of
coupling to guided modes

$$\Gamma_{\text{sub}} \ll \Gamma'$$





Collective waveguide QED

For atoms equally spaced with period d , excitations form **Bloch states**:

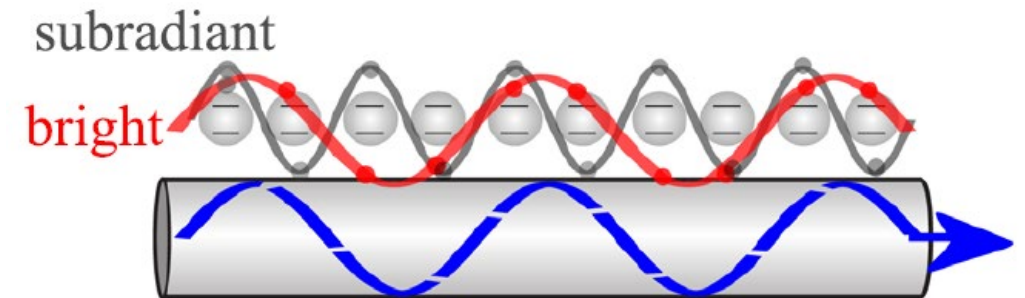
$$|k_B\rangle = \frac{1}{\sqrt{N}} \sum_{j=1}^N e^{ik_B z_j} \sigma_j^- |g \dots g\rangle$$

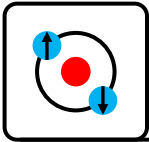
The decay rate of the collective mode is then:

$$\Gamma(k_B) = \Gamma' + \Gamma_{1D} \cos(k_B d)$$

Rich band structure leads to rich physics:

- Photon transport control
- Switching
- Photon localization
- ...





Fidelity in Collective Waveguides

Enhanced coupling for the same OD leads to exponentially better fidelities storing quantum information

Random arrays:

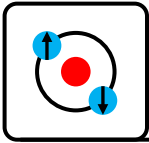
$$\text{Fidelity} \leq 1 - \frac{5.8}{OD} = 1 - \frac{5.8}{2N\beta}$$

Subwavelength defect free arrays:

$$\text{Fidelity} \leq 1 - \frac{15}{N \cdot OD} = 1 - \frac{15}{2N^2\beta}$$

The power of waveguides:

	2D MOT quantum memory	Nanofiber waveguide	Nanofiber waveguide (collective)	
N	3e7	350	30	350
OD	70	70	70	70
β	1e-6 (effective)	0.1	0.1	0.1
Fidelity	91.7%	91.7%	91.7%	99.94%

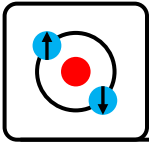


Foundations and Theory

Atom-photon interaction in a cavity (cavity QED)

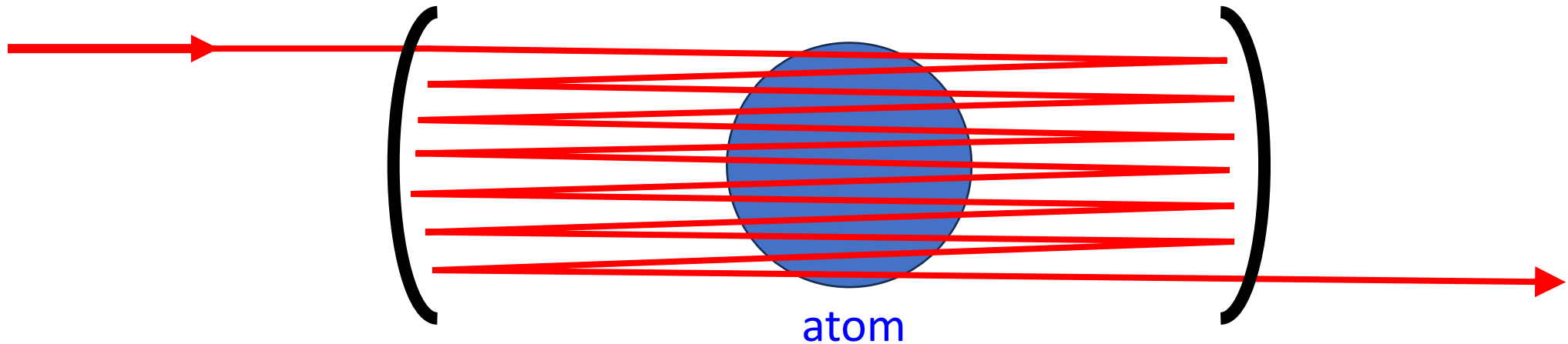
References:

arXiv:1412.2889 “Cavity-based quantum networks with single atoms and optical photons”
Fundamentals of Photonics



Optical Cavities

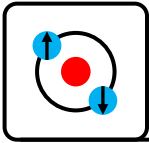
*A cavity bounces a photon so that it **interacts many times** with each atom.*



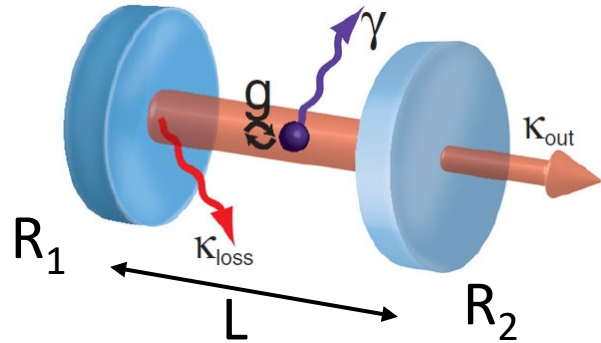
This increases the effective cross section by the number of bounces F/π (F is called the cavity Finesse)

Cavities help by:

- Minimizing mode volume
- High Finesse: many passes



Optical Cavities (recap)



A Fabry–Pérot cavity of length L and mirror reflectivity R_1 and R_2 .

Free spectral range:

$$\text{FSR} = \frac{c}{2L}$$

Finesse:

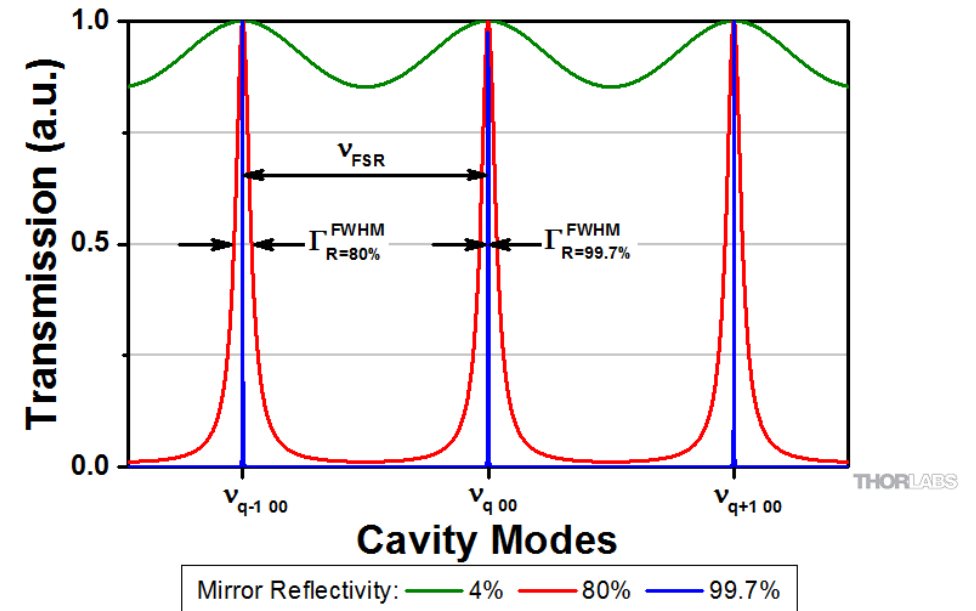
$$\mathcal{F} = \frac{\pi(R_1 R_2)^{1/4}}{1 - \sqrt{R_1 R_2}}$$

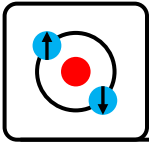
Cavity decay rate:

$$\kappa = \frac{\text{FSR}}{\mathcal{F}}$$

Mode volume:

$$V = \int \epsilon_0 |\mathbf{E}|^2 d^3 r / \max |\mathbf{E}|^2$$





Cooperativity

Deterministic/guaranteed interaction requires:

$$\underbrace{\frac{3\lambda^2}{2\pi}}_{\sigma_{\text{abs}}} \times \underbrace{\frac{\mathcal{F}}{\pi}}_{\text{bounces}} \gg \underbrace{\frac{\pi}{4} w_0^2}_{\text{beam area } A}$$

$$\frac{3\lambda^2}{2\pi} \frac{\mathcal{F}}{\pi} \frac{1}{A} \gg 1$$

This can be rewritten as:
single atom **cooperativity**:

$$C_1 = \frac{g^2}{2\kappa\gamma} \gg 1$$

For N atoms in the cavity we care
about the **collective cooperativity**, C_N :

$$C_N = N C_1 = N \frac{g^2}{2\kappa\gamma} \gg 1$$

Atom-cavity
coupling constant

$$g = \sqrt{\frac{d_{ge}^2 \omega}{\epsilon_0 \hbar V}}$$

Cavity
decay rate:

$$\kappa = \frac{\pi c}{2L\mathcal{F}}$$

Atom
decay rate:

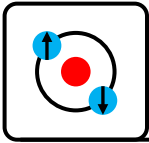
$$\gamma = \frac{d_{ge}^2 \omega^3}{6\pi\epsilon_0 \hbar c^3}$$

Cavity mode volume:

$$V = \int u^2(\vec{r}) d^3x$$

u the dimensionless electric
field mode normalize to 1 at
the field maximum.

****Note some papers/theory uses $C_1 = g^2/\kappa\gamma$, both are common!**



Definitions

$$C_1 = \frac{g^2}{2\kappa\gamma} \gg 1$$

$$g = \sqrt{\frac{d_{ge}^2 \omega}{\epsilon_0 \hbar V}}$$

$$\kappa = \frac{\pi c}{2L\mathcal{F}}$$

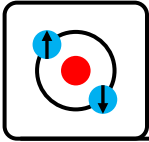
$$\gamma = \frac{d_{ge}^2 \omega^3}{6\pi\epsilon_0 \hbar c^3}$$

- Note: C_1 doesn't depend on L ! ($\kappa \propto 1/L$, $g \propto 1/L^{0.5}$)
- Critical atom number (number of atoms needed in the cavity to achieve strong interaction with a photon $C_N \sim 1$):

$$N_c = 1/C_1$$

- Critical photon number (number of photons needed saturate the atomic transition):

$$n_c = \frac{\gamma^2}{2g^2}$$



Jaynes–Cummings model

A Fabry–Pérot cavity of length L and mirror reflectivity R has:

$$\mathcal{H}_I = -\vec{d} \cdot \vec{E} \rightarrow dE(\sigma_{ce}^\dagger + \sigma_{ce})(a^\dagger + a)$$

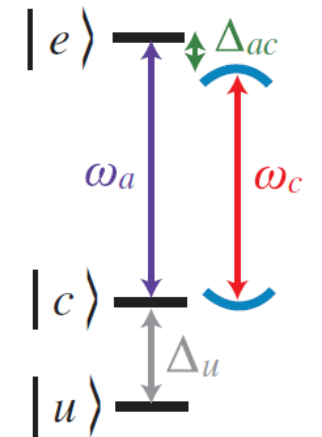
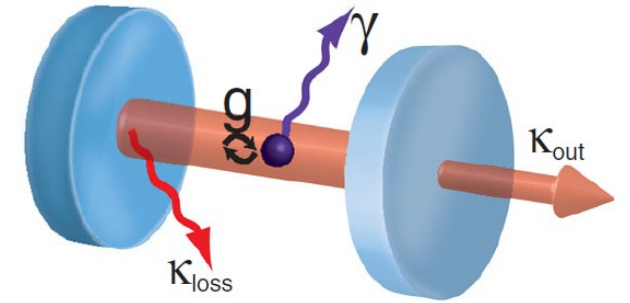
$$= (\sigma_{ce}^\dagger a^\dagger + \sigma_{ce}^\dagger a + \sigma_{ce} a^\dagger + \sigma_{ce} a)$$

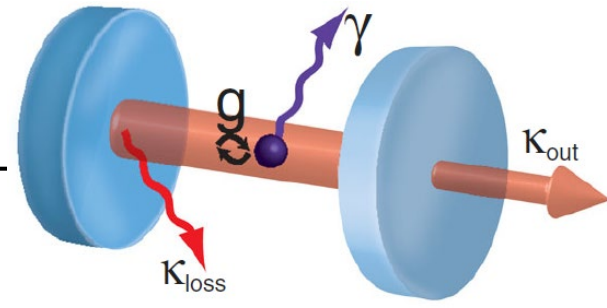
$$\mathcal{H}_{JC} = \hbar\omega_a \sigma_{ce}^\dagger \sigma_{ce} + \hbar\omega_c a^\dagger a + \underbrace{\hbar g(\sigma_{ec} a + \sigma_{ce} a^\dagger)}_{\text{ole coupling}}$$

$$E_0 = 0,$$

$$E_{N,\pm} = N\hbar\omega_c + \hbar\frac{\Delta_{ac}}{2} \pm \frac{\hbar}{2} \sqrt{\Omega_N^2 + \Delta_{ac}^2}$$

$$\Omega_N = 2g\sqrt{N} \quad \Delta_{ac} = \omega_a - \omega_c$$



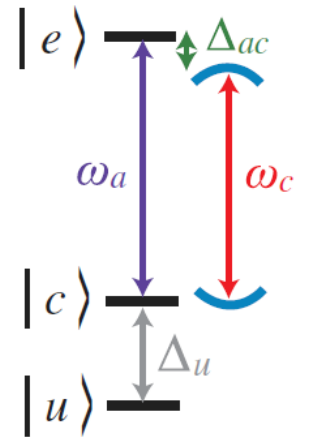
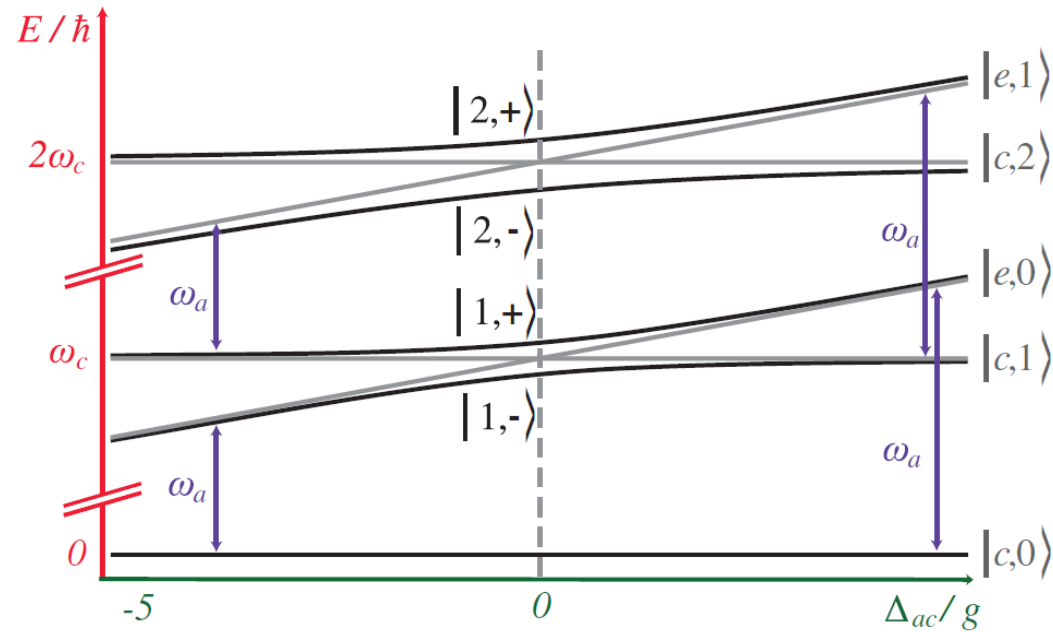


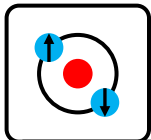
$$E_{N,\pm} = N\hbar\omega_c + \hbar\frac{\Delta_{ac}}{2} \pm \frac{\hbar}{2}\sqrt{\Omega_N^2 + \Delta_{ac}^2}$$

$$\Omega_N = 2g\sqrt{N} \quad \Delta_{ac} = \omega_a - \omega_c$$

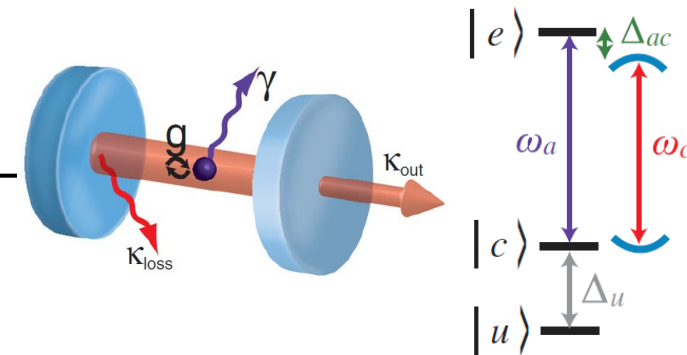
For zero detuning the levels
are separated by Ω_N

For zero initial energy ($N-1=0$)
and zero detuning
 $\Omega_1=2 \text{ g}$





Vacuum Rabi frequency



$$E_0 = 0,$$

$$E_{N,\pm} = N\hbar\omega_c + \hbar\frac{\Delta_{ac}}{2} \pm \frac{\hbar}{2}\sqrt{\Omega_N^2 + \Delta_{ac}^2}$$

$$\Omega_N = 2g\sqrt{N}$$

$$\Delta_{ac} = \omega_a - \omega_c$$

For zero initial energy ($N-1=0$) and zero detuning:

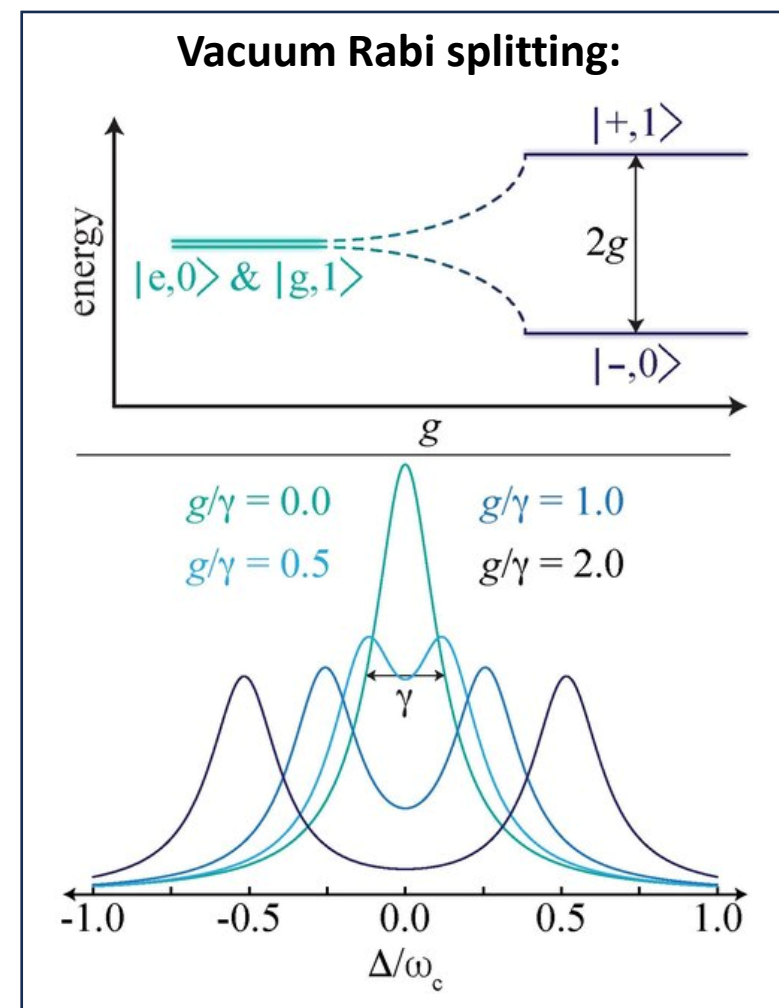
$$\Omega_1 = 2g$$

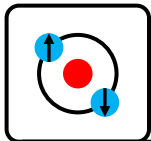
Probability of the atom being in $|e\rangle$ as a function of time:

$$P_{|e\rangle}(t) = \cos^2(\Omega_N t/2)$$

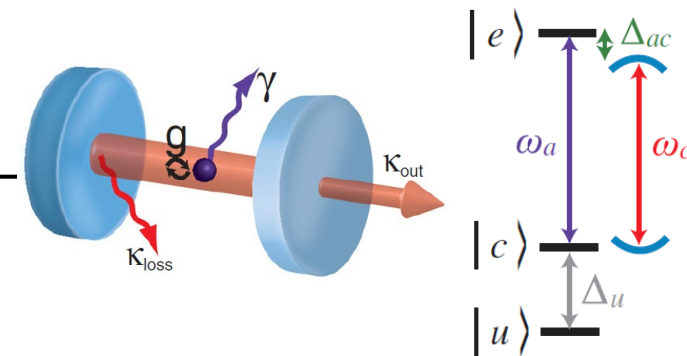
Vacuum Rabi frequency:

$$\Omega_1 = 2g$$





Strong coupling



Strong coupling regime:

$$g \gg (\kappa, \gamma)$$

- Reversible atom-photon coupling is faster than all irreversible processes
- Coherent processes
- Vacuum-Rabi splitting observed for weak probes

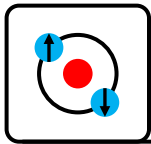
$$g = \sqrt{\frac{d_{ge}^2 \omega}{\epsilon_0 \hbar V}}$$

$$\kappa = \frac{\pi c}{2L\mathcal{F}}$$

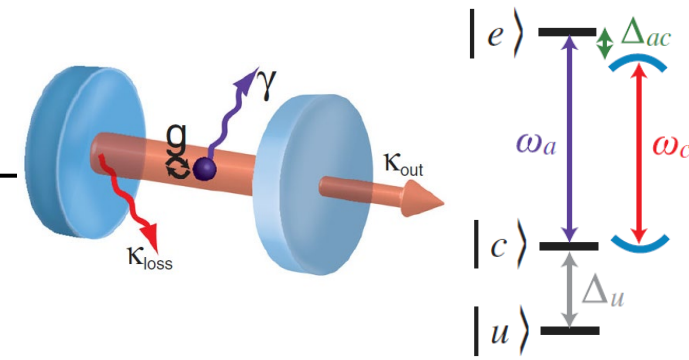
Atom – Cavity exchange energy at a rate of:

$$\Omega_{N=1, \Delta_{ac}} = \sqrt{\Delta_{ac}^2 + 4g^2}$$

$$\gamma = \frac{d_{ge}^2 \omega^3}{6\pi \epsilon_0 \hbar c^3}$$



Purcell/fast cavity regime



Purcell regime:

$$\kappa \gg g \gg \gamma$$

- Vacuum-Rabi oscillations cannot be observed (photon lost from the resonator before reabsorption)
- Decay into cavity near atomic frequency can be **strongly increased** (Purcell, 1946) or **decreased** (Kleppner, 1981)

Decay rate into the cavity mode from an atom:

$$\gamma_c = \frac{g^2 \kappa}{\kappa^2 + \Delta_{ac}^2}$$

Δ_{ac} = detuning between atom and cavity

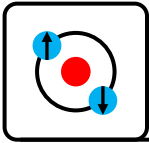
On resonance decay into the cavity mode can be much larger than natural decay rate γ :

$$\gamma_c = 2C\gamma$$

If cavity mode covers a large fraction of the solid angle for an atom in a cavity the spontaneous emission rate into free space is affected:

$$\gamma = (1 - \zeta/4\pi) \frac{\Gamma}{2}$$

ζ = cavity solid angle at the atom



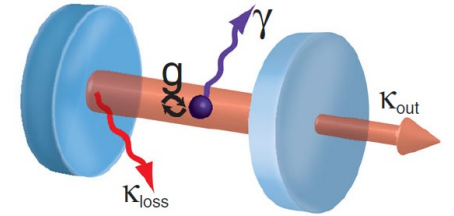
Superradiant lasers

Superradiant laser regime:

$$\kappa \gg g\sqrt{N} \gg \gamma$$

Superradiant lasers are potential next generation atomic clocks.

- Cavity noise is strongly suppressed (eg 1e6)
- Enhanced emission rates **gain $\propto N^2$**
- Allows creation μHz lasers on forbidden transitions



Purcell regime

Cavity controls atomic emission **$C_1 \gg 1$**

Short atomic coherence vs cavity lifetime

Phase defined by cavity

Cavity noise NOT suppressed

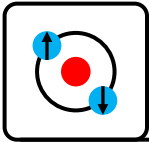
Superradiant laser regime

Cavity has no influence on emission **$C_1 \ll 1$**

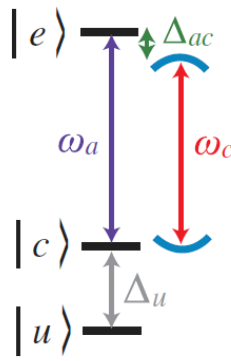
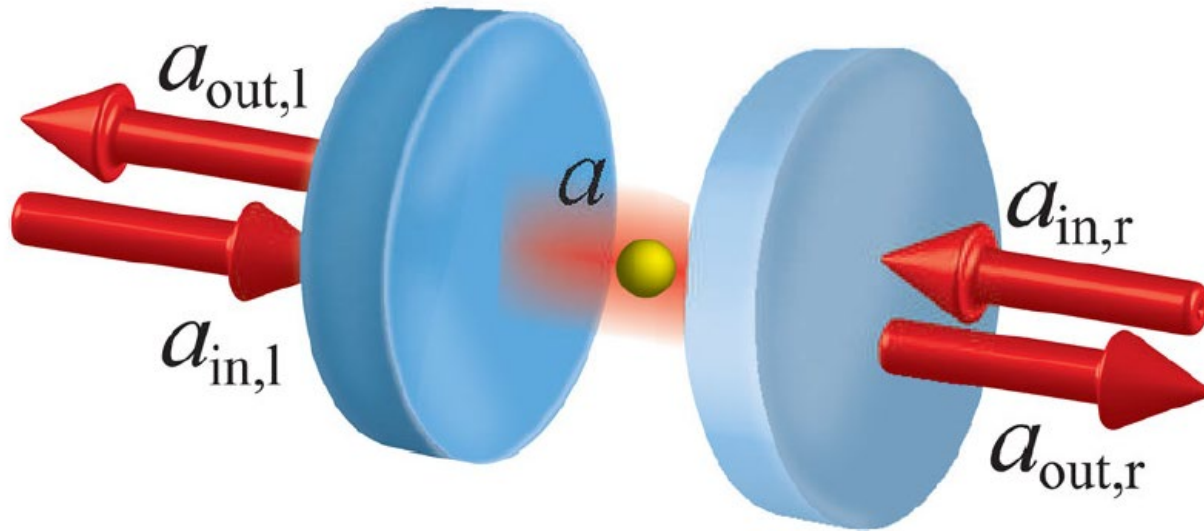
Long atomic coherence vs cavity lifetime

Phase defined by atoms **$C_N \gg 1$**

Cavity noise **strongly** suppressed



Transmission and reflection



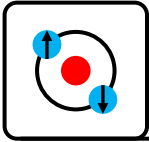
$$r(\omega) = 1 - \frac{2\kappa_l(i\Delta_a + \gamma)}{(i\Delta_c + \kappa)(i\Delta_a + \gamma) + g^2}$$

$$t(\omega) = \frac{2\sqrt{\kappa_l\kappa_r}(i\Delta_a + \gamma)}{(i\Delta_c + \kappa)(i\Delta_a + \gamma) + g^2}$$

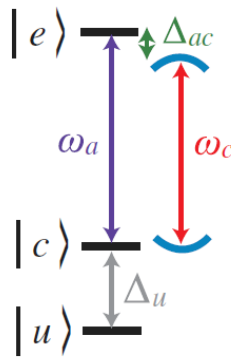
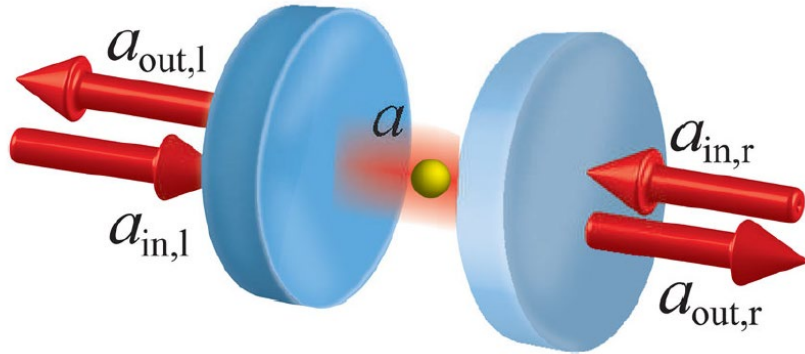
r, t the **amplitude** reflection and transmission coefficients

Intensity $R=|r|^2$, $T=|t|^2$, reflected phase = $\arg(r)$

Important practical note: Any imperfection in modematching between the incoming light and the cavity



Mode-matching



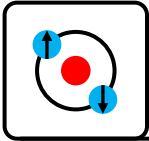
Practical note: Imperfect mode-matching spatial and temporal between the incoming light mode and the cavity mode will be reflected by the cavity.

For free space cavities this can dominate errors.

$$R_{\text{experiment}} = (1 - \xi) + \xi R$$

ξ = mode matching efficiency

****Optical fiber based cavities can have much better spatial mode matching efficiency compared with free space cavities**



Estimating possible fidelities using cavities

For an optical cavity

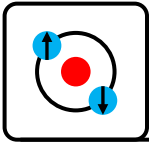
$$OD \sim 4C_N$$

Using our earlier expression for memory fidelity:

$$\text{Fidelity} \leq 1 - \frac{5.8}{OD} = 1 - \frac{5.8}{4NC_1}$$

	2D MOT quantum memory	Nanofiber waveguide	Nanofiber waveguide (collective)		Cavity	Cavity with waveguide	Cavity with waveguide (collective)
N	3e7	350	30	350	1	100	15
OD	70	70	70	70	100	10000	1500
β	1e-6 (effective)	0.1	0.1	0.1			
C_1	6e-7	6e-2	6e-2	6e-2	25	25	25
Fidelity	91.7%	91.7%	91.7%	99.94%	94.2%	99.94%	99.94%

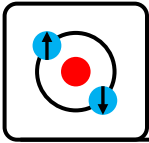
Realistically other factor will probably start to dominate.



Part III: Experimental Progress

1. Quantum memories using atomic clouds (recap)
2. Interfacing to trapped ions
3. Interfacing to neutral atoms
4. Interfacing to NV centers and rare earth ions
5. Interfacing to superconducting qubits
6. Metrics and comparisons





Quantum interfaces for neutral atoms

Seeing a single photon without destroying it

[G. Nogues](#), [A. Rauschenbeutel](#), [S. Osnaghi](#), [M. Brune](#), [J. M. Raimond](#) & [S. Haroche](#)

Nature **400**, 239–242 (1999) | [Cite this article](#)

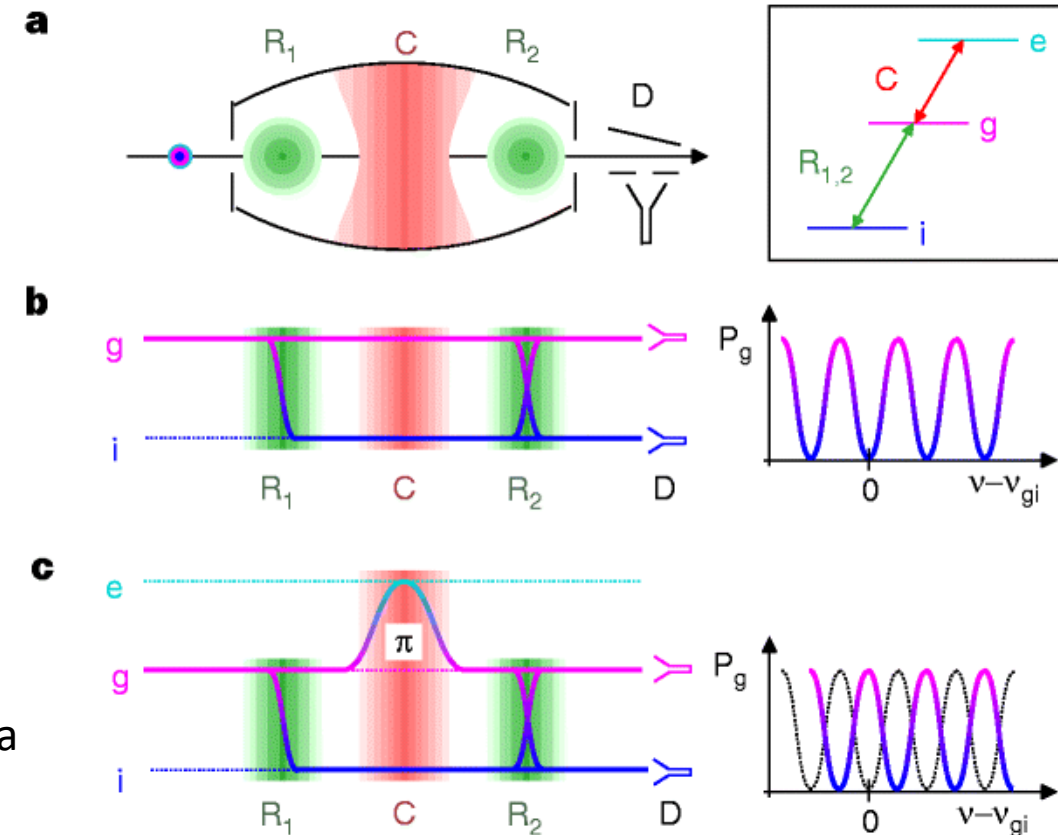
First non-destructive, repeatable detection
of a single photon using cavity QED

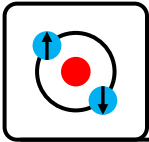
2012 Nobel prize – visiting in Jan 2026

Basis for phase gates/switches

Experiment:

1. Prepare a single photon in the microwave cavity (lifetime 1ms) (use a Rydberg atom as a photon source).
2. Prepare a Rydberg atom in a superposition via Ramsey zone R_1 .
3. Atom flies through cavity $\rightarrow$ picks up π phase only if photon present.
4. Second Ramsey zone R_2 converts phase into population difference.
5. Detect atom $\rightarrow$ reveals whether a photon is in the cavity.
6. Photon remains intact $\rightarrow$ repeat with another atom for confirmation.





Quantum interfaces for neutral atoms

Long-distance quantum communication with atomic ensembles and linear optics

[L.-M. Duan](#), [M. D. Lukin](#), [J. I. Cirac](#) & [P. Zoller](#)

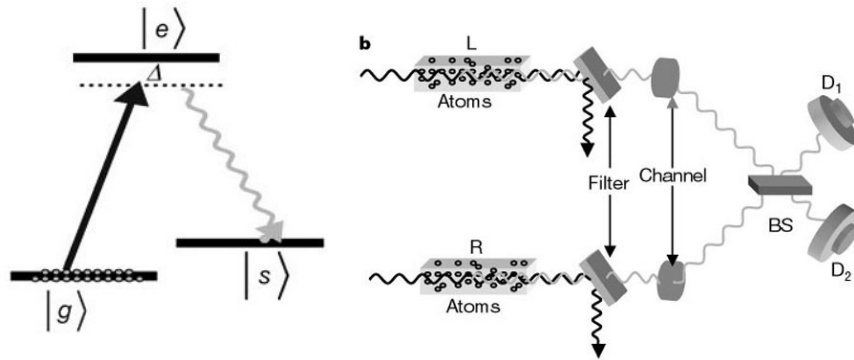
[Nature](#) **414**, 413–418 (2001) | [Cite this article](#)

The basis for virtually all quantum communication

DLCZ protocol / Entanglement swapping

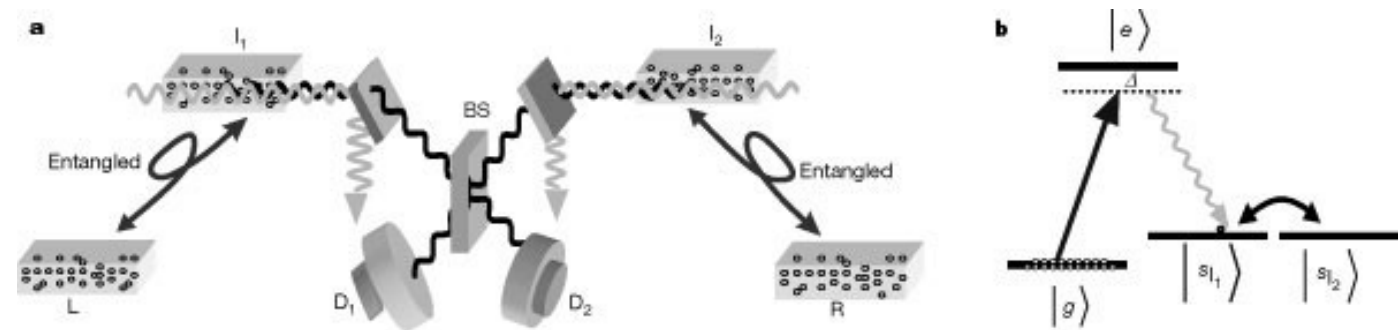
Invented heralded entanglement generation

Step 1: Entangle two remote ensembles



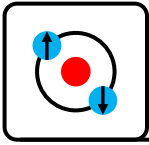
Entanglement protocol

1. Prepare two atomic ensembles in $|g\rangle$
2. Apply a weak write raman pulse to both
3. Interfere stokes output on a beam splitter
4. If only 1 detector clicks one of the ensembles is in $|g\rangle$ and the other is in $|s\rangle$
5. If no clicks repeat.



Entanglement swapping protocol

1. Apply a read pulse ($|s\rangle \rightarrow |e\rangle$) to both I_1 and I_2 ensembles simultaneously and interfere on beamsplitter
2. If one detector clicks ensembles L and R are entangled
3. Otherwise repeat both entanglement generation and swapping protocols



Experimental Progress

Quantum interfaces to ensembles

(recap)

References:

arXiv:1002.4659 Optical quantum memory

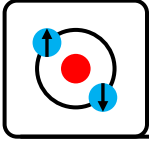
arXiv:0807.3358 Quantum interface between light and atomic ensembles

arXiv:0806.4195 The Quantum Internet

<https://doi.org/10.1126/science.aam9288> Quantum internet: A vision for the road ahead

<https://doi.org/10.6028/jres.125.002> Optical Quantum Memory and its Applications in Quantum Communication Systems

arXiv:1605.08519 Highly Efficient Coherent Optical Memory Based on Electromagnetically Induced Transparency



Quantum interfaces to ensembles (recap)

REVIEWS OF MODERN PHYSICS, VOLUME 82, APRIL–JUNE 2010

Quantum interface between light and atomic ensembles

Klemens Hammerer

Institute for Theoretical Physics, University of Innsbruck, and Institute for Quantum Optics and Quantum Information, Austrian Academy of Science, Technikerstrasse 25, 6020 Innsbruck, Austria

Anders S. Sørensen and Eugene S. Polzik

Niels Bohr Institute, Copenhagen University, Blegdamsvej 17, Copenhagen 2100, Denmark

Optical quantum memory

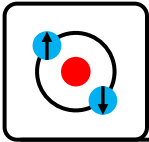
[Alexander I. Lvovsky](#) , [Barry C. Sanders](#) & [Wolfgang Tittel](#)

[Nature Photonics](#) **3**, 706–714 (2009) | [Cite this article](#)

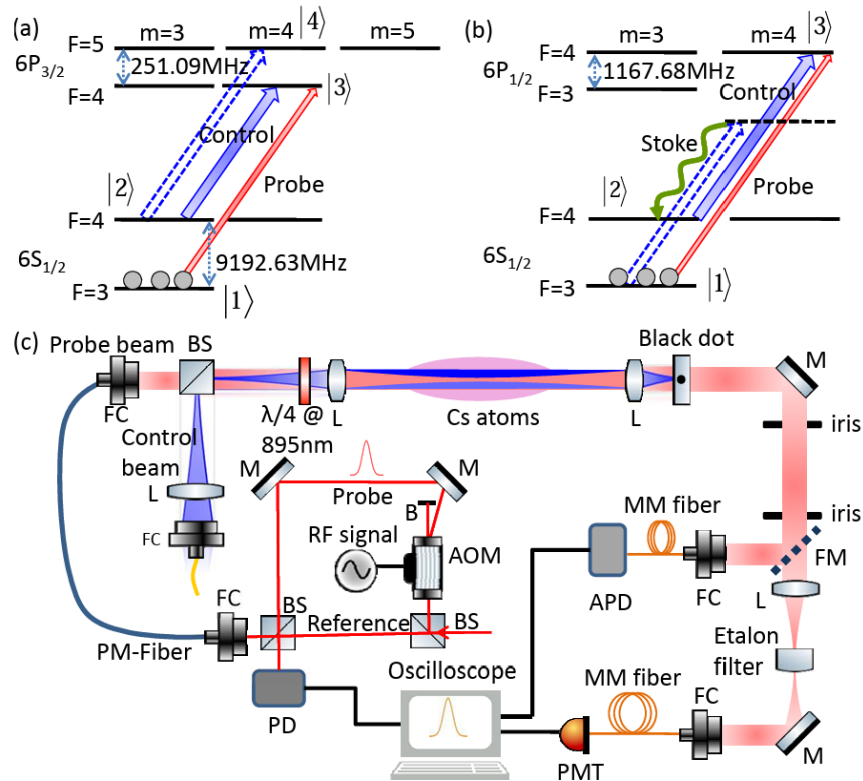
Approaches used for quantum memory in large ensembles

- EIT
- Gradient Echo Memory (GEM)

arXiv:1605.08519 Highly Efficient Coherent Optical Memory Based on Electromagnetically Induced Transparency, PRL 120, 183602, (2018)
Highly-efficient quantum memory for polarization qubits in a spatially-multiplexed cold atomic ensemble, Nature Communications 9, 363 (2018)



Quantum interfaces to ensembles: EIT (recap)

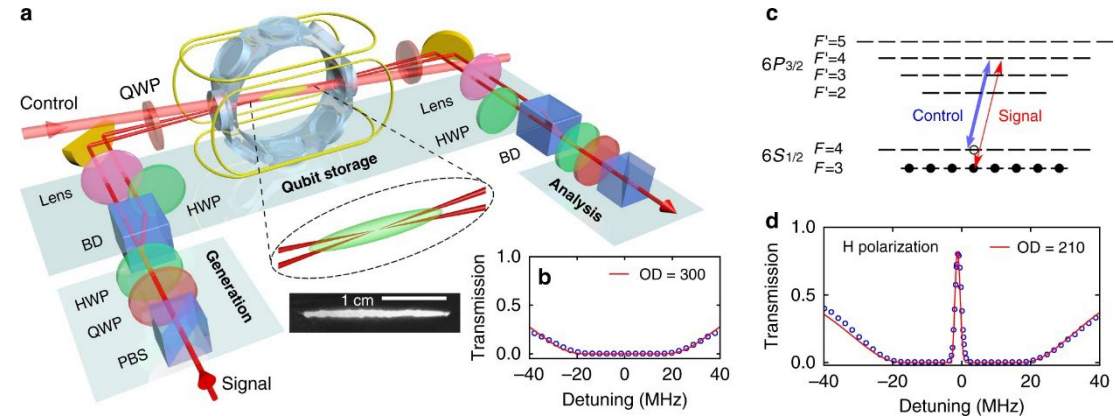


92% storage efficiency

OD: 800

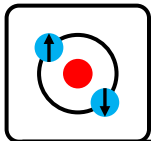
Time-bandwidth product: 1200

N atoms: $5e9$



- Storing and retrieving single photons in cold atom ensembles is well established with world leading performance at IAMS/NTHU (Ying-Cheng Chen)
- This is a leading technology for quantum communication/entanglement distribution networks
- This technology was covered in detail in earlier lectures

**...BUT it requires $>1e8$ atoms to efficiently store a single qubit
 $\Rightarrow$ unsuitable for interfacing to single qubits in quantum processors.**

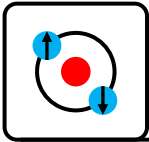


Experimental Progress

Interfacing to trapped ions

References:

Colloquium: Quantum networks with trapped ions, Rev. Mod. Phys. 82, 1209 (2010)

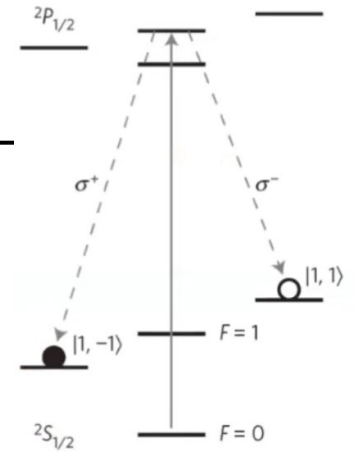
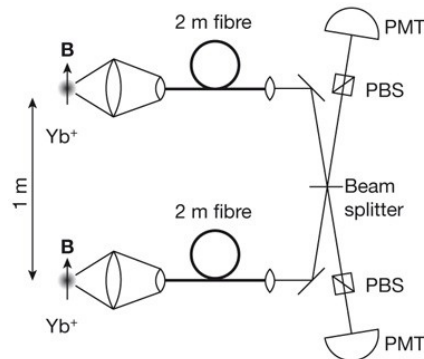
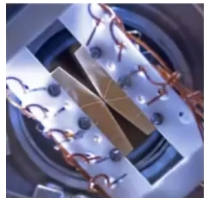
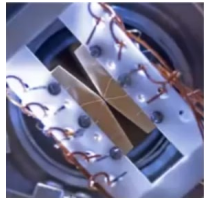


Quantum interfaces for trapped ions

Entanglement of single-atom quantum bits at a distance

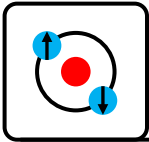
[D. L. Moehring](#) , [P. Maunz](#), [S. Olmschenk](#), [K. C. Younge](#), [D. N. Matsukevich](#), [L.-M. Duan](#) & [C. Monroe](#)

[Nature](#) **449**, 68–71 (2007) | [Cite this article](#)



1. Excite ions in both traps simultaneously to $2P_{1/2}$
2. Ions decay emitting a σ^+/σ^- polarization photon
3. Both photons simultaneously cross on a 50:50 beamsplitter erasing which way information
4. **If photons have opposite polarizations** and click PMTs simultaneously the ions are entangled
5. If photons have same polarization both ions are in a known state and no entanglement is generated (**fails 50% of the time**).

Heralding method allows a deterministic outcome (entanglement) with some (50% in the case) probability

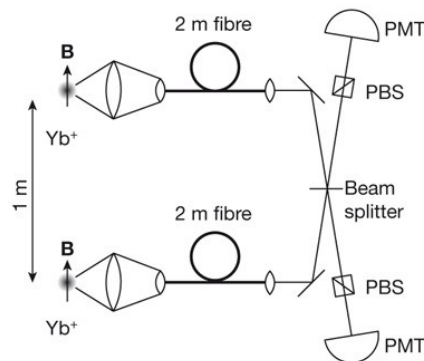
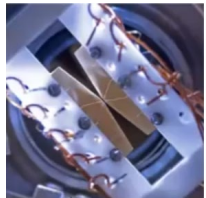
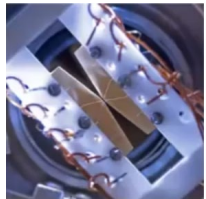


Quantum interfaces for trapped ions

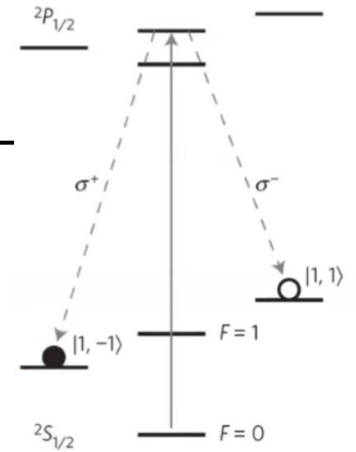
Entanglement of single-atom quantum bits at a distance

[D. L. Moehring](#) , [P. Maunz](#), [S. Olmschenk](#), [K. C. Younge](#), [D. N. Matsukevich](#), [L.-M. Duan](#) & [C. Monroe](#)

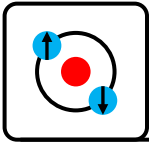
[Nature](#) **449**, 68–71 (2007) | [Cite this article](#)



1. Excitation pulses @ 550kHz
2. Excitation probability 50%
3. Huge losses in collection/detection
4. Success probability: 3.6×10^{-9}



One entangled pair generated every 8.3 minutes

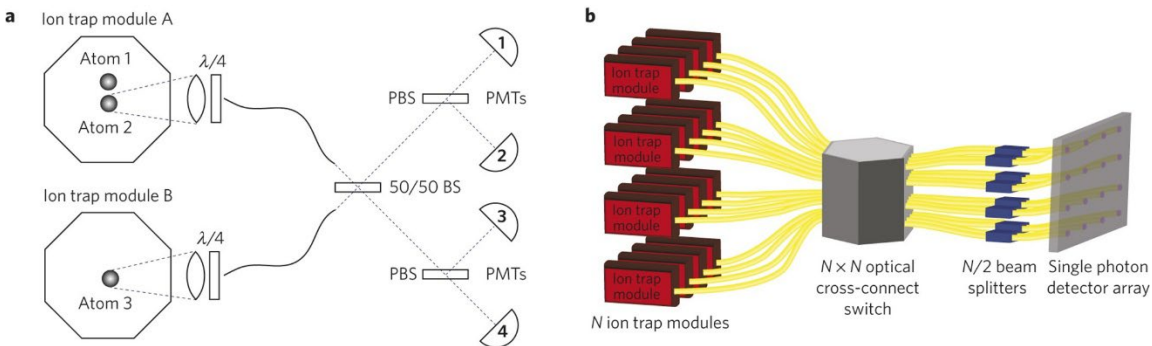
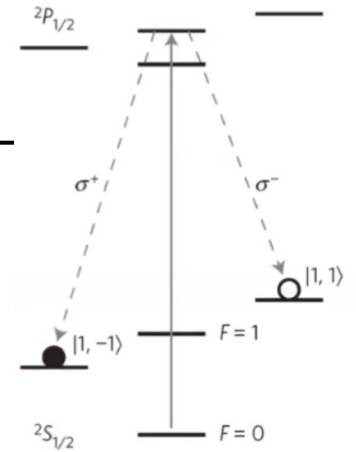


Quantum interfaces for trapped ions

Modular entanglement of atomic qubits using photons and phonons

[D. Hucul](#) , [I. V. Inlek](#), [G. Vittorini](#), [C. Crocker](#), [S. Debnath](#), [S. M. Clark](#) & [C. Monroe](#)

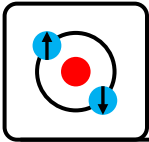
[Nature Physics](#) **11**, 37–42 (2015) | [Cite this article](#)



Most gains through improved collection and detection efficiency

1. Fidelity of ion-photon entangled state wrt Bell state 92%
2. Fidelity of ion-ion entangled state: 78(3)%
3. 4.5Hz Entanglement generation
4. 1.2s coherence time

First to demonstrate entanglement generation faster than decoherence

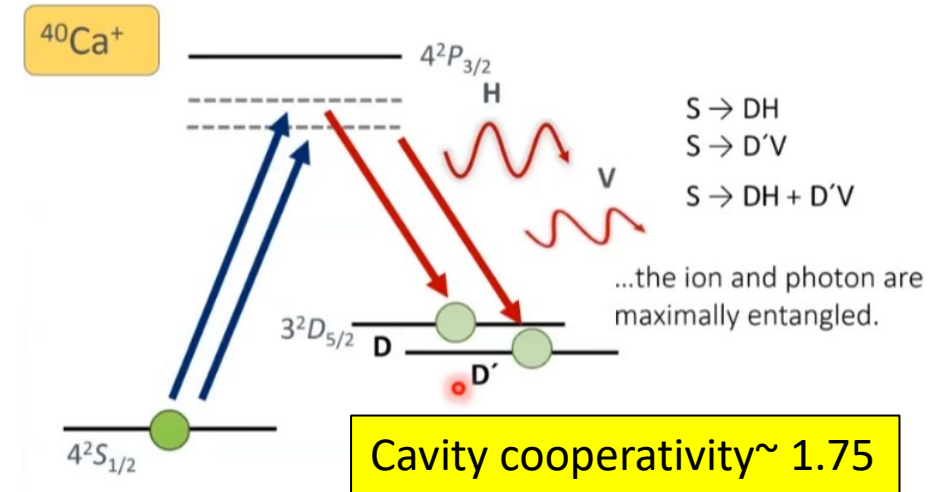
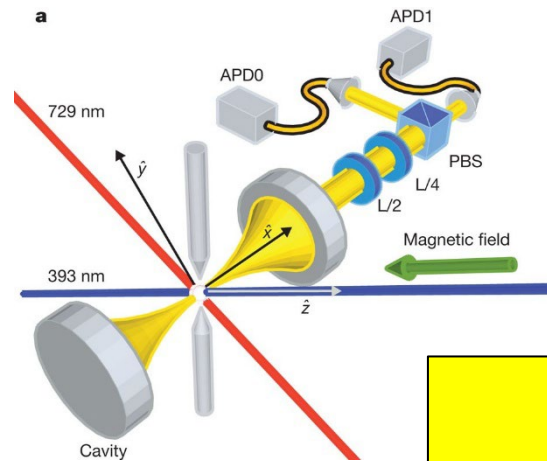
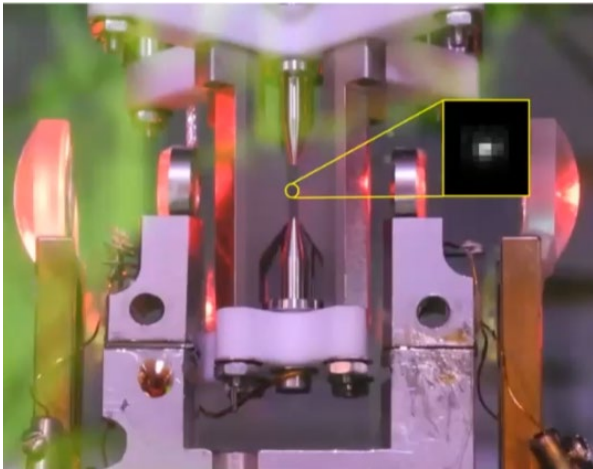


Quantum interfaces for trapped ions

Tunable ion-photon entanglement in an optical cavity

A. Stute, B. Casabone, P. Schindler, T. Monz, P. O. Schmidt, B. Brandstätter, T. E. Northup & R. Blatt

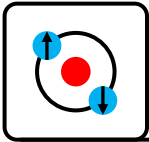
Nature **485**, 482–485 (2012) | [Cite this article](#)



First to use a cavity with ions allowing extremely efficient entanglement generation

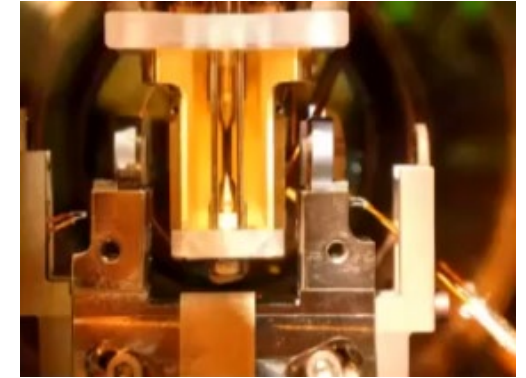
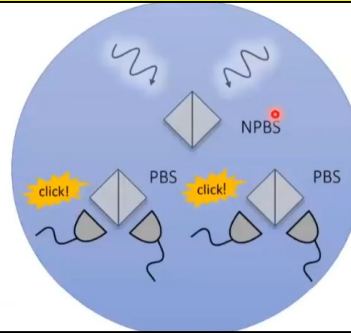
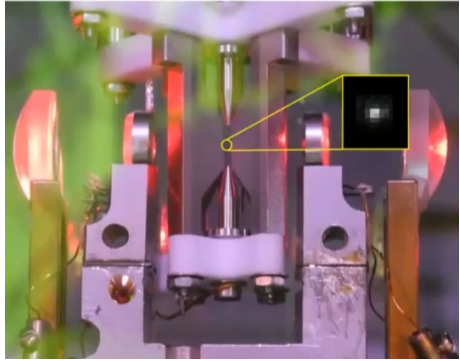
- Cavity allow extremely high entanglement generation probability:
“primarily limited by the probability of the photon leaving the cavity (16%) and photodiode efficiencies (40%)”
- we control **both amplitude and phase** of entangled state using via two simultaneous cavity-mediated Raman transitions.
- Record ion-photon entanglement fidelity 97.4%
- 40.5 Hz entanglement generation

<https://www.youtube.com/watch?v=PLiWZ8RvmkE>



Quantum interfaces for trapped ions

Long distance entanglement generation



PHYSICAL REVIEW LETTERS **130**, 050803 (2023)

Editors' Suggestion Featured in Physics

Entanglement of Trapped-Ion Qubits Separated by 230 Meters

V. Krutyanskiy^{1,2}, M. Galli^{1,2}, V. Krcmaršky^{1,2}, S. Baier^{1,2}, D. A. Fioretto², Y. Pu^{1,2}, A. Mazloom³, P. Sekatski⁴, M. Caneri^{1,2}, M. Teller², J. Schupp^{1,2}, J. Bate², M. Meraner^{1,2}, N. Sangouard⁵, B. P. Lanyon^{1,2,*} and T. E. Northup²

¹Institut für Quantenoptik und Quanteninformation, Österreichische Akademie der Wissenschaften, Technikerstraße 21a, 6020 Innsbruck, Austria

²Institut für Experimentalphysik, Universität Innsbruck, Technikerstraße 25, 6020 Innsbruck, Austria

³Department of Physics, Georgetown University, 37th and O Streets NW, Washington, D.C. 20057, USA

⁴Department of Applied Physics, University of Geneva, 1211 Geneva, Switzerland

⁵Institut de Physique Théorique, Université Paris-Saclay, CEA, CNRS, 91191 Gif-sur-Yvette, France

(Received 3 September 2022; accepted 20 December 2022; published 2 February 2023)

target state:

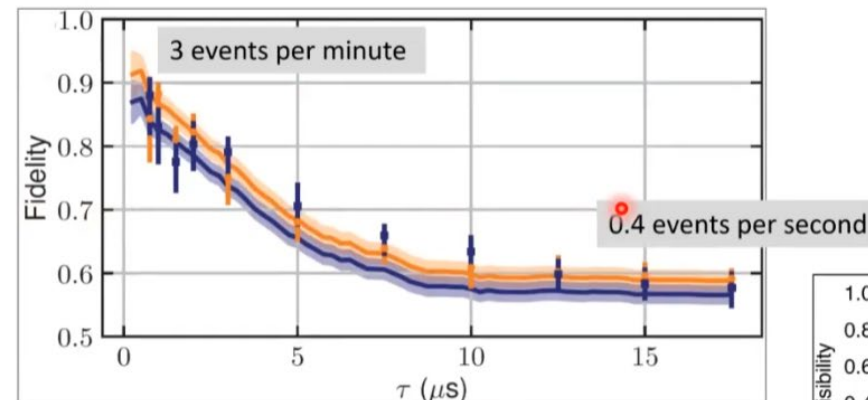
$$|\psi^-\rangle = |\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle$$

fidelity with respect to target state:

$$\langle\psi^-|\rho|\psi^-\rangle = 0.82 \pm 0.02 - 0.06$$

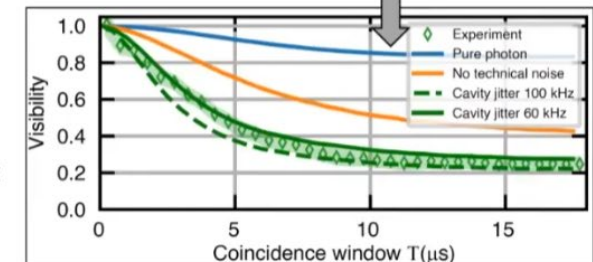
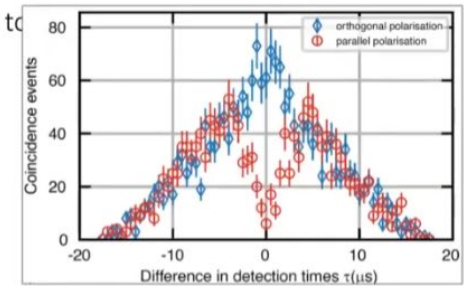
classical bound: 0.5

The fidelity decreases as longer intervals between photons are considered due to photon distinguishability, but it remains nonclassical.

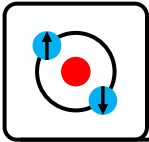


We can understand this trade-off via a master-equation model that reproduces our experimental visibility.

Krutyanskiy, M. Galli et al., *Phys. Rev. Lett.* **130**, 050803 (2023)

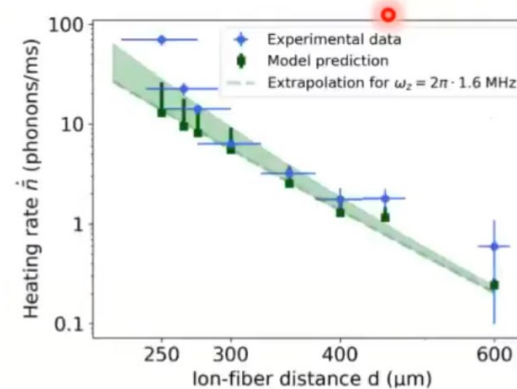
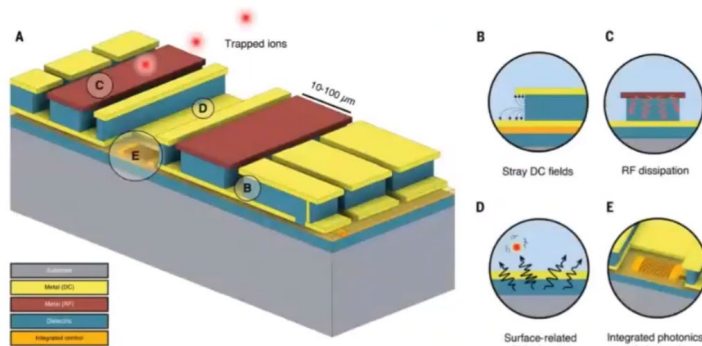
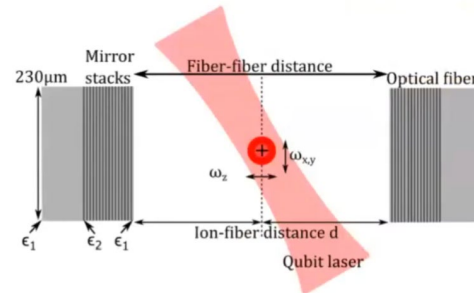
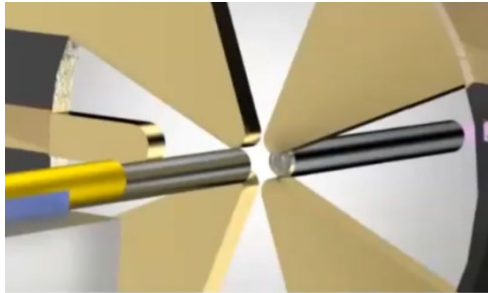


<https://www.youtube.com/watch?v=PlIWZ8RvmkE>



Quantum interfaces for trapped ions - Outlook

Miniaturization – fiber cavities

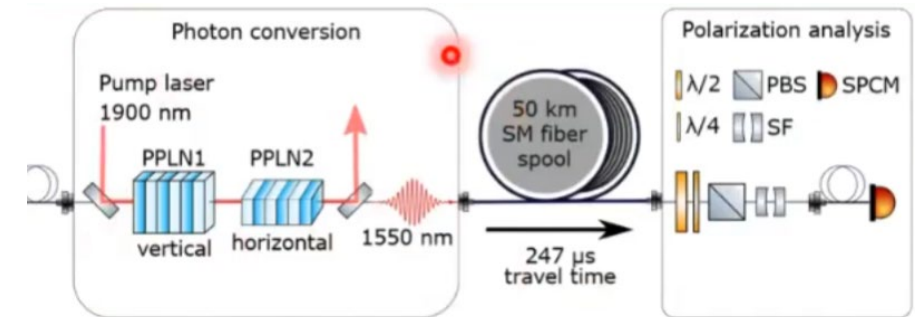
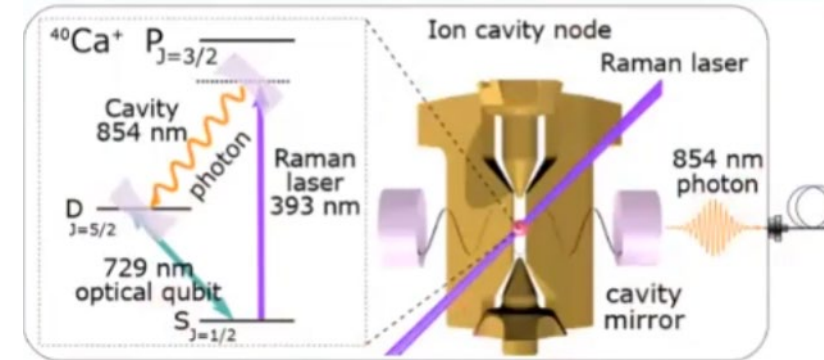


Challenge: heating and surface charges!

M. Teller et al. PRL 126, 230505 (2021)

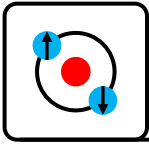
<https://www.youtube.com/watch?v=PlIWZ8RvmkE>

Frequency conversion



Transmission $\gg 100\text{m}$ in optical fiber **requires conversion to telecom wavelengths** (1.3 μm /1.55 μm) otherwise losses are too high

V. Krutyanskiy et al. NPJ Quantum Inf 5 72 (2019)

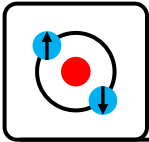


Experimental Progress

Interfacing to neutral atoms

References:

- Cavity-based quantum networks with single atoms and optical photons, Rev. Mod. Phys. 87, 1379 (2015)
- HJ Kimble, The quantum internet. Nature 453, 1023–1030 (2008).
- Quantum networks with neutral atom processing nodes, npj Quantum Information **9**, 90 (2023)
- Quantum matter built from nanoscopic lattices of atoms and photons, Rev. Mod. Phys. 90, 031002 (2018)



Quantum interfaces for neutral atoms

Nanophotonic quantum phase switch with a single atom

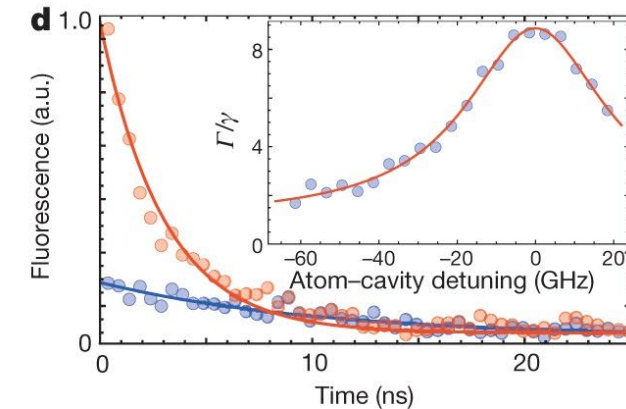
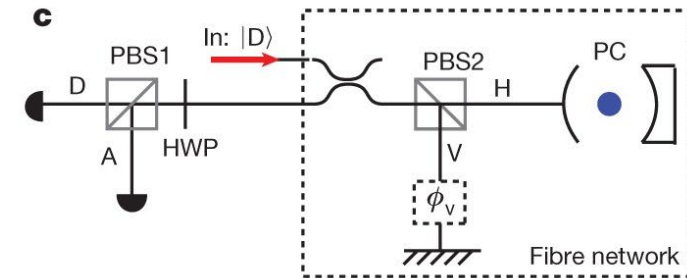
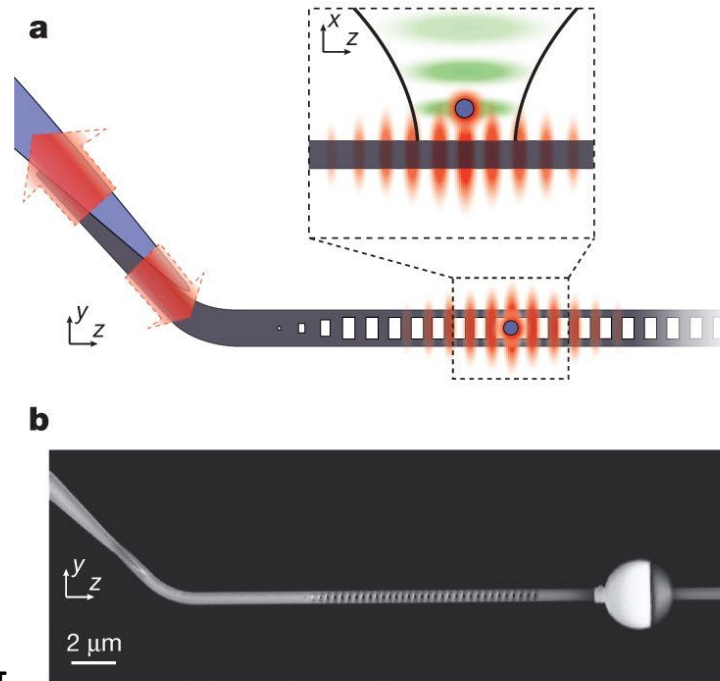
[T. G. Tiecke](#), [J. D. Thompson](#), [N. P. de Leon](#), [L. R. Liu](#), [V. Vuletić](#) & [M. D. Lukin](#)

Nature **508**, 241–244 (2014) | [Cite this article](#)

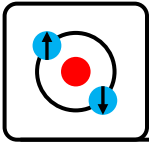
First atomic state - photonic cavity
controlled phase gate interface

- Prepare atom in superposition $|u\rangle + |c\rangle$
- Reflect one single photon
- Atomic superposition is shifted by nearly π
- Verified via Ramsey interference

- A single atom can flip the phase of a photon by $(1.1)\pi$



Cooperativity $C=7.7$



Quantum interfaces for neutral atoms

Entanglement transport and a nanophotonic interface for atoms in optical tweezers

TAMARA ĐORĐEVIĆ , POLNOP SAMUTPRAPHOOT , PALOMA L. OCOLA , HANNES BERNIEN , BRANDON GRINKEMEYER , IVANA DIMITROVA ,

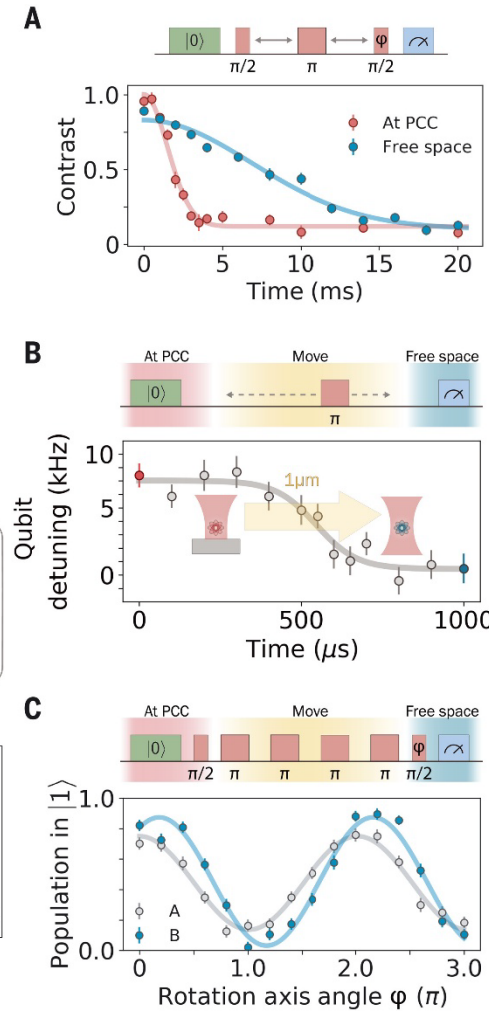
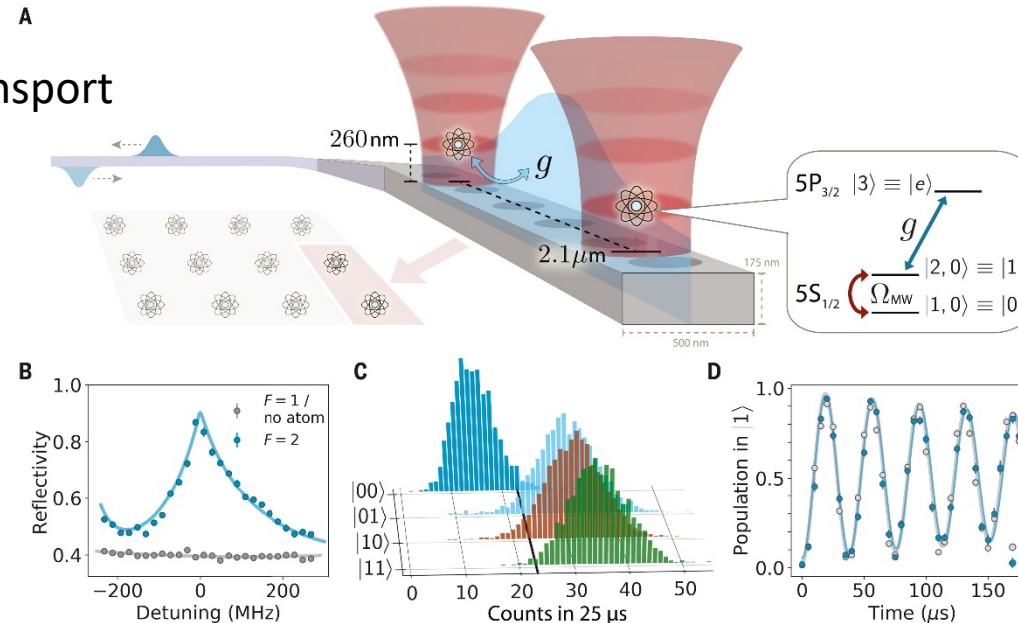
VLADAN VULETIĆ , AND MIKHAIL D. LUKIN [Authors Info & Affiliations](#)

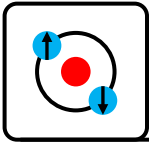
SCIENCE • 13 Aug 2021 • Vol 373, Issue 6562 • pp. 1511-1514 • DOI: 10.1126/science.abi9917

First demonstration of entanglement preservation after transporting atoms away from cavity into free space

Cooperativity $C=27$

- 15% loss probability for each atom during transport
- Fidelity 72% near cavity
- Fidelity 65% after 1μm transport
- Much longer lifetime away from cavity





Quantum interfaces for neutral atoms

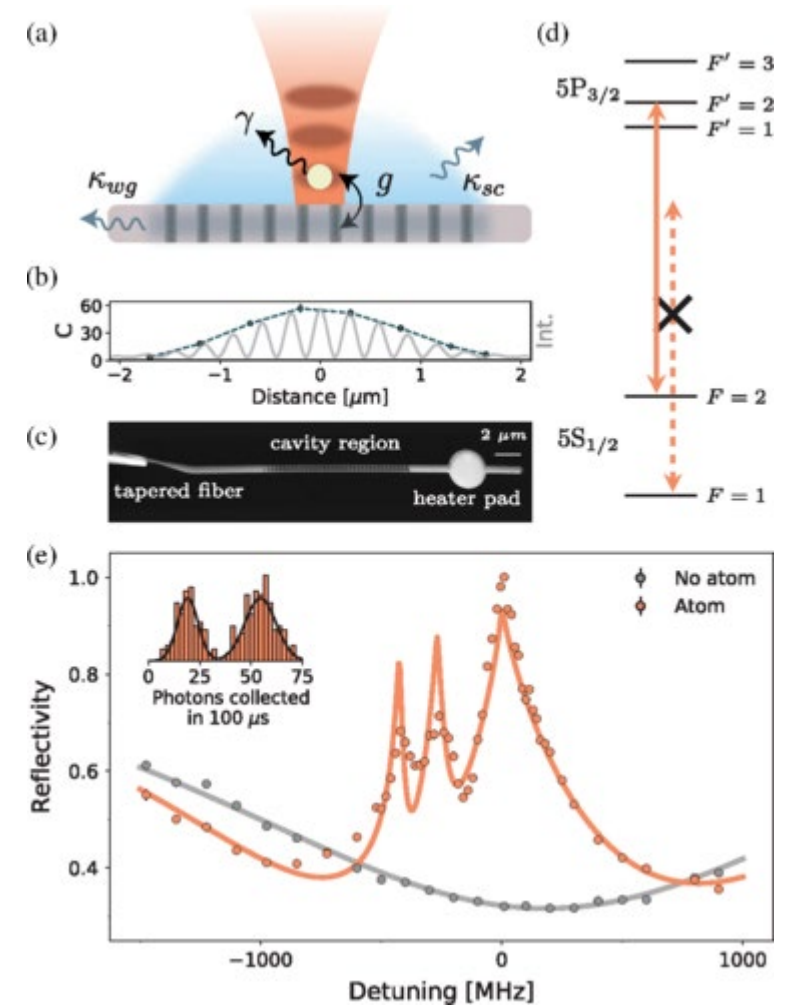
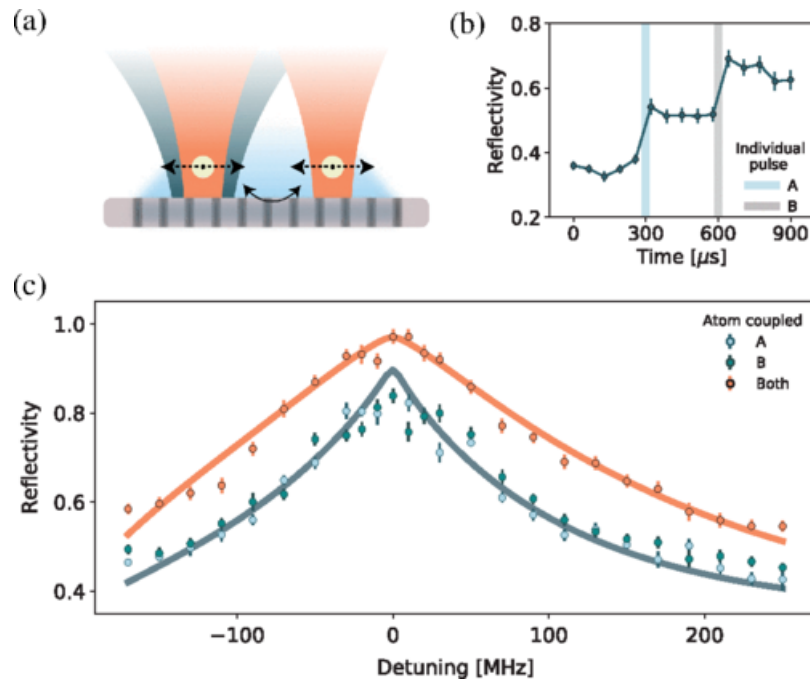
Strong Coupling of Two Individually Controlled Atoms via a Nanophotonic Cavity

[Polnop Samutpraphoot](#)^{1,*}, [Tamara Đorđević](#)^{1,*}, [Paloma L. Ocola](#)^{1,*}, [Hannes Bernien](#)², [Crystal Senko](#)^{3,4}, [Vladan Vuletić](#)⁵, and [Mikhail D. Lukin](#)^{1,†}

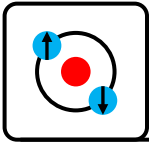
Show more

Phys. Rev. Lett. **124**, 063602 – Published 11 February, 2020

Long range cavity mediated interactions between 2 atoms in PCC



Cooperativity $C=71$
Single photon Rabi frequency 1.24GHz!



Quantum interfaces for neutral atoms

Atom-atom interactions around the band edge of a photonic crystal waveguide

Jonathan D. Hood, Akihisa Goban, Ana Asenjo-Garcia, ⁺³, and H. J. Kimble [Authors Info & Affiliations](#)

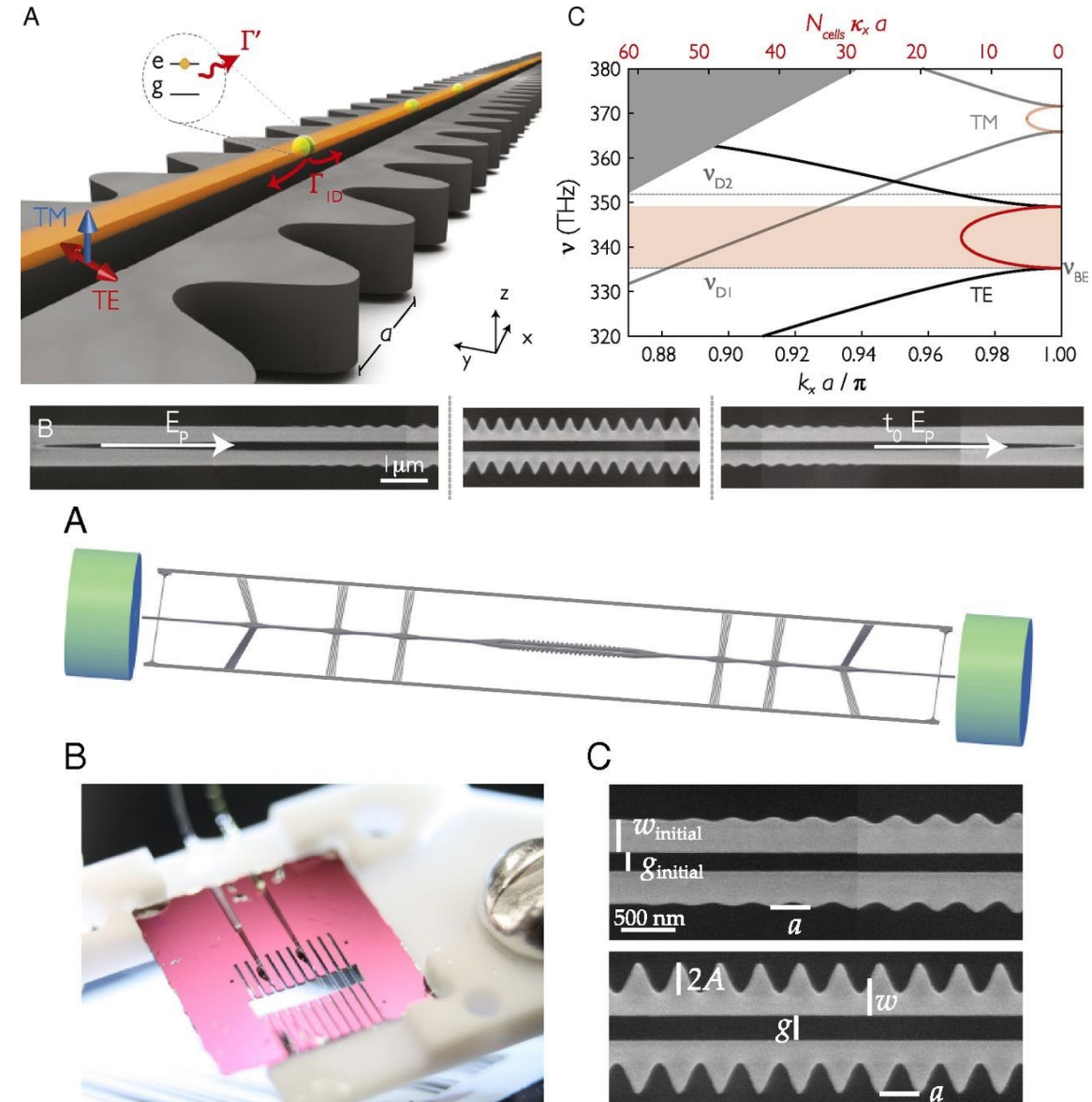
Contributed by H. Jeffrey Kimble, June 17, 2016 (sent for review March 7, 2016; reviewed by Eugene Polzik and Dan Stamper-Kurn)

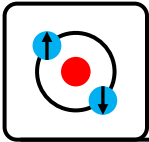
August 31, 2016 | 113 (38) 10507-10512 | <https://doi.org/10.1073/pnas.1603788113>

Long range strong coupling in a waveguide (NOT cavity!!)

Famous “alligator waveguide”

$$\beta = 0.3$$



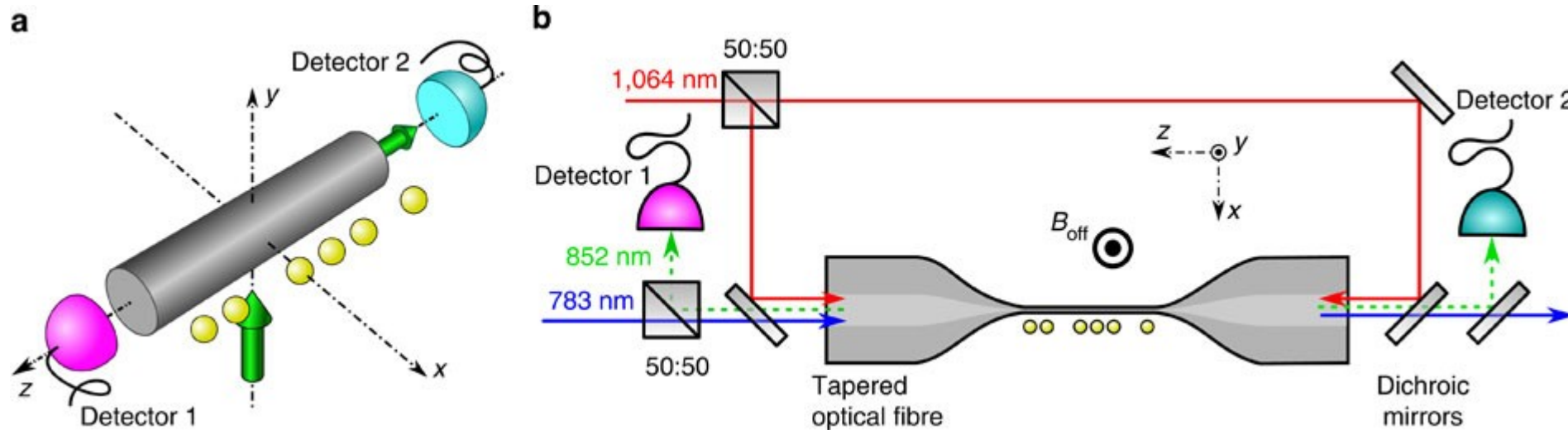


Quantum interfaces for neutral atoms

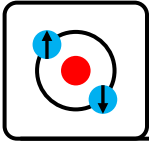
Quantum state-controlled directional spontaneous emission of photons into a nanophotonic waveguide

[R. Mitsch](#), [C. Sayrin](#), [B. Albrecht](#), [P. Schneeweiss](#) ✉ & [A. Rauschenbeutel](#) ✉

[Nature Communications](#) **5**, Article number: 5713 (2014) | [Cite this article](#)



strong coupling in a
de (NOT cavity!!)



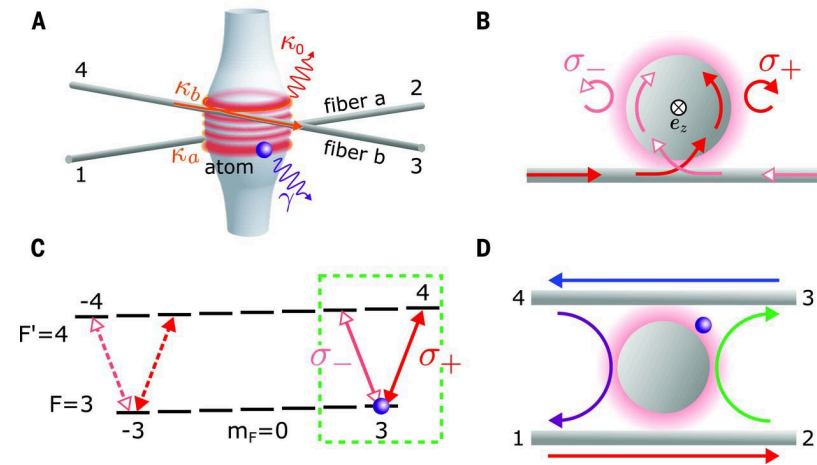
Quantum interfaces for neutral atoms

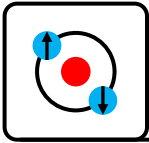
Quantum optical circulator controlled by a single chirally coupled atom

MICHAEL SCHEUCHER, ADÈLE HILICO, ELISA WILL, JÜRGEN VOLZ, AND ARNO RAUSCHENBEUTEL [Authors Info & Affiliations](#)

SCIENCE • 8 Dec 2016 • Vol 354, Issue 6319 • pp. 1577-1580 • DOI: 10.1126/science.aaj2118

Long range strong coupling in a waveguide (NOT cavity!!)



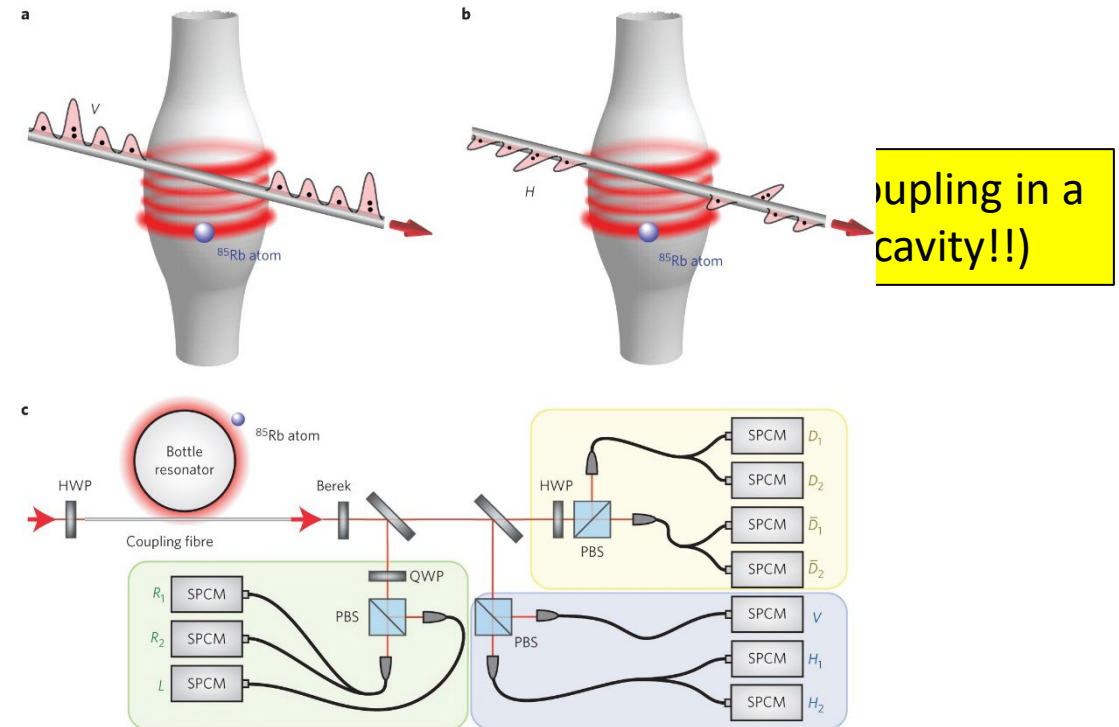


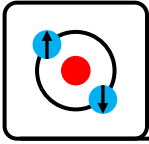
Quantum interfaces for neutral atoms

Nonlinear π phase shift for single fibre-guided photons interacting with a single resonator-enhanced atom

[Jürgen Volz](#) , [Michael Scheucher](#), [Christian Junge](#) & [Arno Rauschenbeutel](#) 

[Nature Photonics](#) **8**, 965–970 (2014) | [Cite this article](#)





Quantum interfaces for neutral atoms

Error-detected quantum operations with neutral atoms mediated by an optical cavity

BRANDON GRINKEMEYER , ELMER GUARDADO-SANCHEZ , IVANA DIMITROVA , DANILO SHCHEPANOVICH , G. EIRINI MANDOPOULOU ,

JOHANNES BORREGAARD , VLADAN VULETIĆ , AND MIKHAIL D. LUKIN [Authors Info & Affiliations](#)

SCIENCE • 20 Mar 2025 • Vol 387, Issue 6740 • pp. 1301-1305 • DOI: 10.1126/science.adr7075

Demonstrates high fidelity operations between 2 atoms in a cavity with built in error detection

Entanglement generation

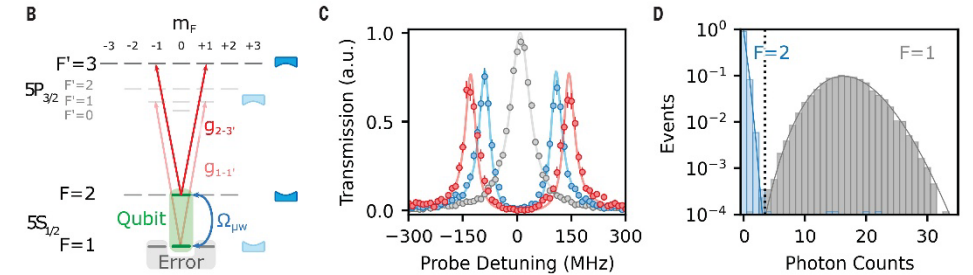
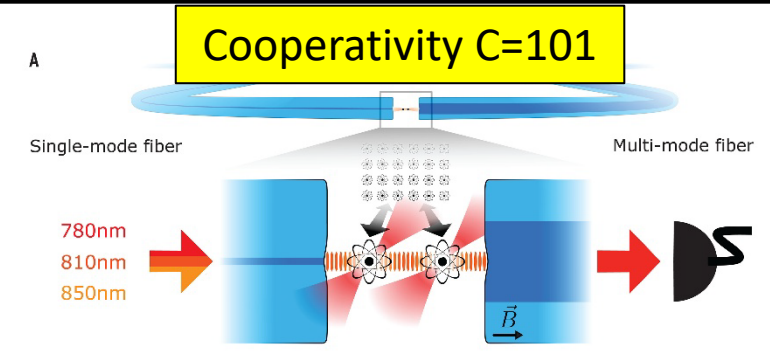
- Fidelity = 91%
- 32% deterministic success (no heralding!)

Deterministic controlled phase gates

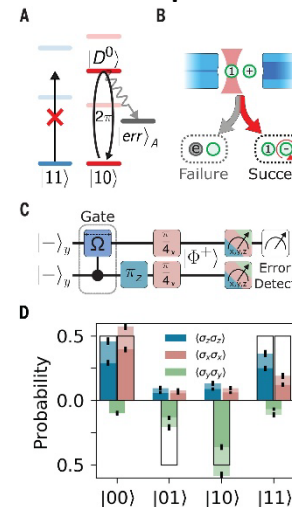
- Deterministic fidelity 52%
- Error detected fidelity 76%

High fidelity readout

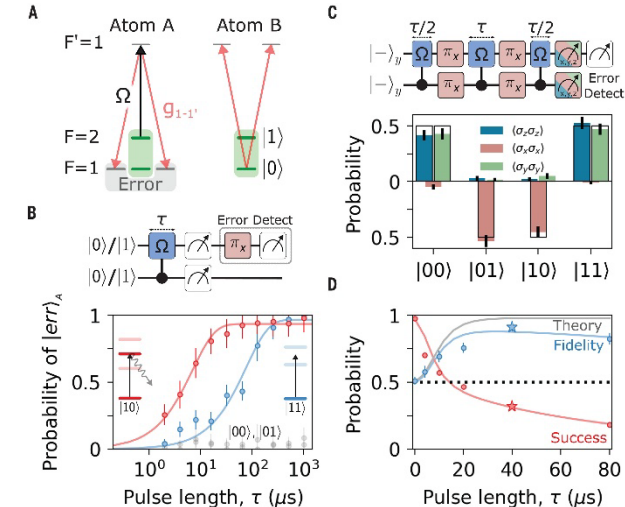
- 99.96% readout fidelity in 10us
- Just 0.034% loss

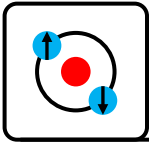


Deterministic phase gate:



Entanglement:





Quantum interfaces for neutral atoms

Heralded Entanglement Between Widely Separated Atoms

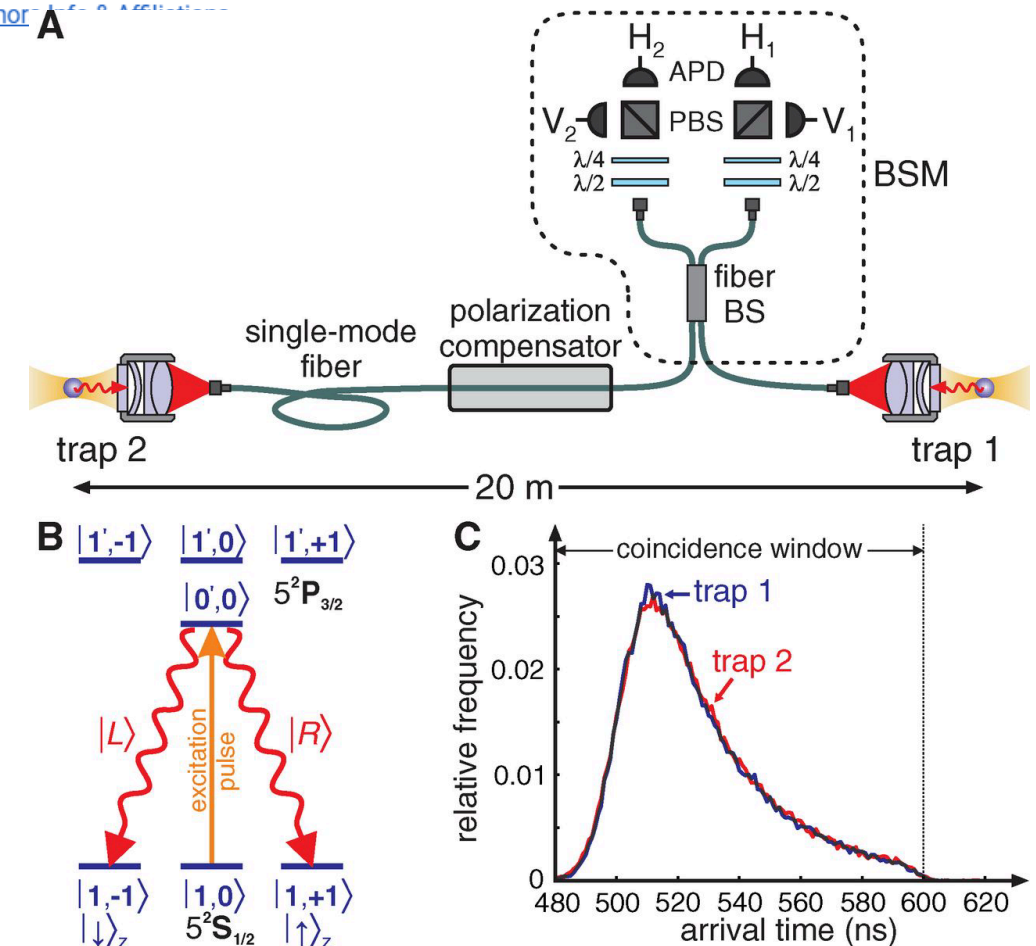
JULIAN HOFMANN, MICHAEL KRUG, NORBERT ORTEGEL, LEA GÉRARD, MARKUS WEBER, WENJAMIN ROSENFELD, AND HARALD WEINFURTER [Author information](#)

SCIENCE • 6 Jul 2012 • Vol 337, Issue 6090 • pp. 72-75 • DOI: 10.1126/science.1221856

First high-fidelity, heralded entanglement between two individually trapped remote atoms

Performance:

- Detection efficiency $1e-3$ (no cavity, low solid angle)
- Operate at 50kHz excitation rate
- $4.4e-7$ Heralded success probability (2 photons of different polarizations)
- 1 entanglement generated every 100 seconds



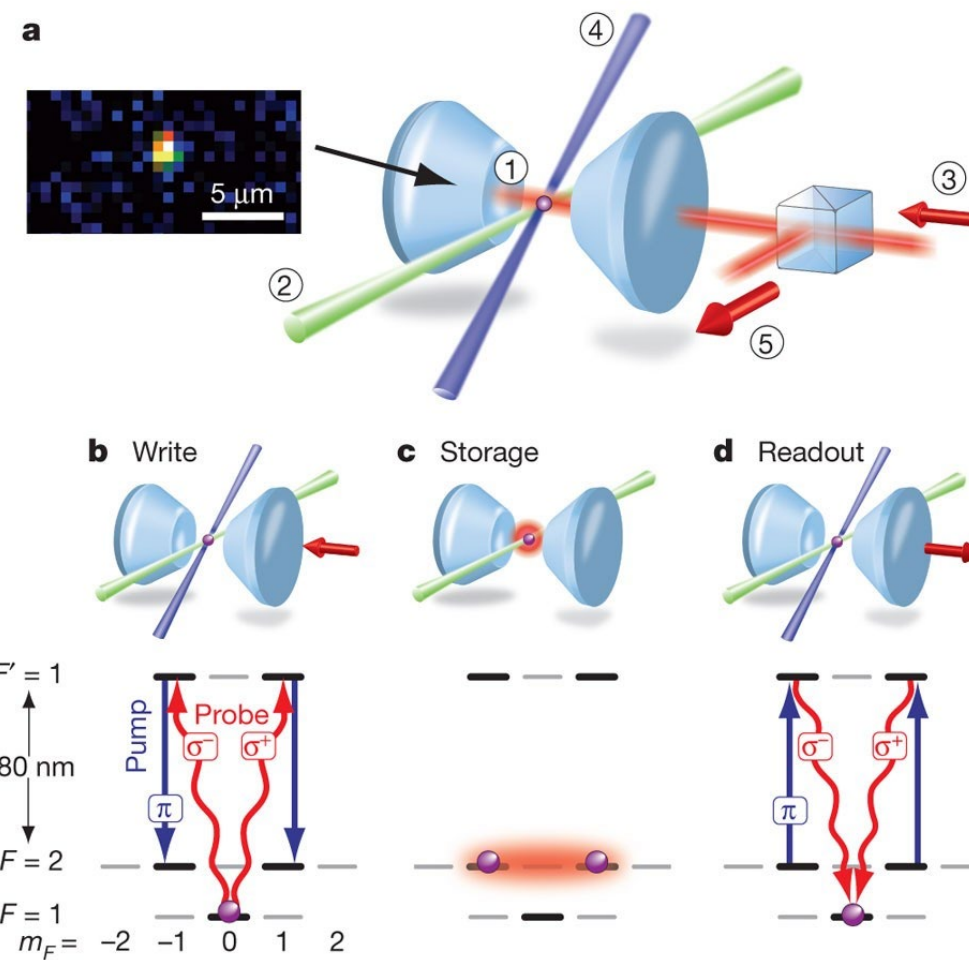


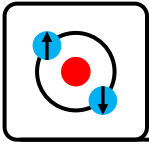
[Holger P. Specht](#), [Christian Nölleke](#), [Andreas Reiserer](#), [Manuel Uphoff](#), [Eden Figueroa](#), [Stephan Ritter](#) & [Gerhard Rempe](#)

First single atom quantum memory

Deterministic storage, not heralded

- 92.7% fidelity
- 184us storage time
- Uses control beam to store photon to atomic state
- 56% retrieval efficiency
- Storage+retrieval efficiency 9.3%



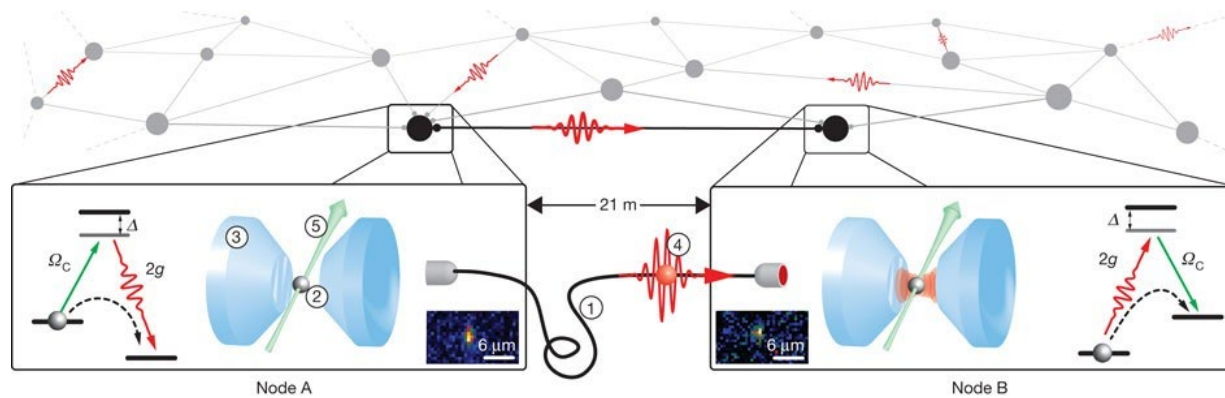


Quantum interfaces for neutral atoms

An elementary quantum network of single atoms in optical cavities

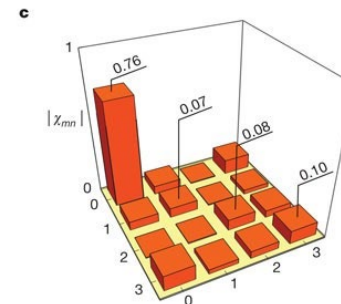
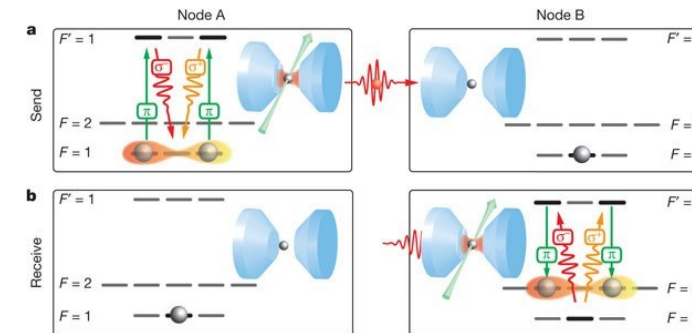
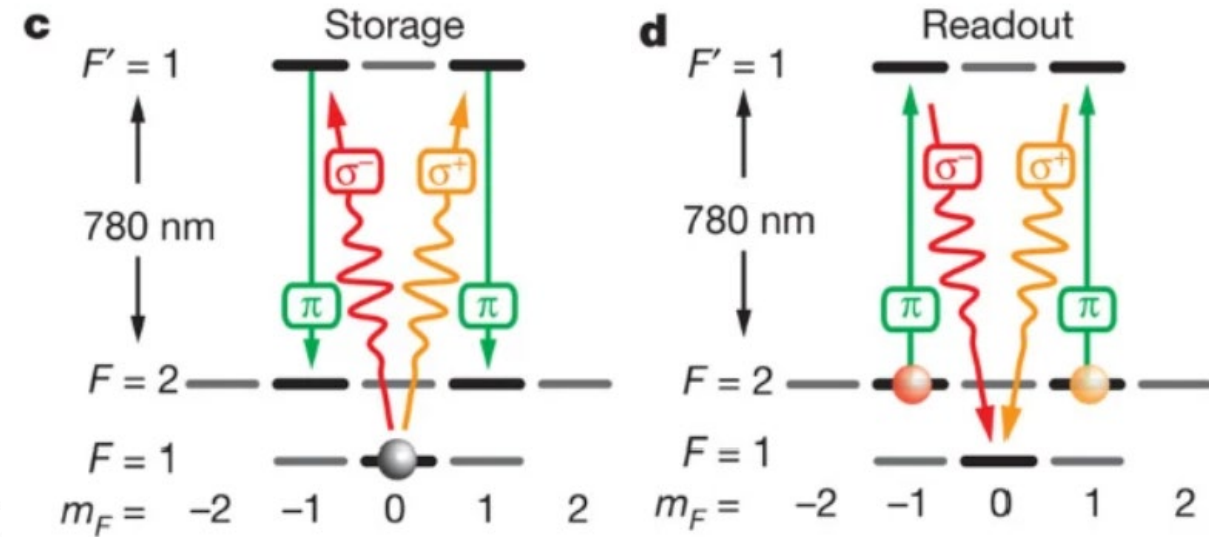
Stephan Ritter , Christian Nölleke, Carolin Hahn, Andreas Reiserer, Andreas Neuzner, Manuel Uphoff, Martin Mücke, Eden Figueroa, Joerg Bochmann & Gerhard Rempe

Nature **484**, 195–200 (2012) | [Cite this article](#)



First quantum network

- Able to send, receive and store quantum information between two remote nodes
- 60% photon creation efficiency
- 17% photon storage efficiency (STIRAP)
- State transfer fidelity 84%

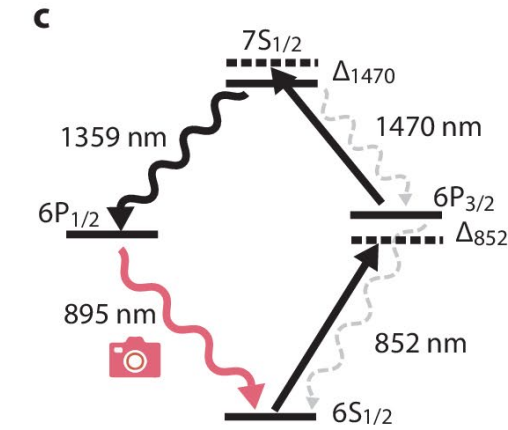
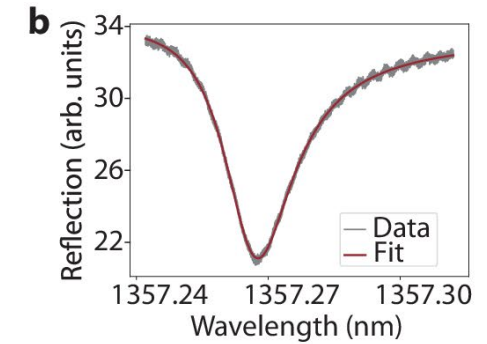
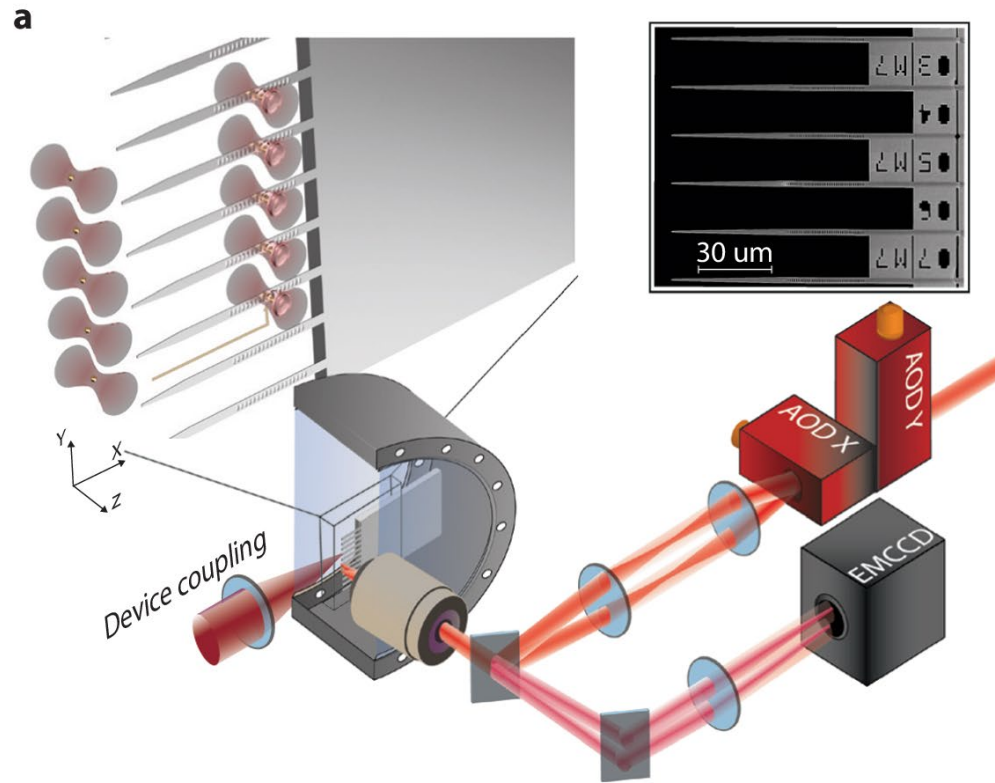
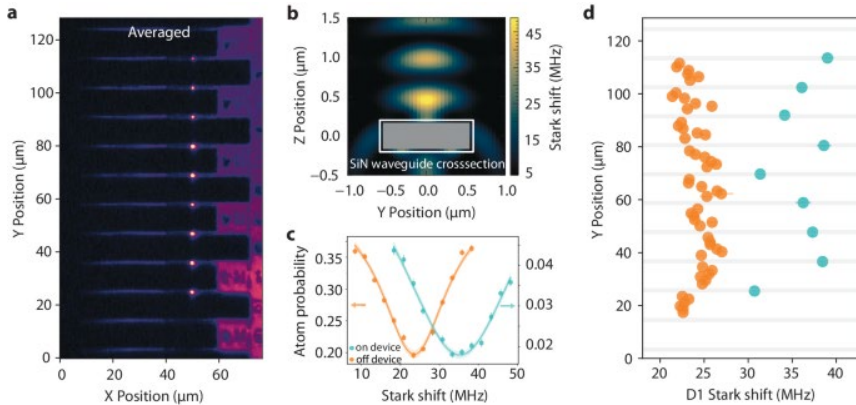


Cooperativity = 1.4



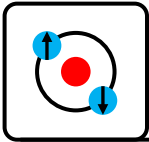
Nature Communications **15**, Article number: 6156 (2024) | [Cite this article](#)

Cavity arrays (proof-of-principle)



- Demonstrated scalable approach able to make hundreds of cavities
- 29% single atom loading probability

Estimated cooperativity $C \sim 50$



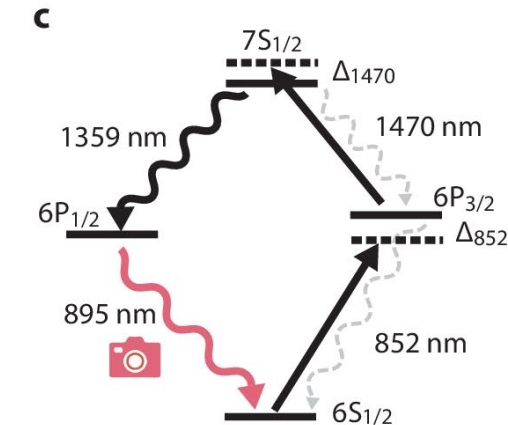
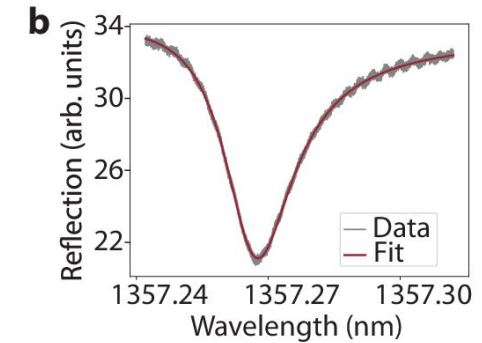
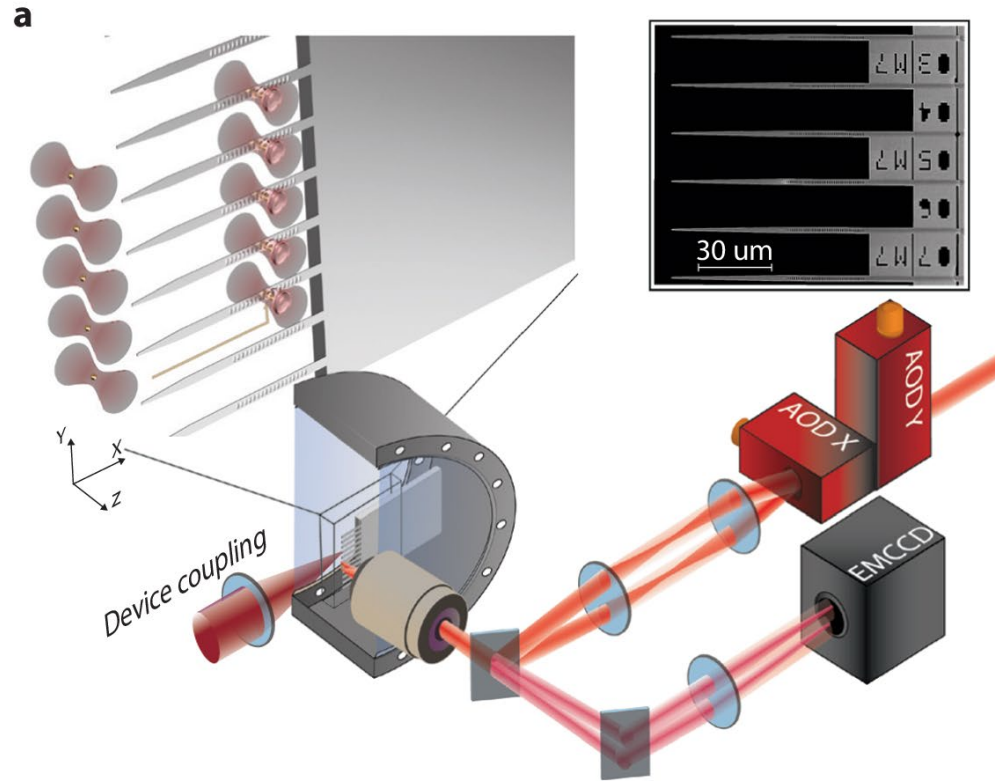
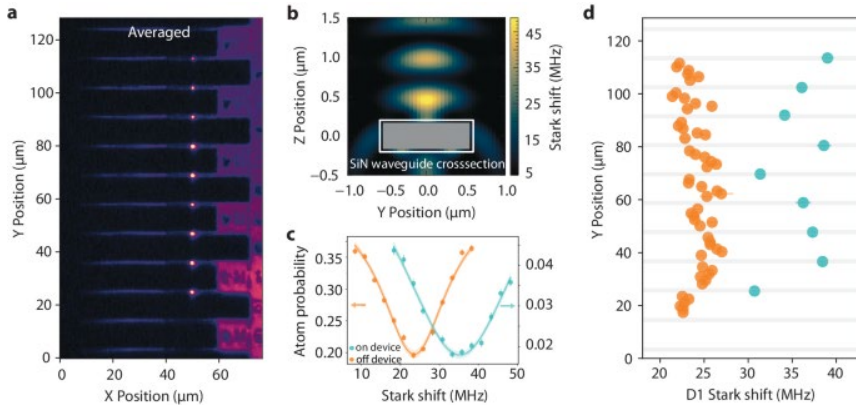
Quantum interfaces for neutral atoms

An integrated atom array-nanophotonic chip platform with background-free imaging

[Shankar G. Menon](#), [Noah Glachman](#), [Matteo Pompili](#), [Alan Dibos](#) & [Hannes Bernien](#)

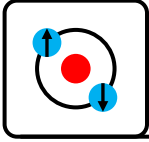
[Nature Communications](#) **15**, Article number: 6156 (2024) | [Cite this article](#)

Cavity arrays (proof-of-principle)c



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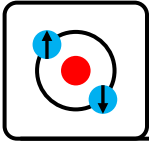


Quantum interfaces for neutral atoms

A quantum gate between a flying optical photon and a single trapped atom

[Andreas Reiserer](#), [Norbert Kalb](#), [Gerhard Rempe](#)  & [Stephan Ritter](#)

[Nature](#) **508**, 237–240 (2014) | [Cite this article](#)

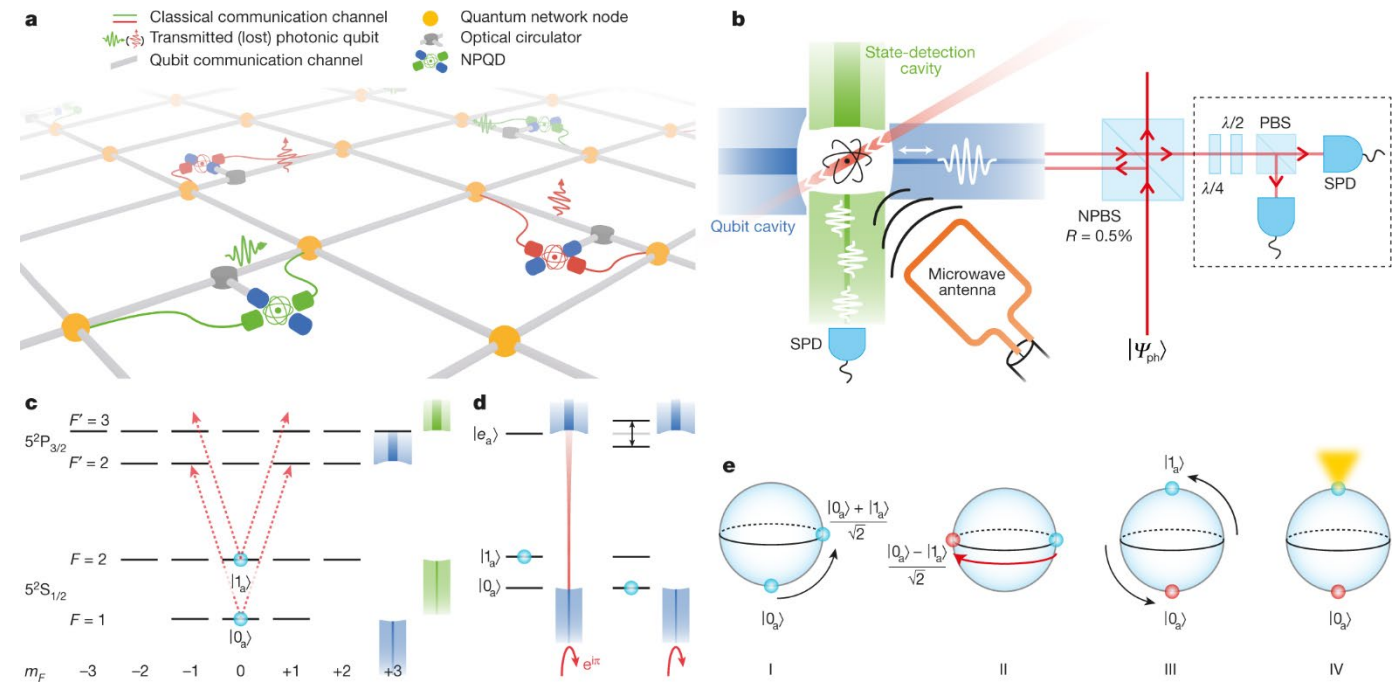


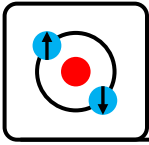
Quantum interfaces for neutral atoms

Nondestructive detection of photonic qubits

[Dominik Niemietz](#) , [Pau Farrera](#), [Stefan Langenfeld](#) & [Gerhard Rempe](#)

[Nature](#) **591**, 570–574 (2021) | [Cite this article](#)



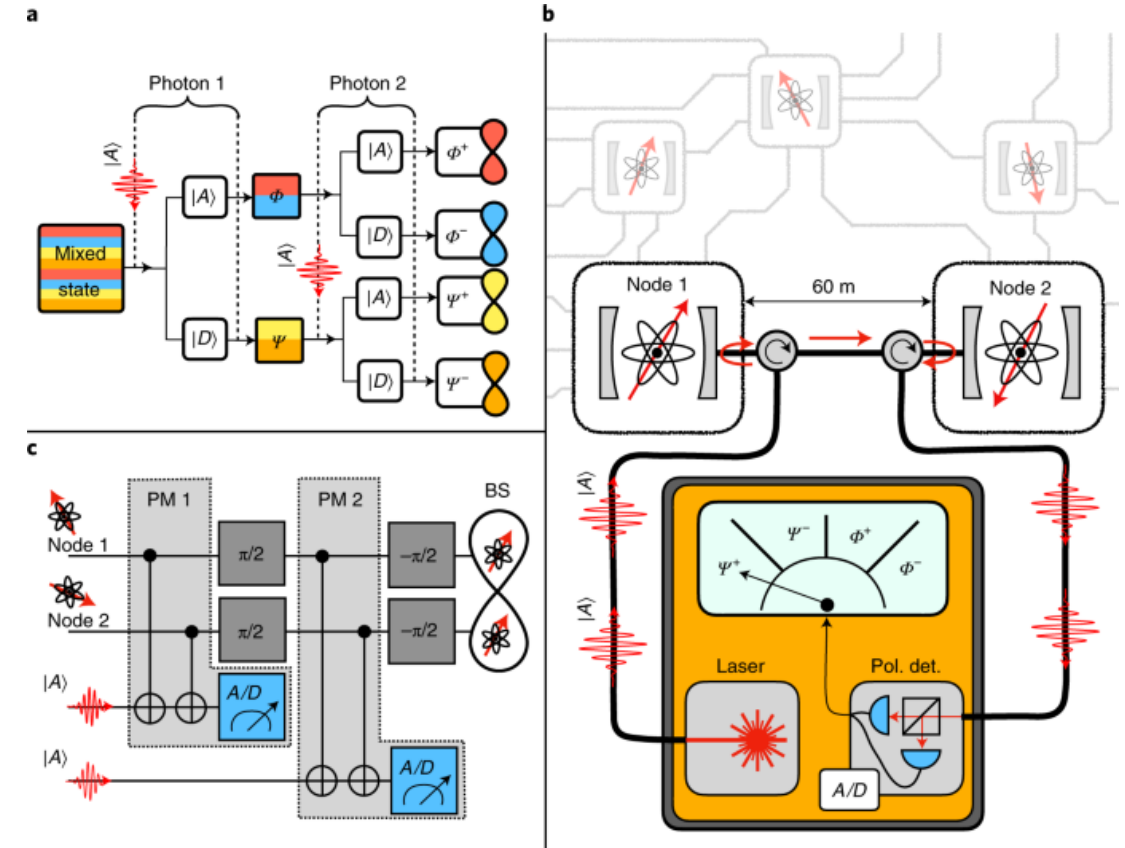


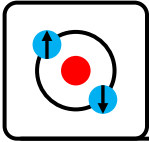
Quantum interfaces for neutral atoms

A nondestructive Bell-state measurement on two distant atomic qubits

[Stephan Welte](#), [Philip Thomas](#), [Lukas Hartung](#), [Severin Daiss](#), [Stefan Langenfeld](#), [Olivier Morin](#), [Gerhard Rempe](#) & [Emanuele Distante](#) ✉

[Nature Photonics](#) **15**, 504–509 (2021) | [Cite this article](#)





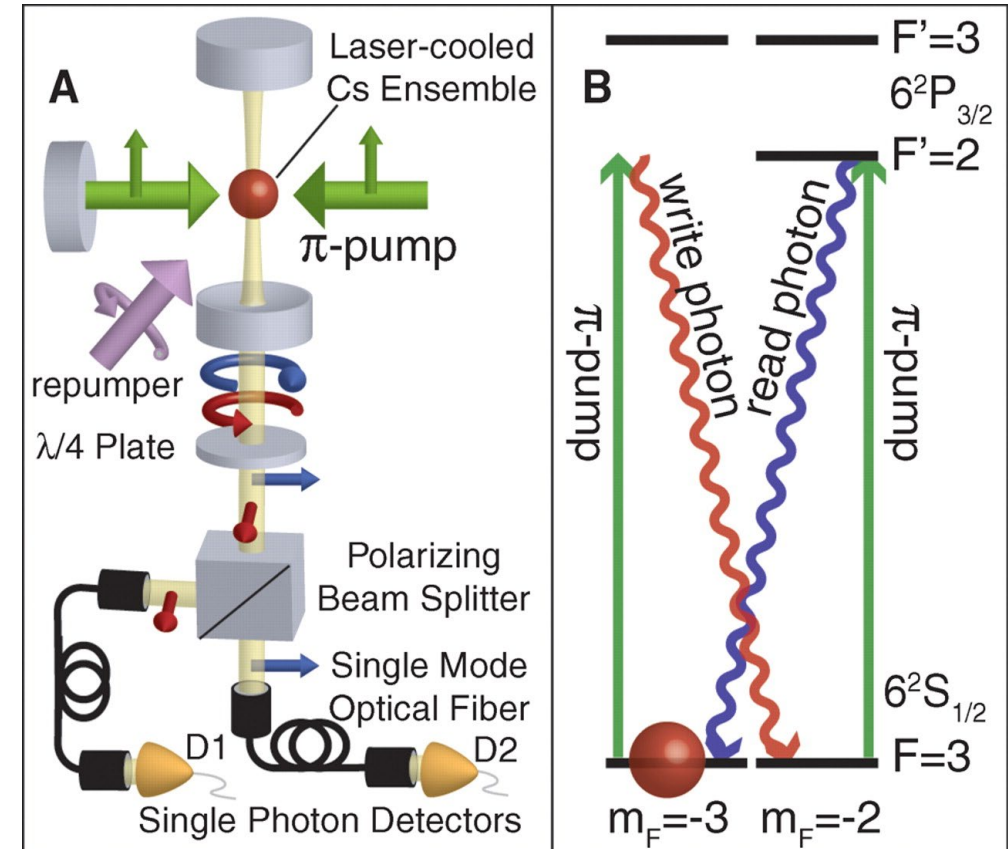
Quantum interfaces for neutral atoms

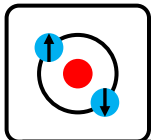
A High-Brightness Source of Narrowband, Identical-Photon Pairs

JAMES K. THOMPSON, JONATHAN SIMON, HUANQIAN LOH, AND VLADAN VULETIĆ [Authors Info & Affiliations](#)

SCIENCE • 7 Jul 2006 • Vol 313, Issue 5783 • pp. 74-77 • DOI: 10.1126/science.1127676

Cooperativity = 1.4



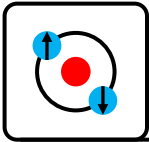


Experimental Progress

Interfacing to NV centers and quantum dots

References:

P Lodahl, S Mahmoodian, S Stobbe, Interfacing single photons and single quantum dots with photonic nanostructures. *Rev Mod Phys* **87**, 347 (2015).



Quantum interfaces for NV centers

Quantum entanglement between an optical photon and a solid-state spin qubit

[E. Togan](#), [Y. Chu](#), [A. S. Trifonov](#), [L. Jiang](#), [J. Maze](#), [L. Childress](#), [M. V. G. Dutt](#), [A. S. Sørensen](#), [P. R. Hemmer](#), [A. S. Zibrov](#) & [M. D. Lukin](#)

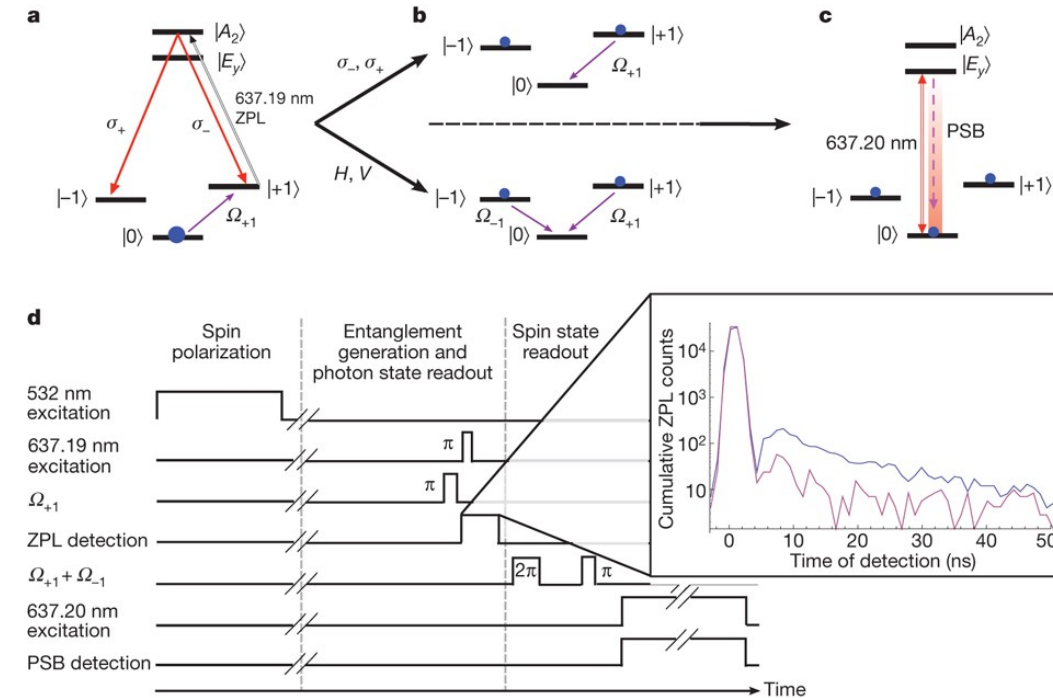
[Nature](#) **466**, 730–734 (2010) | [Cite this article](#)

First entanglement between a NV center and a photon

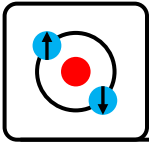
1. Prepare the NV in $|0\rangle$ then flip to $|+1\rangle$ with a microwave π pulse
2. Excite to $|A_2\rangle$ with optical π pulse
3. If it decays to $|-1\rangle$ a σ^- photon is emitted, if it decays to $|+1\rangle$ a σ^+ photon is emitted.

$$\frac{1}{\sqrt{2}} (|\sigma^+\rangle|-1\rangle + |\sigma^-\rangle|+1\rangle)$$

4. Collect photon and send to a PBS optionally with or without waveplate rotation to select measurement basis (H/V vs σ^+/σ^- measurement basis).
5. Apply microwave pulse to map the measured state to $|0\rangle$
6. Confirm NV is in $|0\rangle$ by measuring fluorescence on cycling $|0\rangle \rightarrow |E_y\rangle$



- Per event success probability $1e-6$
- 100kHz repetition rate
- Entanglement every 10 seconds



Quantum interfaces for NV centers

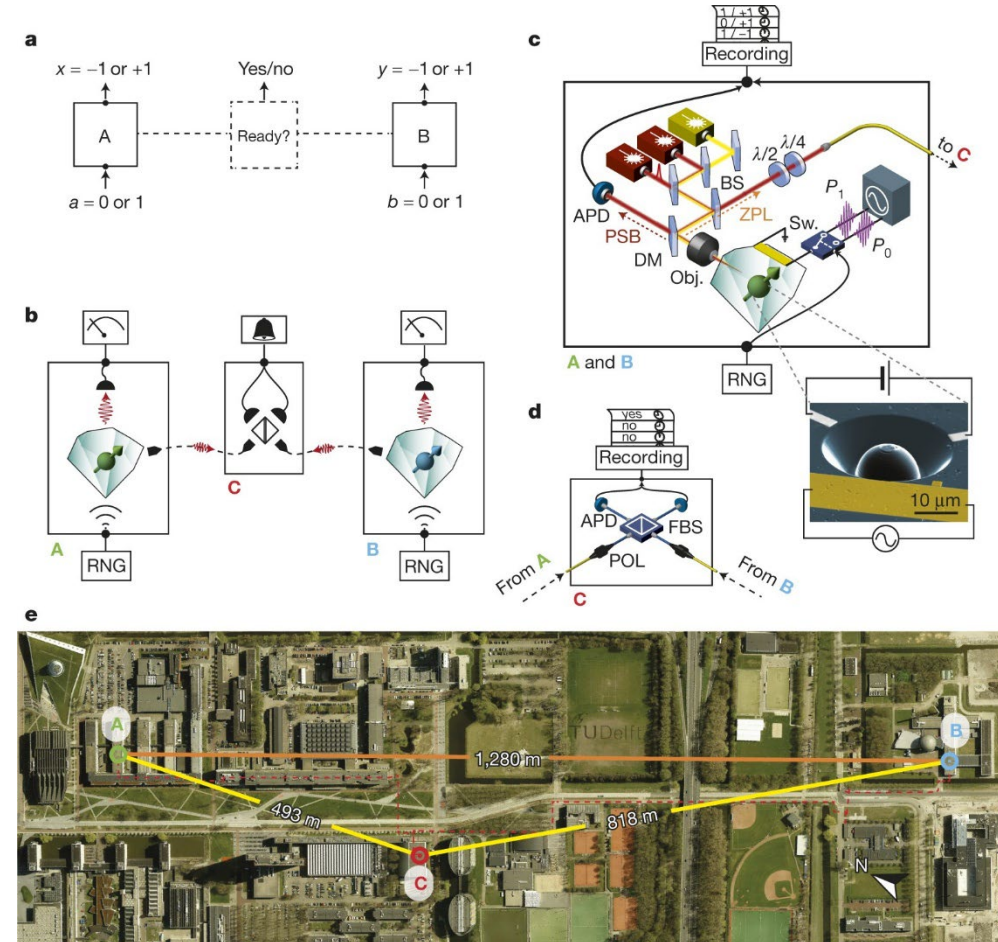
Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres

[B. Hensen](#), [H. Bernien](#), [A. E. Dréau](#), [A. Reiserer](#), [N. Kalb](#), [M. S. Blok](#), [J. Ruitenber](#), [R. F. L. Vermeulen](#), [R. N. Schouten](#), [C. Abellán](#), [W. Amaya](#), [V. Pruneri](#), [M. W. Mitchell](#), [M. Markham](#), [D. J. Twitchen](#), [D. Elkouss](#), [S. Wehner](#), [T. H. Taminiau](#) & [R. Hanson](#)

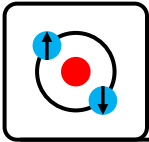
[Nature](#) 526, 682–686 (2015) | [Cite this article](#)

First fully loophole-free Bell test

1. Start with two identical remote setups NV centers in diamond in 4K cryostats.
2. Pump each in a known spin $|m_s=0\rangle$ state.
3. Use a microwave $\pi/2$ pulse to put each into a superposition state $|m_s=0\rangle + |m_s=-1\rangle$.
4. Apply two time separated resonant optical pulses first one for $|m_s=0\rangle$ and later $|m_s=-1\rangle$, if in $|m_s=0\rangle$ emits photon earlier, if in $|m_s=-1\rangle$ fluorescent photon emitted later.
5. The fluorescence decay photon is collected by a microscope and sent to a beamsplitter at C 500m/800m away from the two setups.
6. Use a random number to choose one of two microwave pulses. These pulses rotate the spin so that reading out in the usual $|m_s=0\rangle$ or $|m_s=-1\rangle$ basis effectively measures the spin either along 0 or tilted by $\pm 45^\circ$.
7. Apply a resonant optical π pulse and read the resulting fluorescence photon.
8. At the beamsplitter if photon is early and one is late the two NV centres are in an entangled state (**heralded**).
9. Post select only those measurements in which entanglement was heralded.



- Per event success probability $6e-9$
- Ran experiment for 220 hours with 245 valid Bell tests



Quantum interfaces for NV centers

An integrated diamond nanophotonics platform for quantum-optical networks

A. SIPAHIGIL, R. E. EVANS, D. D. SUKACHEV, M. J. BUREK, J. BORREGAARD, M. K. BHASKAR, C. T. NGUYEN, J. L. PACHECO, H. A. ATIKIAN, [...], AND M. D. LUKIN

[Authors Info & Affiliations](#)

SCIENCE • 13 Oct 2016 • Vol 354, Issue 6314 • pp. 847-850 • DOI: 10.1126/science.aah6875

+6 authors

Strong coupling - Cooperativity $C \sim 1$

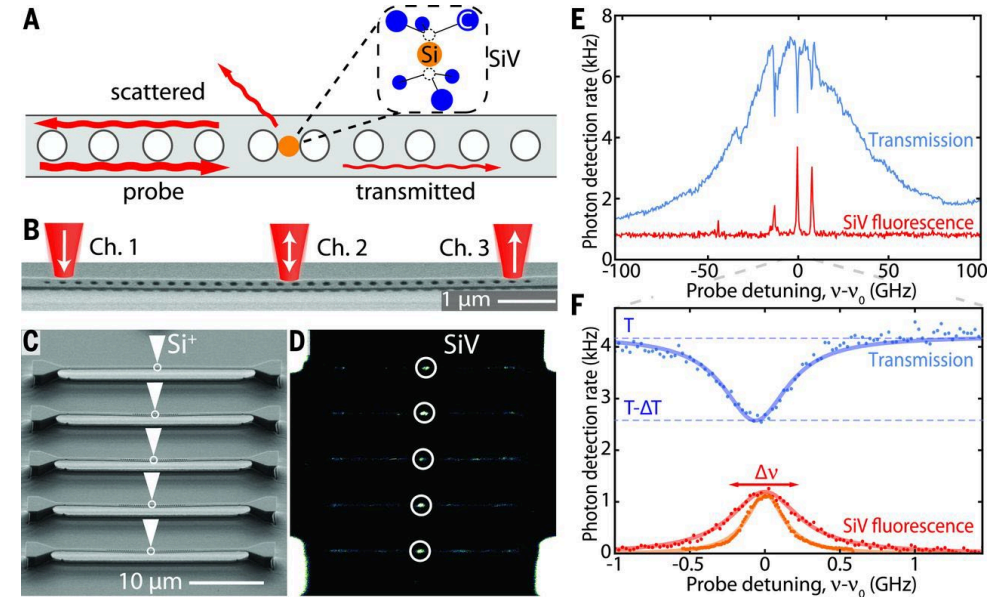
Spectrally tunable single photon source (Raman)

Entanglement between NV centers fiber coupled

Single photon switching

This single platform had 2000 NV center/cavity nodes and demonstrated:

- deterministic emitter placement
- cavity-enhanced emission
- single-photon switching
- tunable single-photon source
- waveguide-mediated two-emitter entanglement



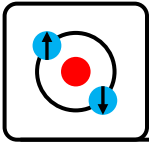
Raman photons ~ 15 kHz

Entanglement ~ 0.5 Hz (entanglement lifetime $T_2 \sim 2.5$ ns)

Challenges:

- 4K operation
- Coherence time < 50 ns
- SiV lifetime ~ 1.8 ns
- SiV centers demonstrated ~ 100 MHz variation (vs GHz/10 GHz for NV centers), ~ 0 for atoms/ions

See also: High quality-factor optical nanocavities in bulk single-crystal diamond, Nature Comms **5, 5718 (2014)



Quantum interfaces for NV centers

Entanglement of nanophotonic quantum memory nodes in a telecom network

[C. M. Knaut, A. Suleymanzade, Y.-C. Wei, D. R. Assumpcao, P.-J. Stas, Y. Q. Huan, B. Machielse, E. N. Knall, M. Sutula, G. Baranes, N. Sinclair, C. De-Eknamkul, D. S. Levonian, M. K. Bhaskar, H. Park, M. Lončar & M. D. Lukin](#)

[Nature](#) **629**, 573–578 (2024) | [Cite this article](#)

Entanglement distribution @ 1350nm over 40km deployed fiber

Deterministic! Cooperativity=12

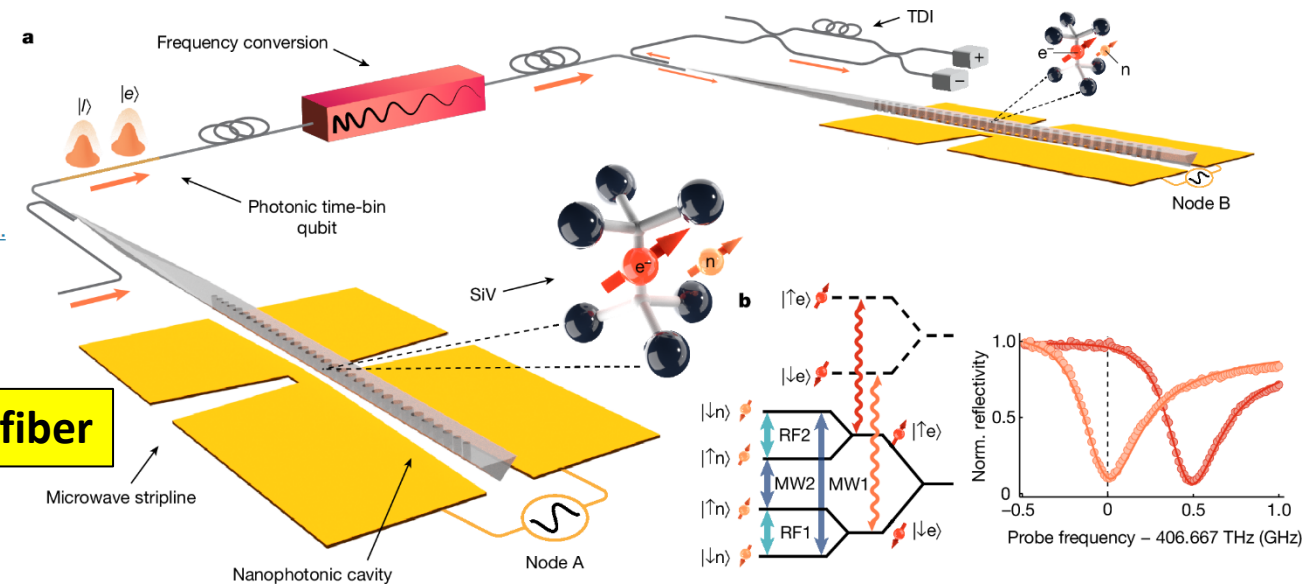
Each node is a **multi-qubit register**:

- 1 electron spin (communication qubit, fast, cavity-coupled)
- 1 nuclear spin (long-lived quantum memory, >2 s coherence)

Entanglement of electron spins at ~1Hz, fidelity 86%

Transfer to nuclear spins which have a lifetime up to 2s

Nuclear spin entanglement rate 1 per 170s (6mHz)



See also:

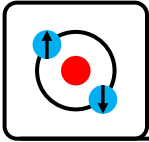
Experimental demonstration of memory-enhanced quantum communication

[M. K. Bhaskar, R. Riedinger, B. Machielse, D. S. Levonian, C. T. Nguyen, E. N. Knall, H. Park, D. Englund, M. Lončar, D. D. Sukachev & M. D. Lukin](#)

[Nature](#) **580**, 60–64 (2020) | [Cite this article](#)

Demonstrated memory assisted

Deterministic! Cooperativity=105



Quantum interfaces for quantum dots

Controlling the dynamics of spontaneous emission from quantum dots by photonic crystals

Peter Lodahl , A. Floris van Driel, Ivan S. Nikolaev, Arie Imman, Karin Overgaag,

Willem L. Vos

Nature **430**, 654–657 (2004) |

Deterministic photon–emitter coupling in chiral photonic circuits

Near-Unity Coupling a Photonic Crystal

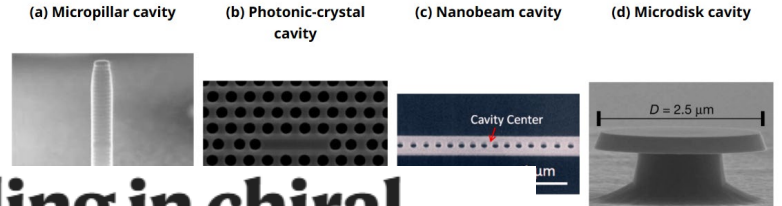
M. Arcari¹, I. Söllner^{1,*}, A. Javadi¹, S. L. Lee², J. D. Song² et al.

Show more

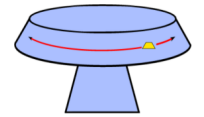
Immo Söllner , Sahand Mahmoodian, Sofie Lindskov Hansen, Leonardo Midolo, Alisa Javadi, Gabija Kiršanskė, Tommaso Pregonato, Haitham El-Ella, Eun Hye Lee, Jin Dong Song, Søren Stobbe & Peter Lodahl

Phys. Rev. Lett. **113**, 093603 – Published 21

Nature Nanotechnology **10**, 775–778 (2015) | [Cite this article](#)

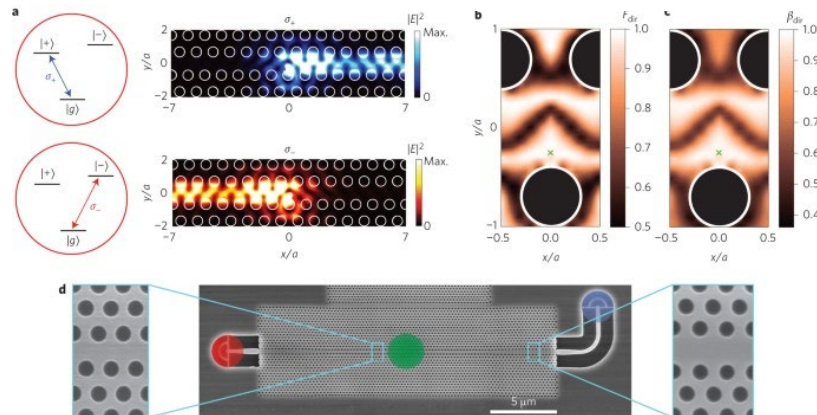


(
Srinivasan and Painter,
2007
)

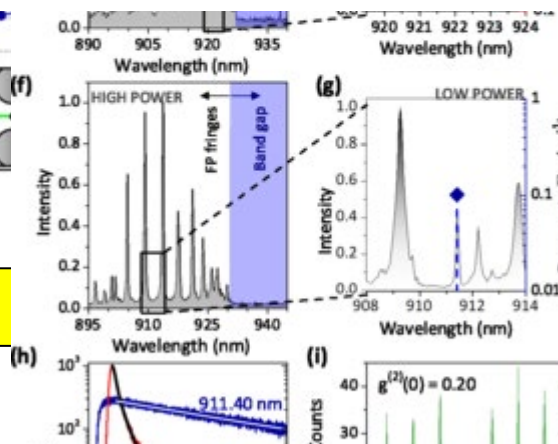


$g/2\pi = 3$ GHz
 $\kappa/2\pi = 1$ GHz
 $\gamma/2\pi \sim 0.6$ GHz

single quantum dots



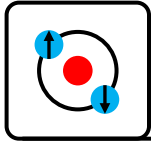
Entanglement



Photonic nanostructures
Sahand Mahmoodian, and Søren Stobbe

347 – Published 11 May, 2015

40km deployed fiber

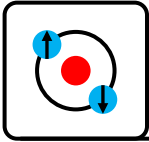


Experimental Progress

Interfacing to superconducting qubits

References:

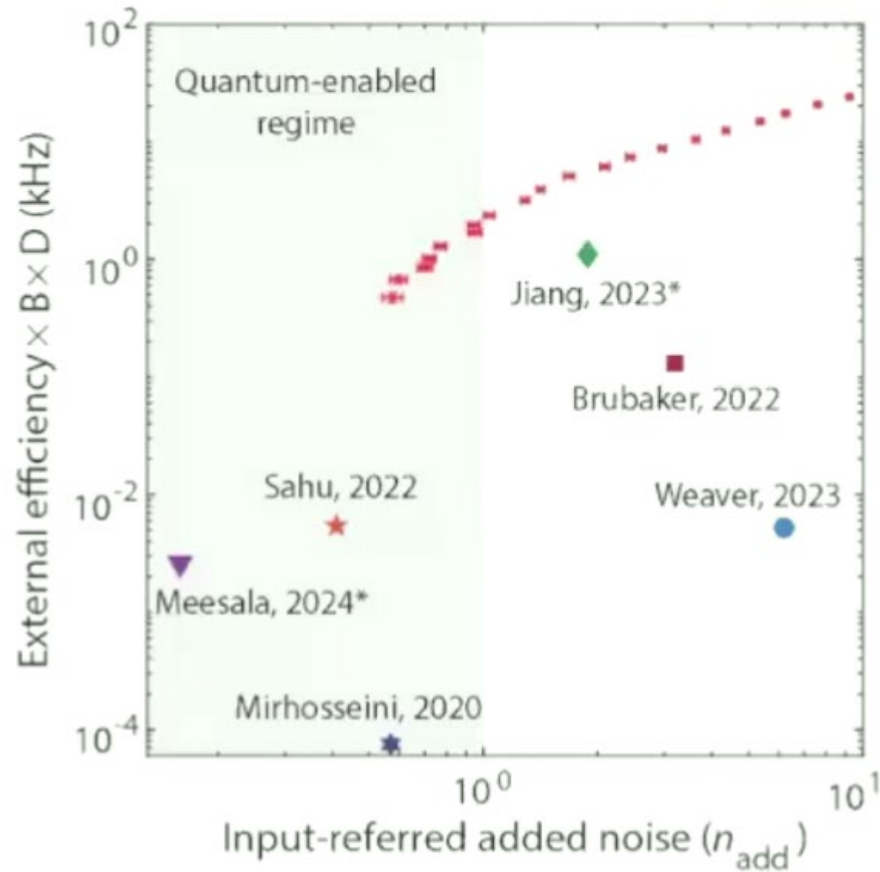
Optomechanics for quantum technologies, Nature Physics **18**, 15 (2022)



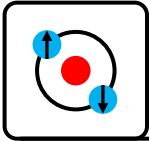
Progress in quantum interfaces for superconducting qubits

Microwave-optical conversion rate:

Microwave–Optical photon conversion rate
Efficiency*Bandwidth*Duty Cycle



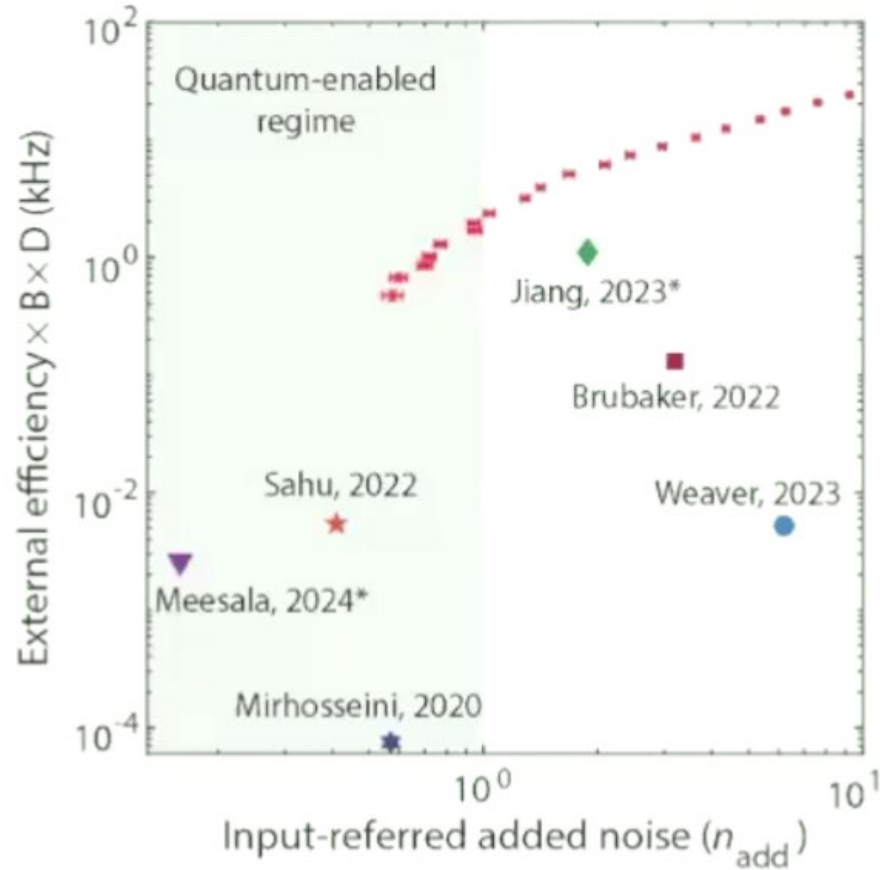
Should be able to operate close to the “clock speed of the system you want to interface to: MHz for superconducting qubits, kHz for ions



Progress in quantum interfaces for superconducting qubits

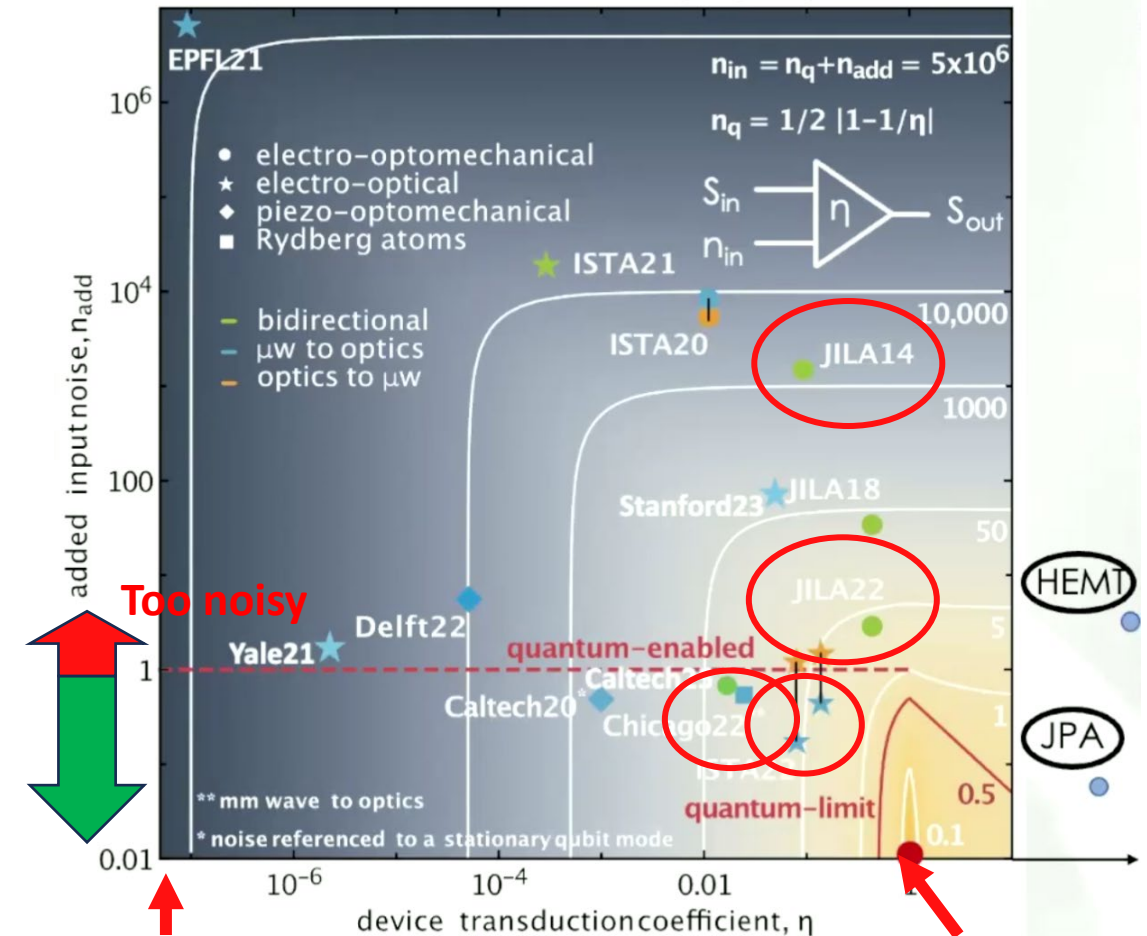
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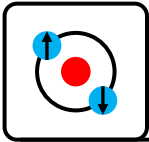
Should be able to operate close to the “clock speed of the system you want to interface to: MHz for superconducting qubits, kHz for ions

Total input noise:



If added noise is < 1 photon it becomes possible to

- Distribute entanglement and teleport
- Herald on successful events



Opto-mechanical microwave-optical transducers

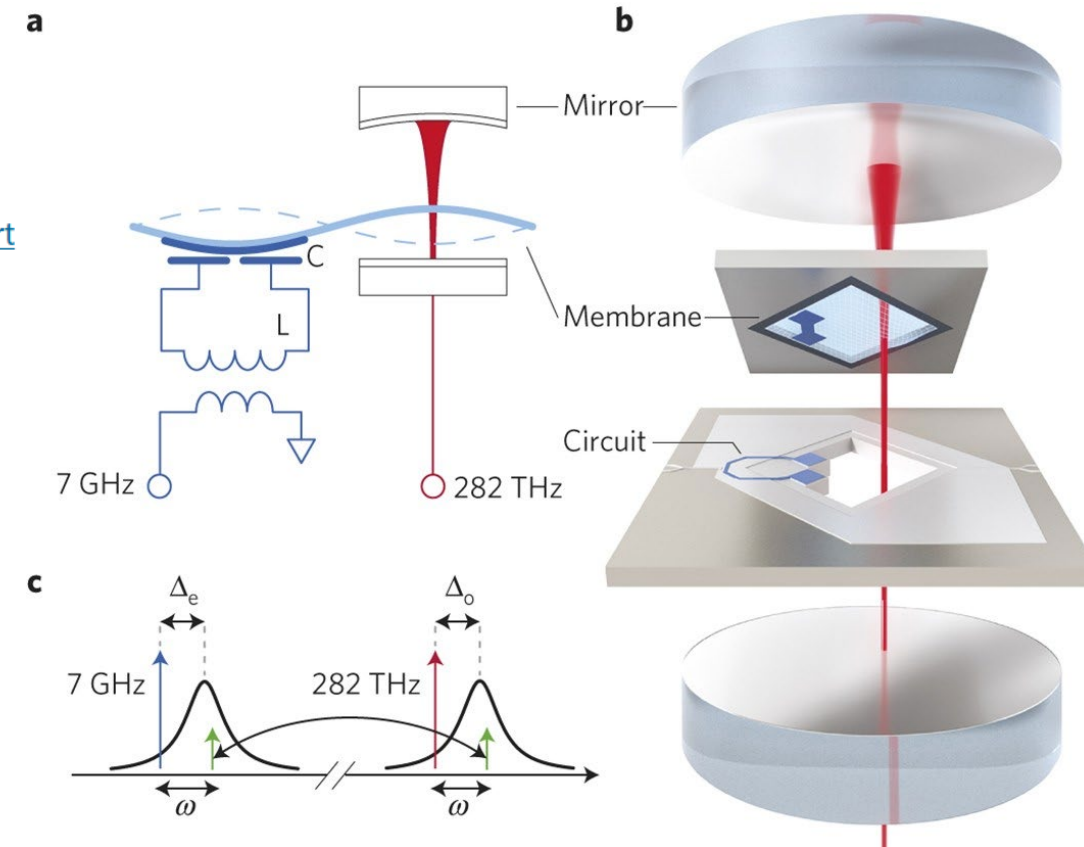
Bidirectional and efficient conversion between microwave and optical light

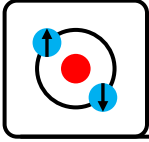
[R. W. Andrews](#) , [R. W. Peterson](#), [T. P. Purdy](#), [K. Cicak](#), [R. W. Simmonds](#), [C. A. Regal](#) & [K. W. Lehnert](#)

[Nature Physics](#) **10**, 321–326 (2014) | [Cite this article](#)

Proof-of-principle demonstration
($\sim 10^3$ photons of added noise)

First bidirectional electro-optomechanical
microwave-optical transducer





Opto-mechanical microwave-optical transducers

PHYSICAL REVIEW X 12, 021062 (2022)

Optomechanical Ground-State Cooling in a Continuous and Efficient Electro-Optic Transducer

B. M. Brubaker^{1,2,*} J. M. Kindem^{1,2,*} M. D. Urmey^{1,2,*†} S. Mittal^{1,2} R. D. Delaney^{1,2}
P. S. Burns^{1,2} M. R. Vissers³ K. W. Lehnert^{1,2,3} and C. A. Regal^{1,2}

¹JILA, National Institute of Standards and Technology and the University of Colorado,
Boulder, Colorado 80309, USA

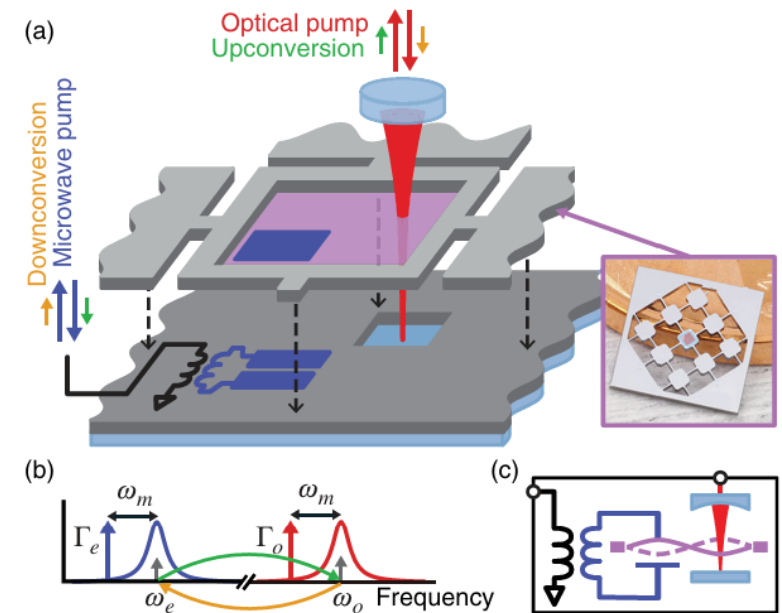
²Department of Physics, University of Colorado, Boulder, Colorado 80309, USA

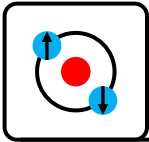
³National Institute of Standards and Technology, Boulder, Colorado 80305, USA

First continuous, high-efficiency, nearly ground-state,
low-noise electro-optomechanical transducer

Key improvements:

- Operating temperature 4K -> 40mK
- Mechanical mode $1e5$ phonon population -> 0.34 phonons
- Added noise: $1e3$ photons -> 3.2 photons
- Efficiency 8.6% -> 47% by reducing optical cavity losses and better modematching
- Heating of superconductors from optical scatter largely eliminated
- Much higher Q mechanical membrane $Q=1.3e7$
- Use of NbTiN superconductor low loss to optical light (very low optical loss)





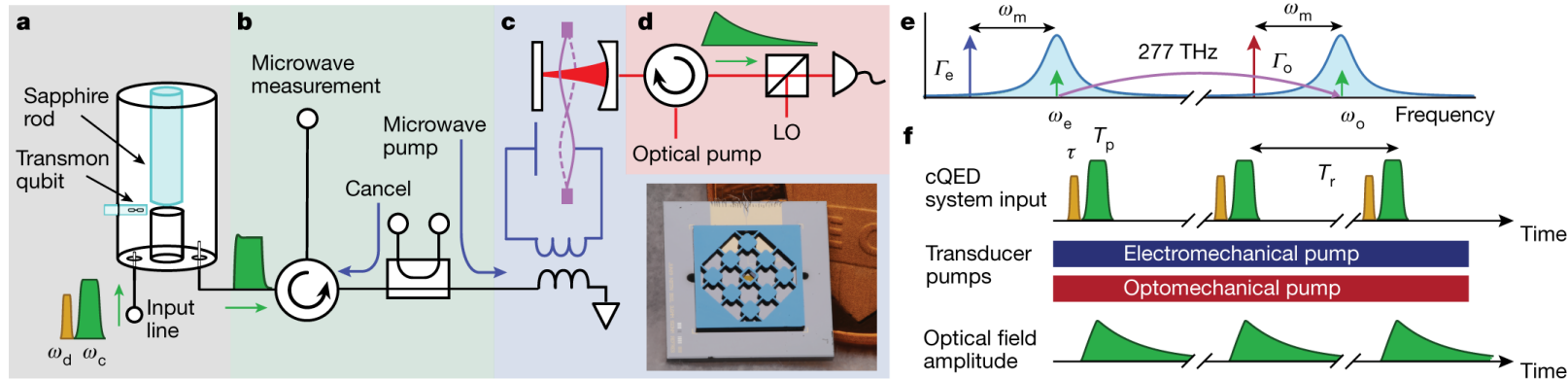
Opto-mechanical microwave-optical transducers

Superconducting-qubit readout via low-backaction electro-optic transduction

R. D. Delaney , M. D. Urmey, S. Mittal, B. M. Brubaker, J. M. Kindem, P. S. Burns, C. A. Regal & K. W.

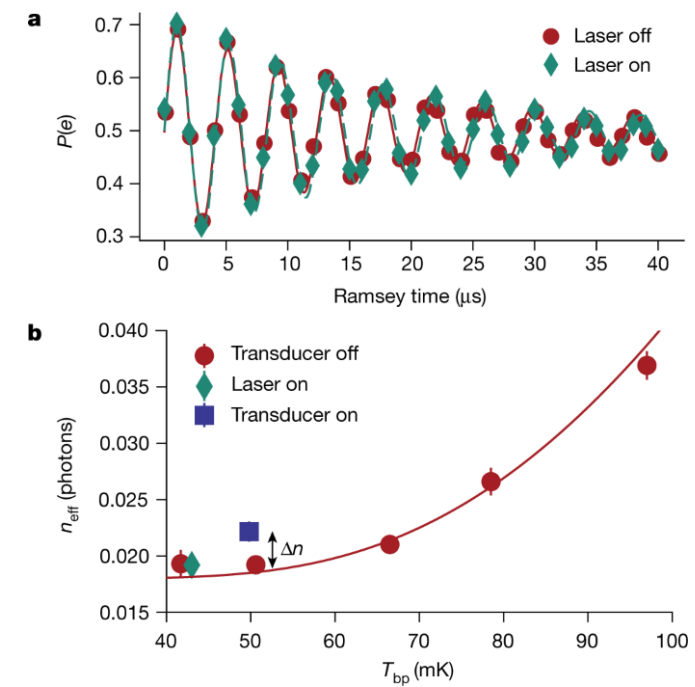
Lehnert

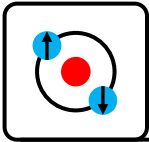
Nature **606**, 489–493 (2022) | [Cite this article](#)



First demonstration of a superconducting qubit readout via optical upconversion.

Demonstrated low backaction on superconducting qubit ($\Delta n \sim 3 \times 10^{-3}$ photons)





Piezo-optomechanical microwave-optical transducers

Superconducting qubit to optical photon transduction

[Mohammad Mirhosseini](#), [Alp Sipahigil](#), [Mahmoud Kalaei](#) & [Oskar Painter](#)

Nature **588**, 599–603 (2020) | [Cite this article](#)

First demonstration of a quantum enabled superconducting microwave to optical transducer (noise added < 1photon)
Added noise just 0.57photons!

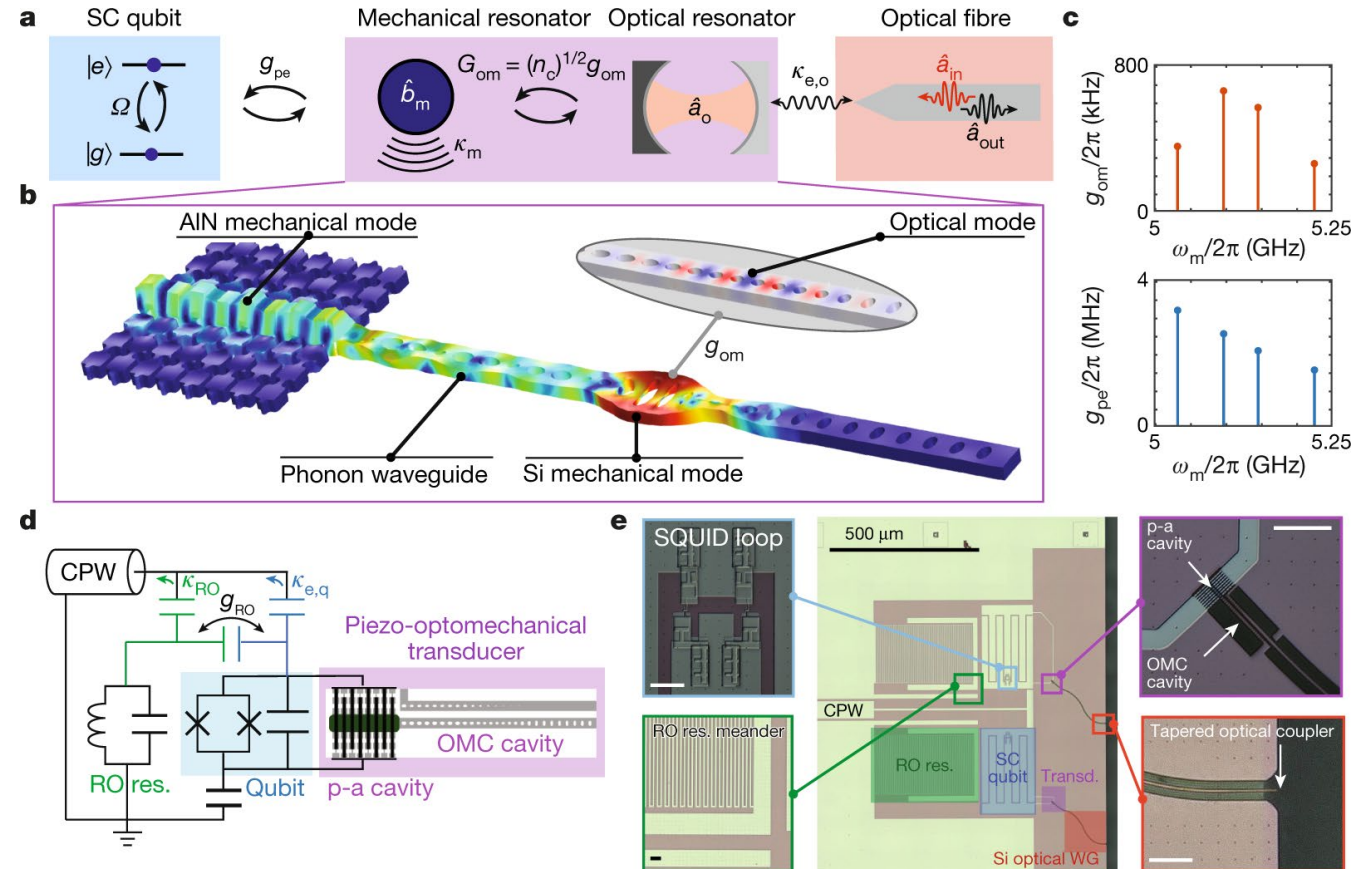
Transduction **efficiency just 1e-5 so very lossy!**

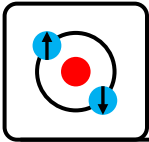
First demonstration of low noise transducer using piezo-optomechanical resonator and directly to telecom band.

Made possible by:

- Operating temperature 15mK
- Tiny optical cavity mode volume with few photons 2uW peak power giving very low heating
- Very low readout duty cycle (1e-5) to reduce heating from laser
- Fast qubit-mechanical phonon swap to minimize loss

Cooperativity ~ 1
(not higher to reduce heating)





Rydberg atom microwave-optical transducers

Quantum-enabled millimetre wave to optical transduction using neutral atoms

[Aishwarya Kumar](#) , [Aziza Suleymanzade](#), [Mark Stone](#), [Lavanya Taneja](#), [Alexander Anferov](#), [David I. Schuster](#) & [Jonathan Simon](#)

Nature **615**, 614–619 (2023) | [Cite this article](#)

Cold Rydberg atoms simultaneously coupled to both a mm-wave (100GHz) superconducting resonator and a high-finesse optical cavity

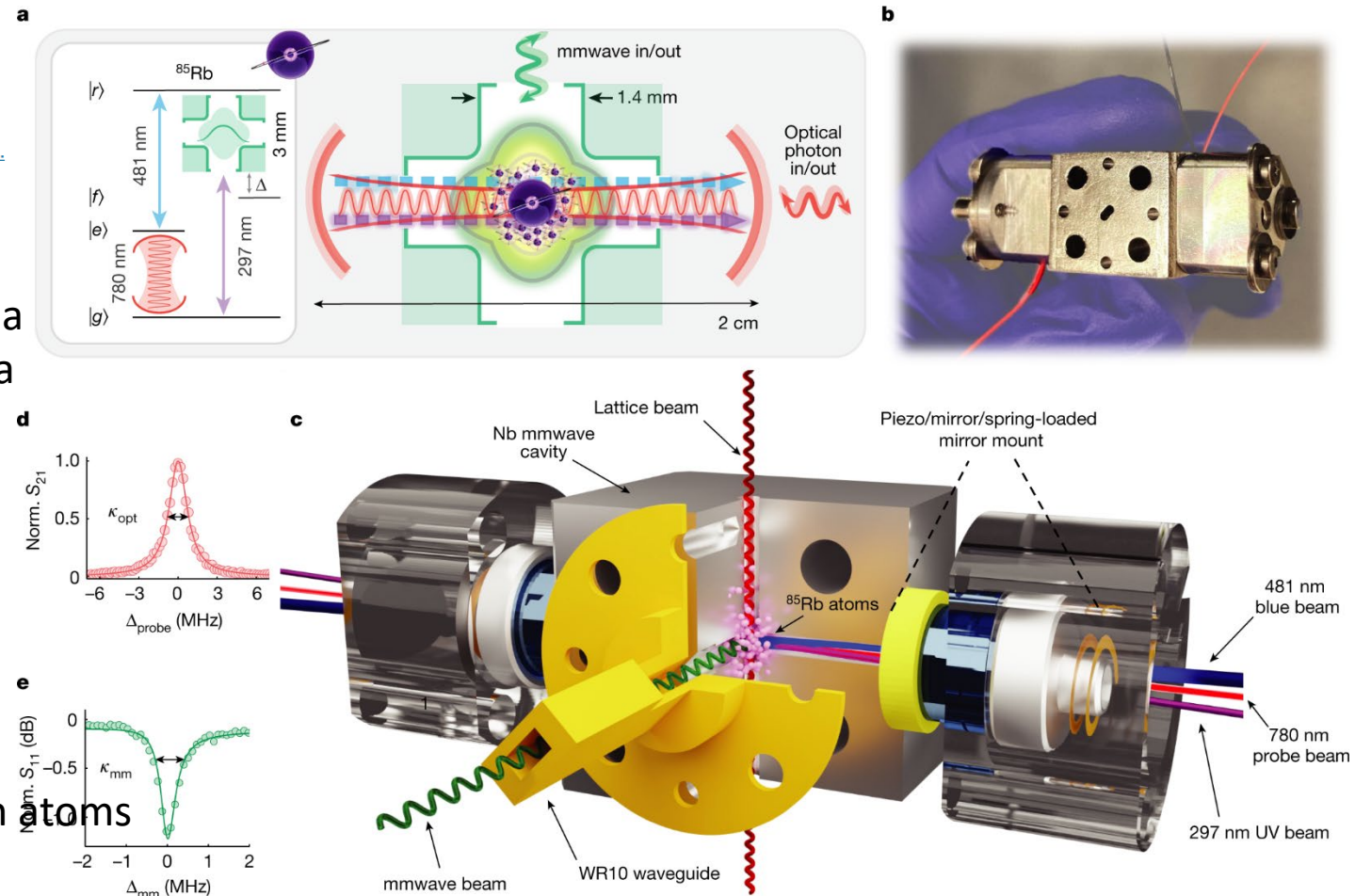
Added noise 0.6photons

360kHz bandwidth

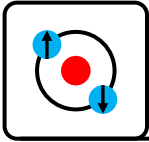
58% internal conversion efficiency (2.5% conversion)

Possible because:

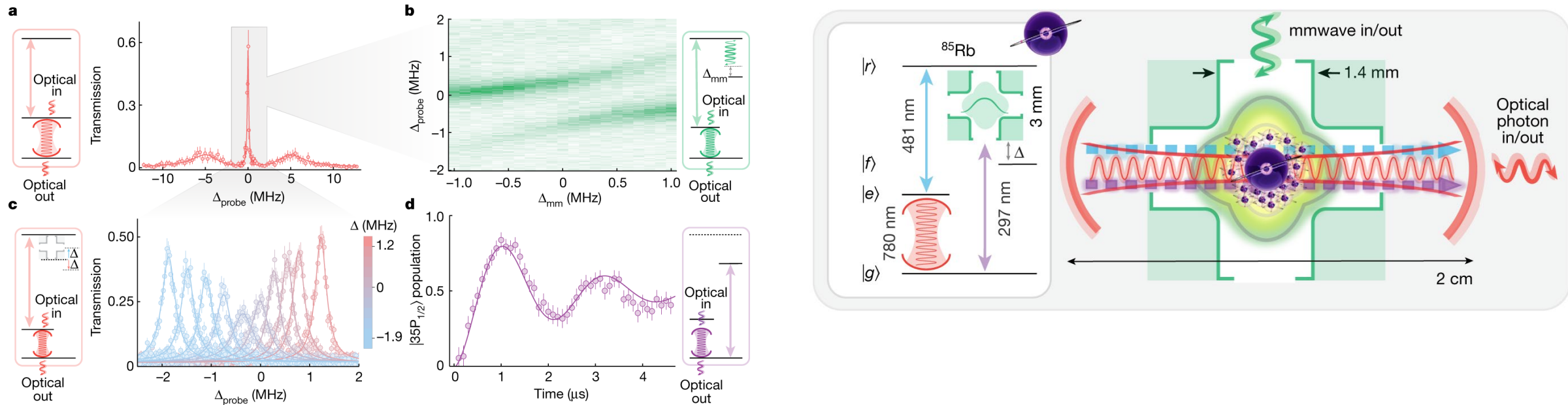
- Optical cavity creates superradiant emission from atoms
- $N \sim 500$ -600 atoms in cavity couple collectively
- Optical cooperativity ~ 3
- Mm-wave cooperativity 3-6
- 481nm cavity cooperativity 20-40
- Impedance matching gives near ideal efficiency



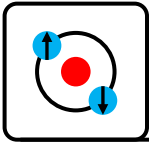
Quantum enabled noise with dramatically improved conversion efficiency!



Rydberg atom microwave-optical transducers



1. 780 nm probe couples $|g\rangle \rightarrow |e\rangle$, injecting optical cavity photons into the atomic ensemble.
2. 481 nm control beam couples $|e\rangle \rightarrow |r\rangle$, creating Rydberg-EIT and forming a dark polariton ($|e\rangle \leftrightarrow |r\rangle$ hybrid).
3. EIT dark polariton suppresses loss via 5P decay and provides a long-lived coherence linking optical photons to Rydberg excitations.
4. mm-wave cavity couples $|r\rangle \rightarrow |f\rangle$ via the $36S \leftrightarrow 35P$ transition, hybridizing the dark polariton with the 100 GHz resonator field.
5. 297 nm UV beam couples $|f\rangle \rightarrow |g\rangle$, closing the four-wave-mixing loop and returning the atom to the ground state.
6. Cycle converts a mm-wave photon into an optical cavity photon (and is fully reversible).
7. Impedance matching via EIT (tuning $\Omega_{(481)}$) ensures optimal mixing of optical and Rydberg components $\rightarrow$ high-efficiency transduction.



Quantum interfaces for superconducting qubits

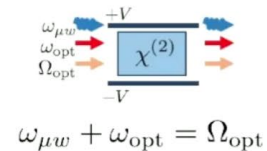
Entangling microwaves with light

R. SAHU , L. QIU , W. HEASE , G. ARNOLD , Y. MINOGUCHI, P. RABL , AND J. M. FINK

SCIENCE • 18 May 2023 • Vol 380, Issue 6646 • pp. 718-721 • DOI: 10.1126/science.adg3812

Mankei Tsang, Cavity Quantum Electro-Optics, Phys. Rev. A 81 (2010)

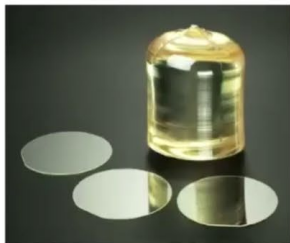
3-wave mixing via Pockels effect:



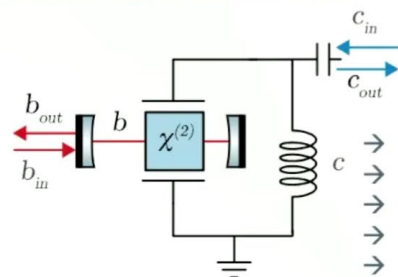
$$\omega_{\mu w} + \omega_{\text{opt}} = \Omega_{\text{opt}}$$

→ conversion, cooling, amplification, SPDC,...

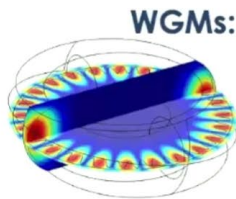
→ Need crystal like LN



Resonant enhancement:



- Spatial mode matching
- Phase matching
- Corr. polarization / tensor
- Power efficient
- Select desired process



WGMs:



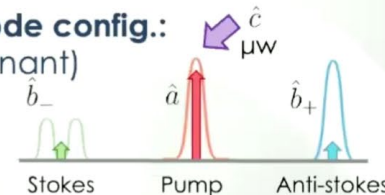
Cavity optomechanics physics!

$$x \rightarrow v_{\mu w}, p \rightarrow I_{\mu w} \propto g(c + c^\dagger)b^\dagger b$$

$$\text{Cooperativity: } C = 4|\alpha_p|^2 g^2 / (\kappa_o \kappa_e)$$

→ So far: $C \ll 1$

Actual mode config.: (triply resonant)



TMS interaction

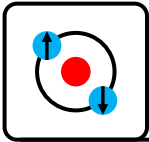
Beam splitter

$$\hat{H}_{\text{int}} = \hbar g(\hat{a}\hat{b}_-^\dagger\hat{c}^\dagger + \hat{a}^\dagger\hat{b}_-\hat{c}) + \hbar g(\hat{a}\hat{b}_+^\dagger\hat{c} + \hat{a}^\dagger\hat{b}_+\hat{c}^\dagger)$$

Use a pulsed (1W) pump (for 100ns) to achieve efficient (99%) conversion of microwave photon to an optical photon
15% effective conversion efficiency due to in/out coupling losses

Only 0.4Hz entanglement at these performances due to heating

Cooperativity ~ 1 (close to strong interaction between optical and microwave photons)

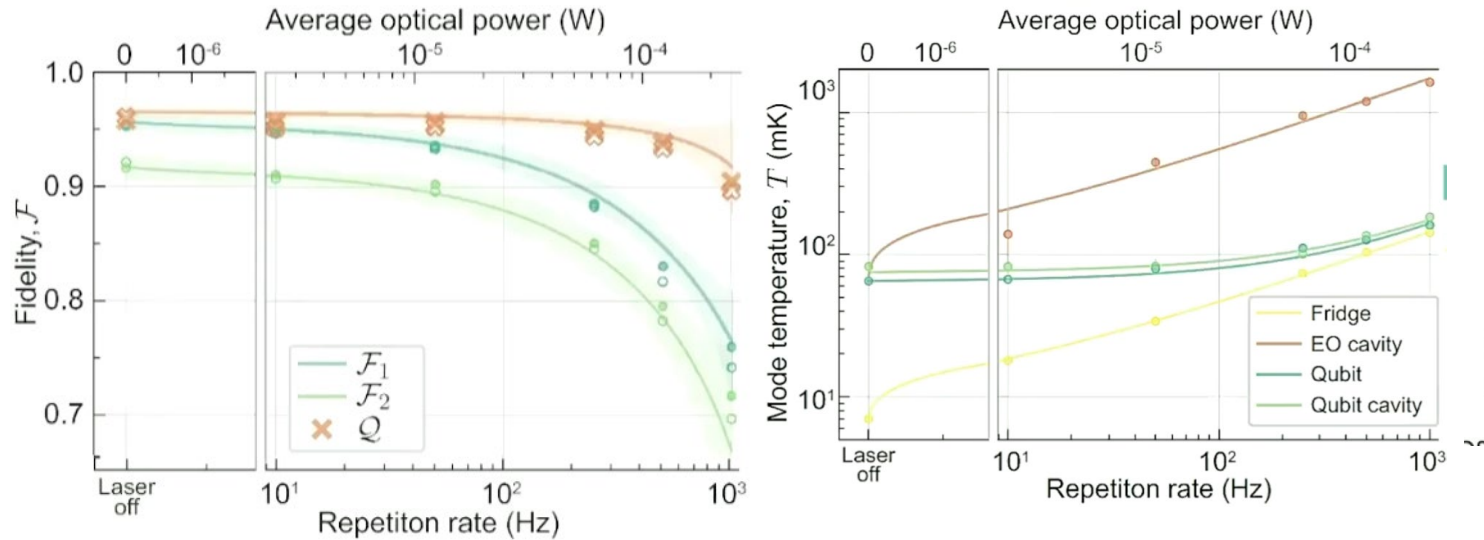


Quantum interfaces for superconducting qubits

All-optical superconducting qubit readout

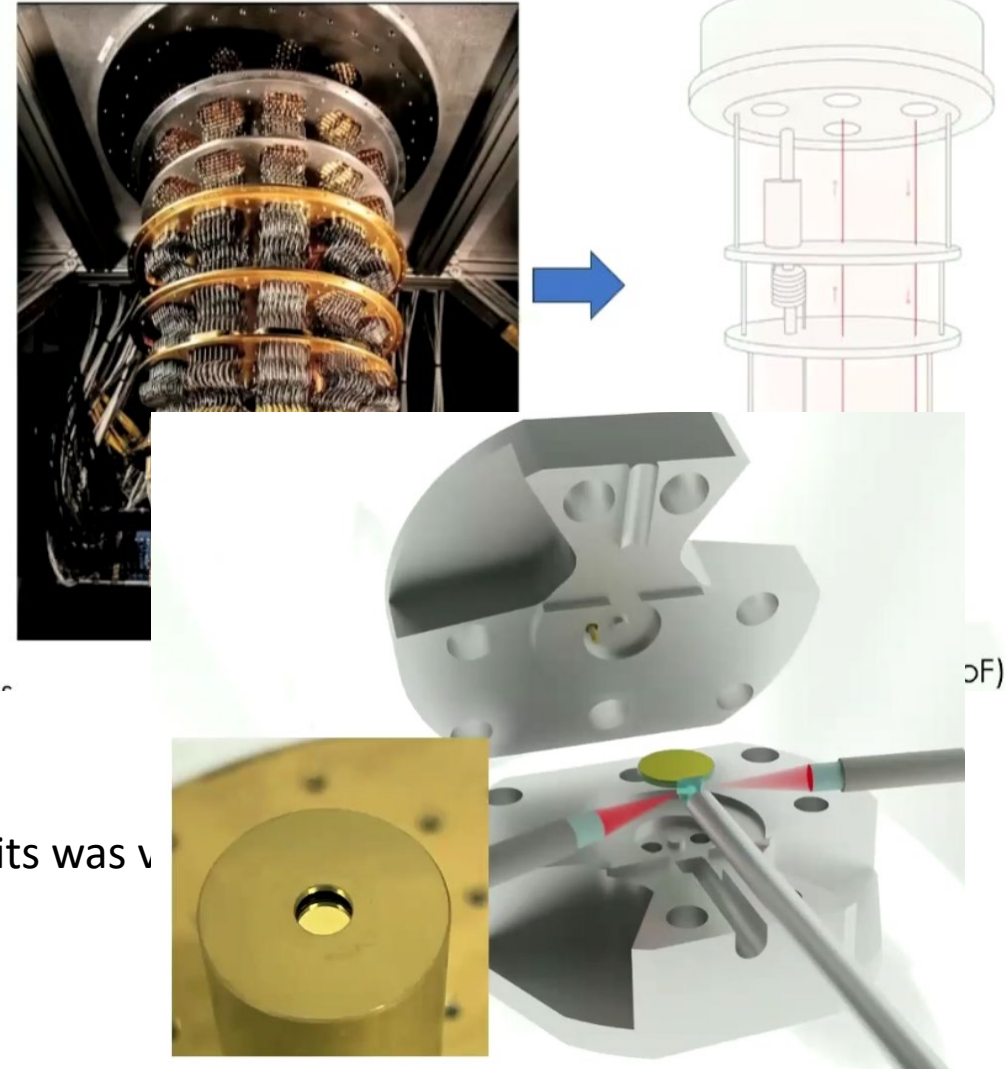
[Georg Arnold](#) , [Thomas Werner](#), [Rishabh Sahu](#), [Lucky N. Kapoor](#), [Liu Qiu](#) & [Johannes M. Fink](#) 

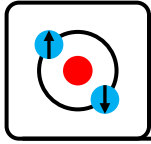
[Nature Physics](#) **21**, 393–400 (2025) | [Cite this article](#)



Demonstrated that the loss in coherence time of superconducting qubits was v
-> Challenge is how to reduce the pump power to reduce heating...

How about **removing the clutter** in the fridge with **cryogenic microwave photonics**?





Experimental Progress

Progress and metrics

References: